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Eitschberger et al.

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(54) **BALLISTICALLY ACTUATED WELLBORE TOOL**

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Troisdorf (DE)

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patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

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PCT Pub. Date: **May 4, 2023**

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Related U.S. Application Data

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31, 2022, provisional application No. 63/271,466,
filed on Oct. 25, 2021.

(51) **Int. Cl.**

E21B 23/04 (2006.01)

E21B 33/12 (2006.01)

E21B 43/116 (2006.01)

(52) **U.S. Cl.**

CPC **E21B 23/0414** (2020.05); **E21B 33/12**
(2013.01); **E21B 43/116** (2013.01)

(58) **Field of Classification Search**

CPC **E21B 23/0414**; **E21B 33/134**; **E21B 29/10**;
E21B 23/065; **E21B 43/116**; **E21B 33/12**
See application file for complete search history.

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166/123
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2020/0115978 A1 * 4/2020 Mickey **E21B 23/065**

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* cited by examiner

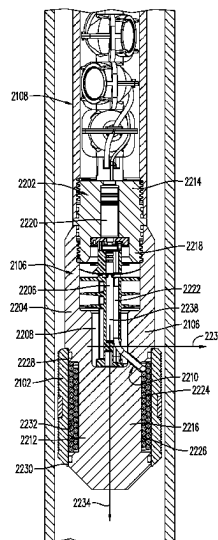
Primary Examiner — Yong-Suk (Philip) Ro

(74) *Attorney, Agent, or Firm* — Womble Bond Dickinson
(US) LLP

(57) **ABSTRACT**

A ballistically actuated plug (2100) for being deployed in a wellbore casing includes a ballistic housing (2106) and an outer seal housing (2102). The ballistic housing includes a first housing portion including a chamber (2208), and a second housing portion having a noncompressible core. The outer seal housing is circumferentially disposed around the noncompressible core (2216). According to an aspect, the initiator (2206) is positioned within the chamber and a detonation extender (2222) is in ballistic communication with the initiator and is circumferentially disposed around the noncompressible core. The detonation extender is configured to, upon detonation, expand the outer seal housing (2102) from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing.

14 Claims, 36 Drawing Sheets



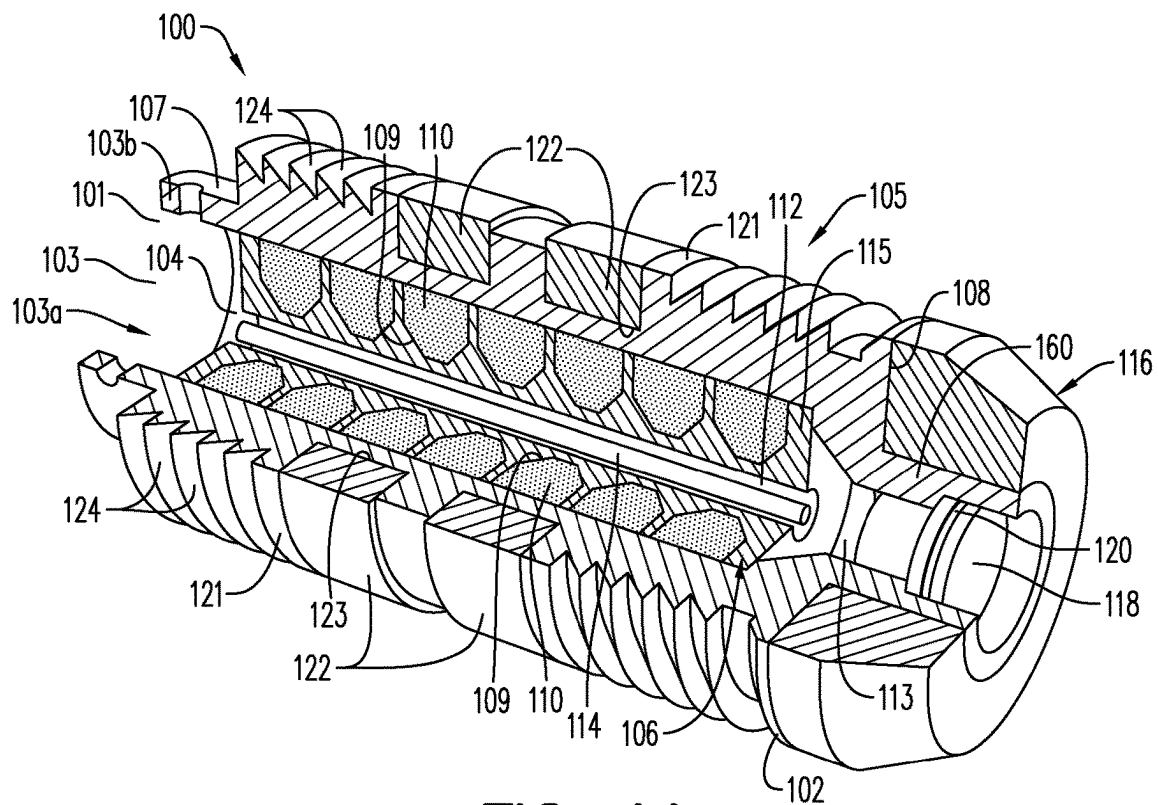


FIG. 1A

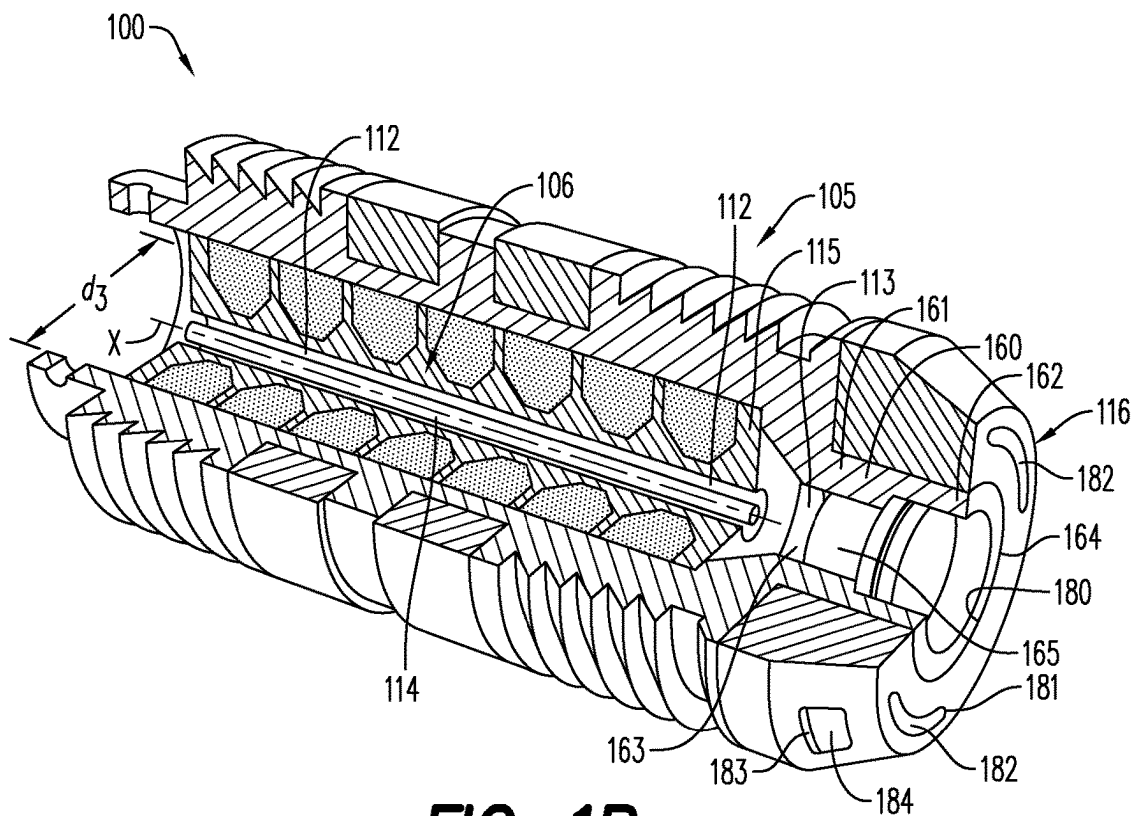


FIG. 1B

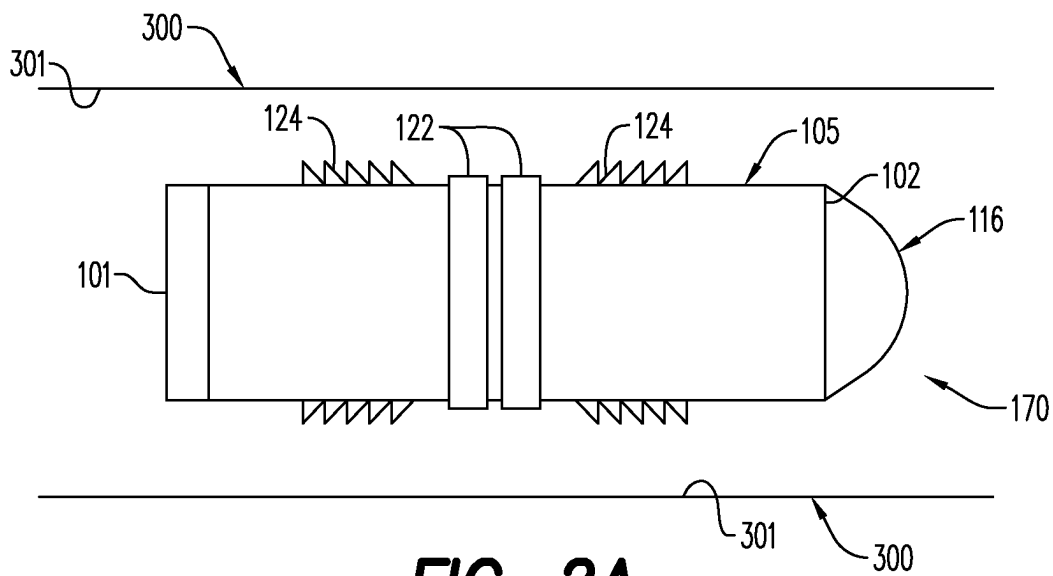


FIG. 2A

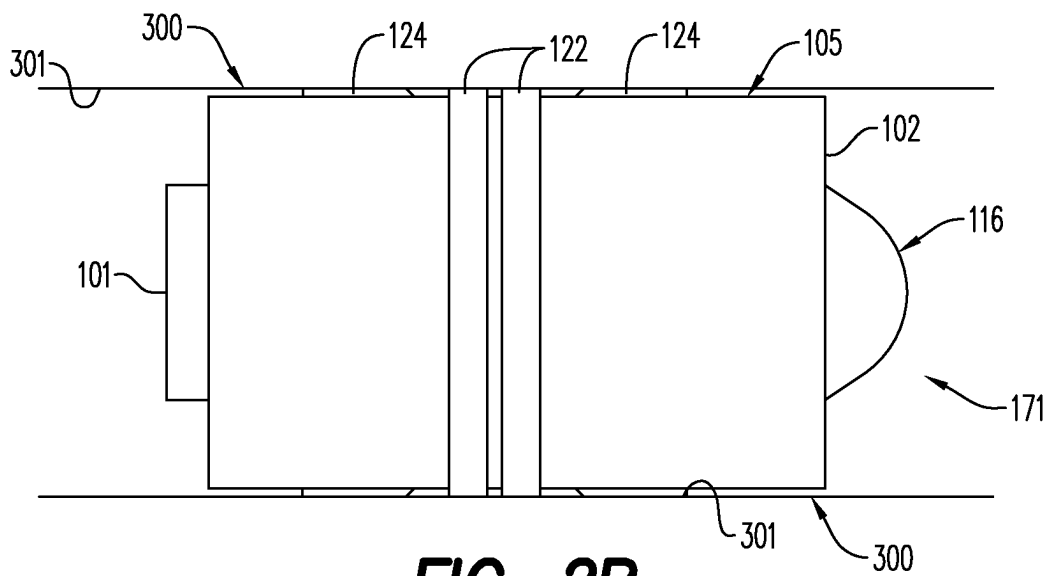


FIG. 2B

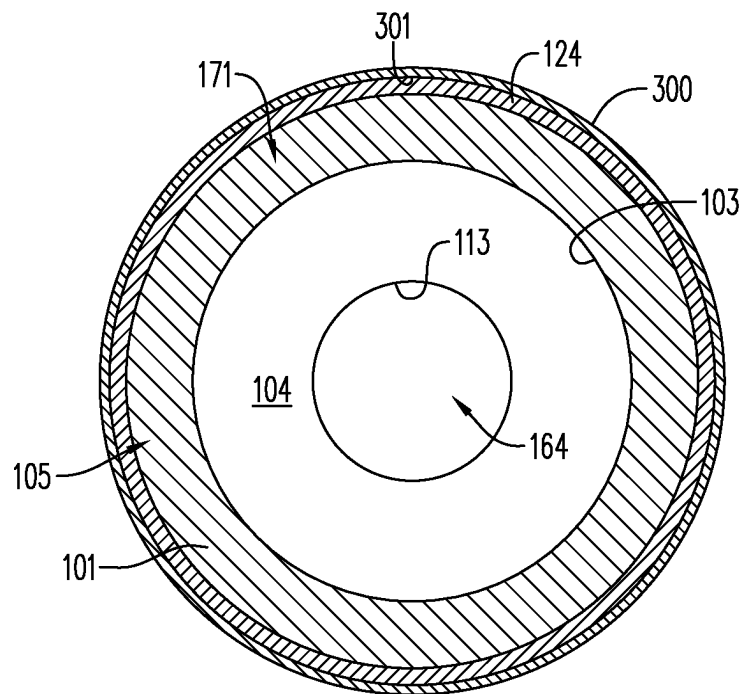


FIG. 2C

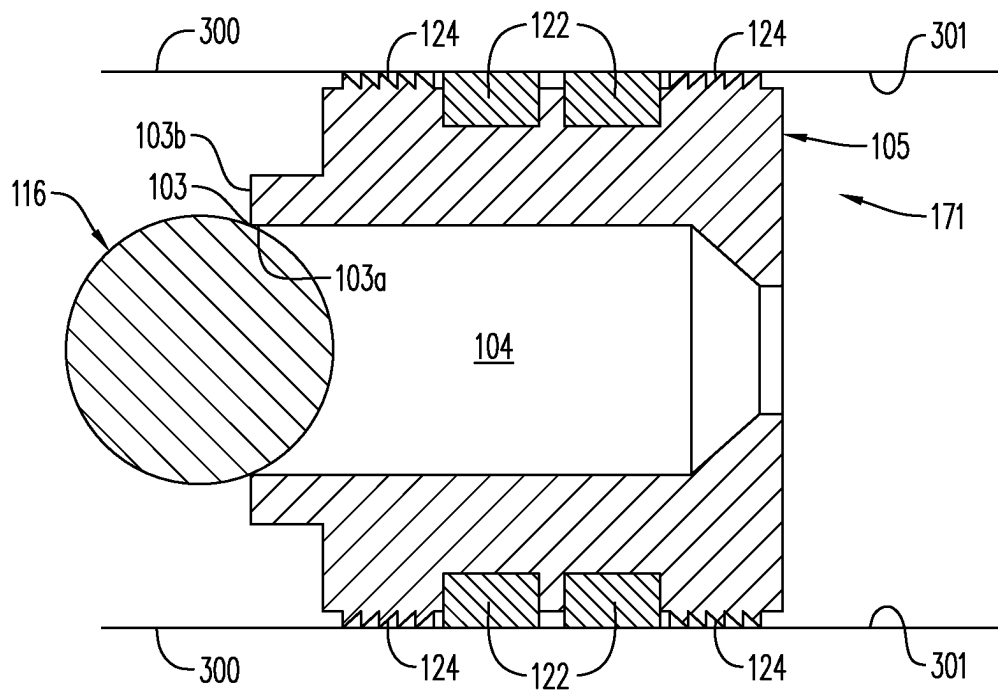


FIG. 2D

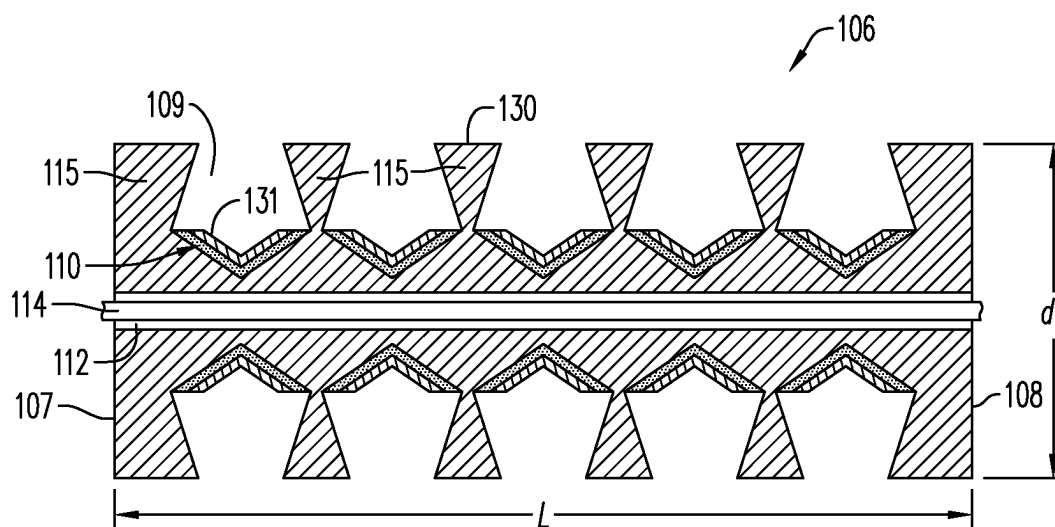


FIG. 3

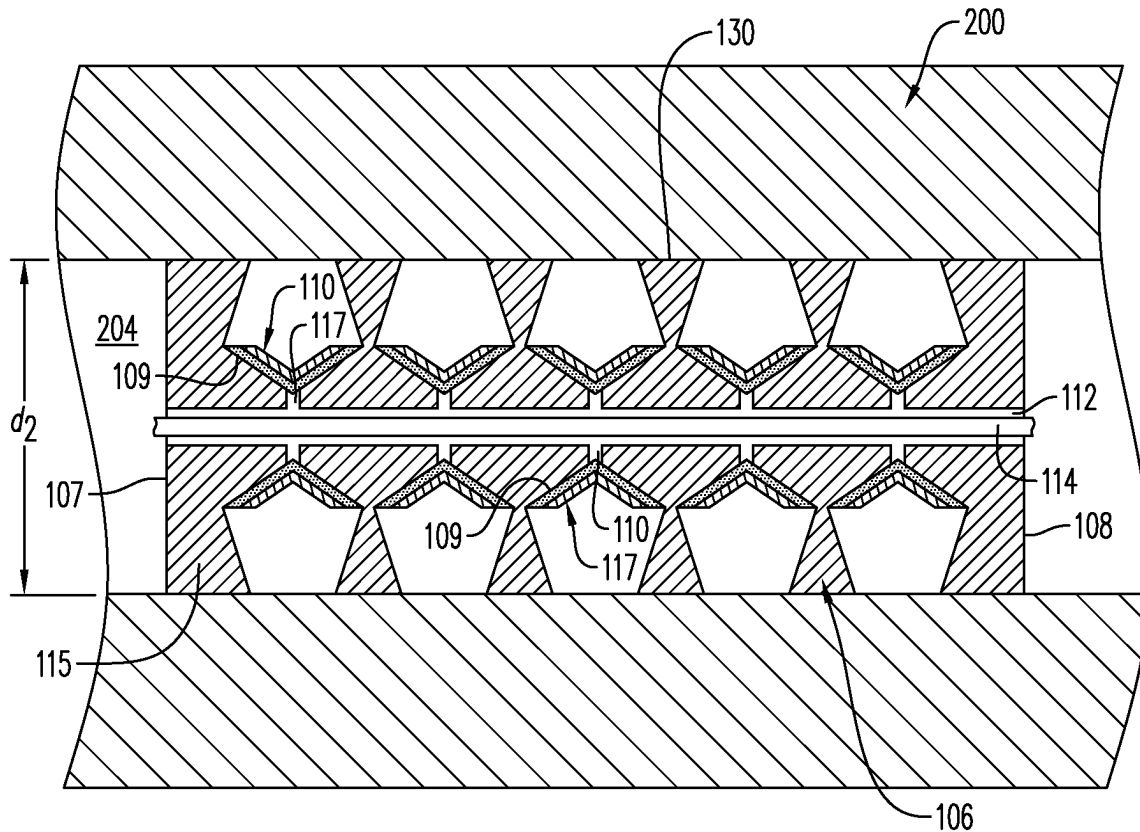


FIG. 4

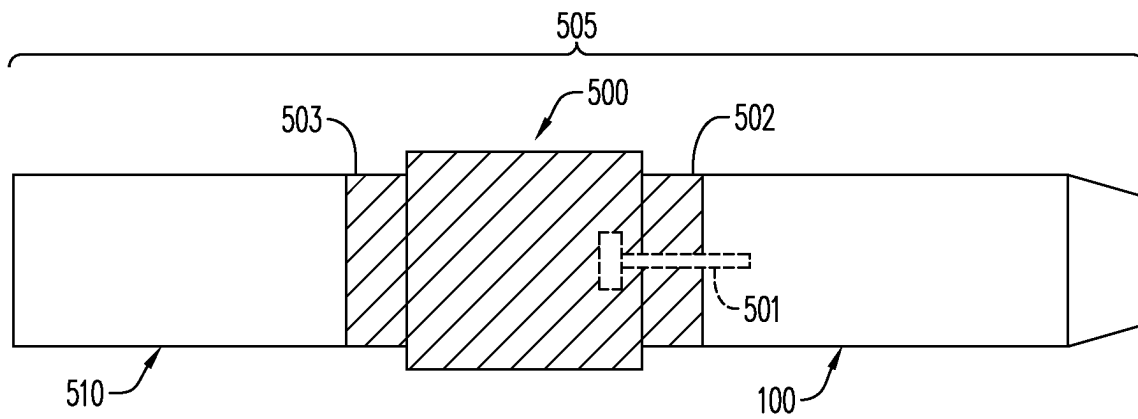


FIG. 5A

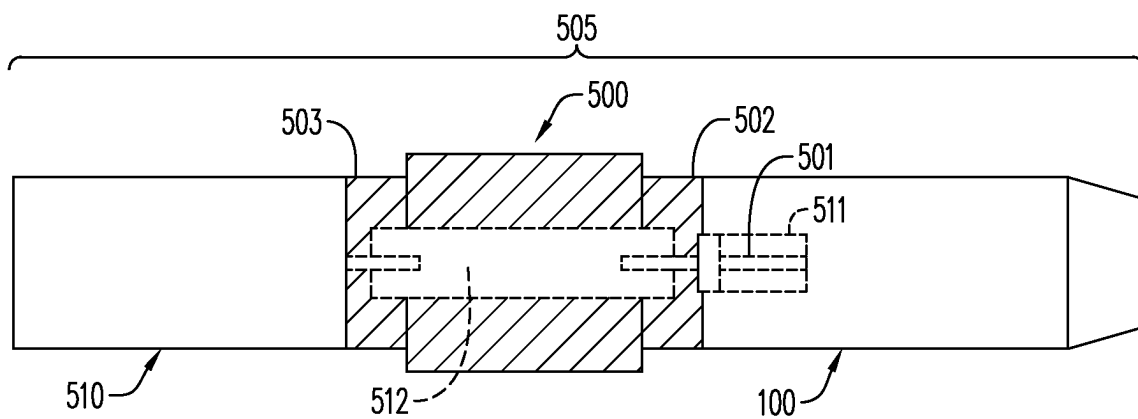
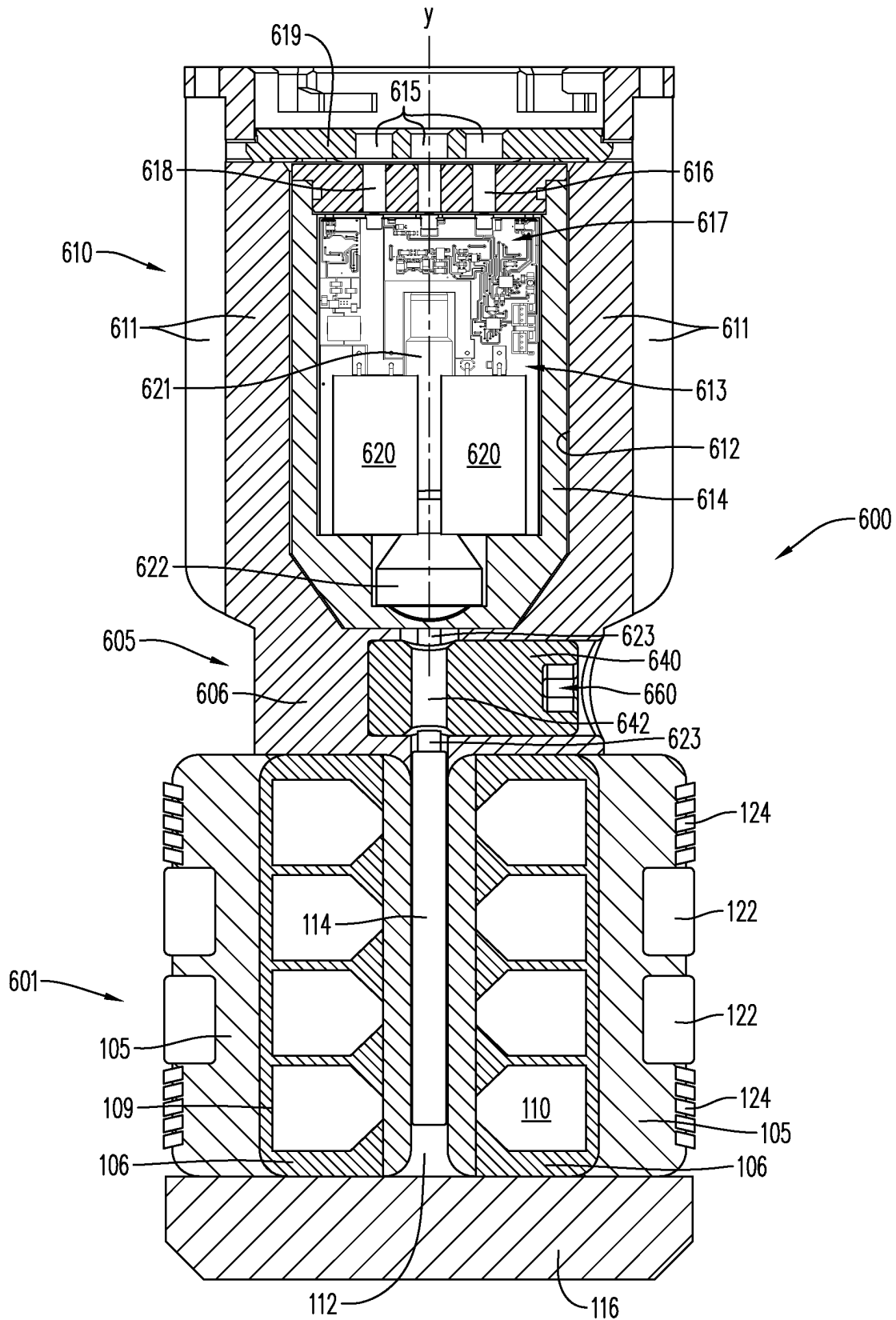


FIG. 5B

FIG. 5C

**FIG. 6**

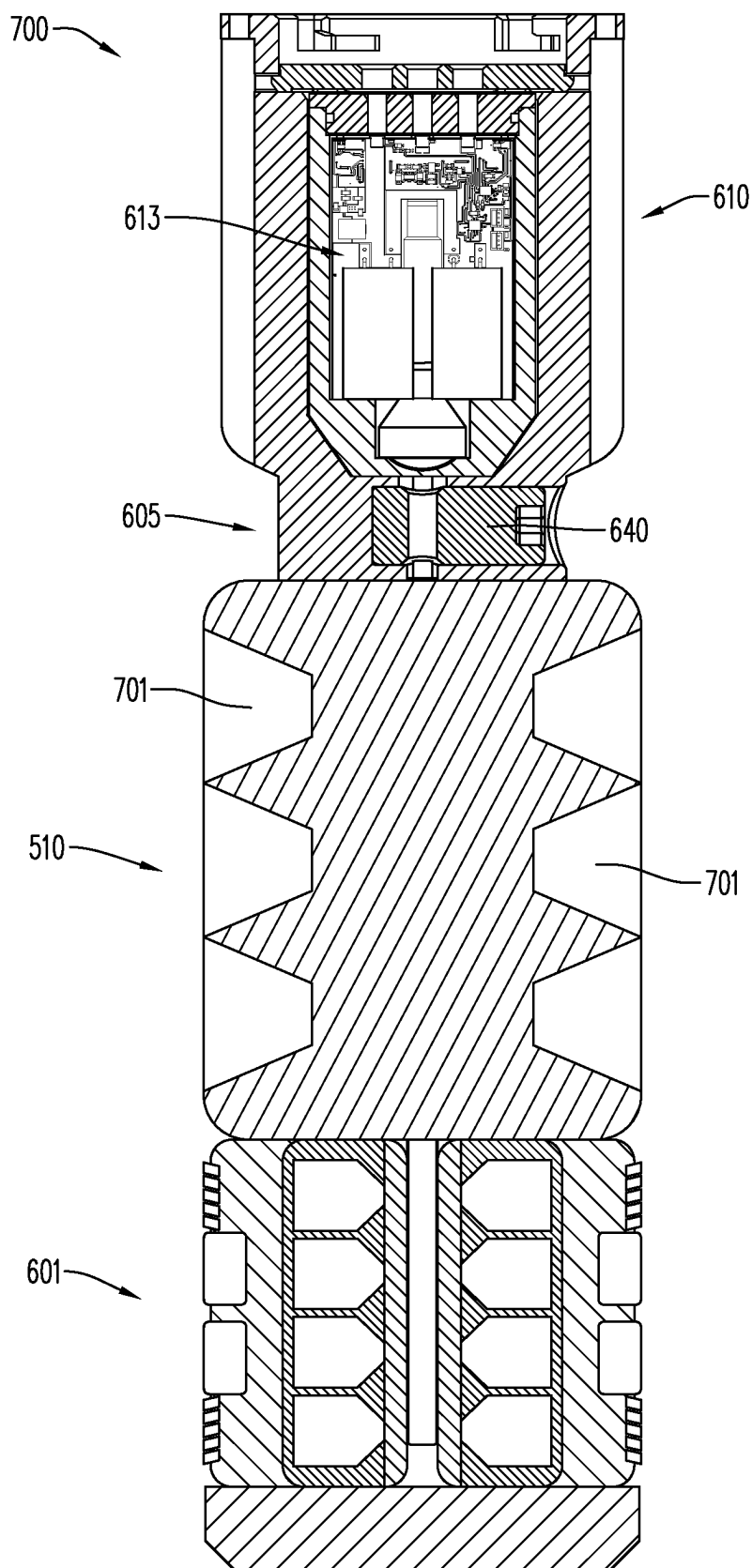


FIG. 7

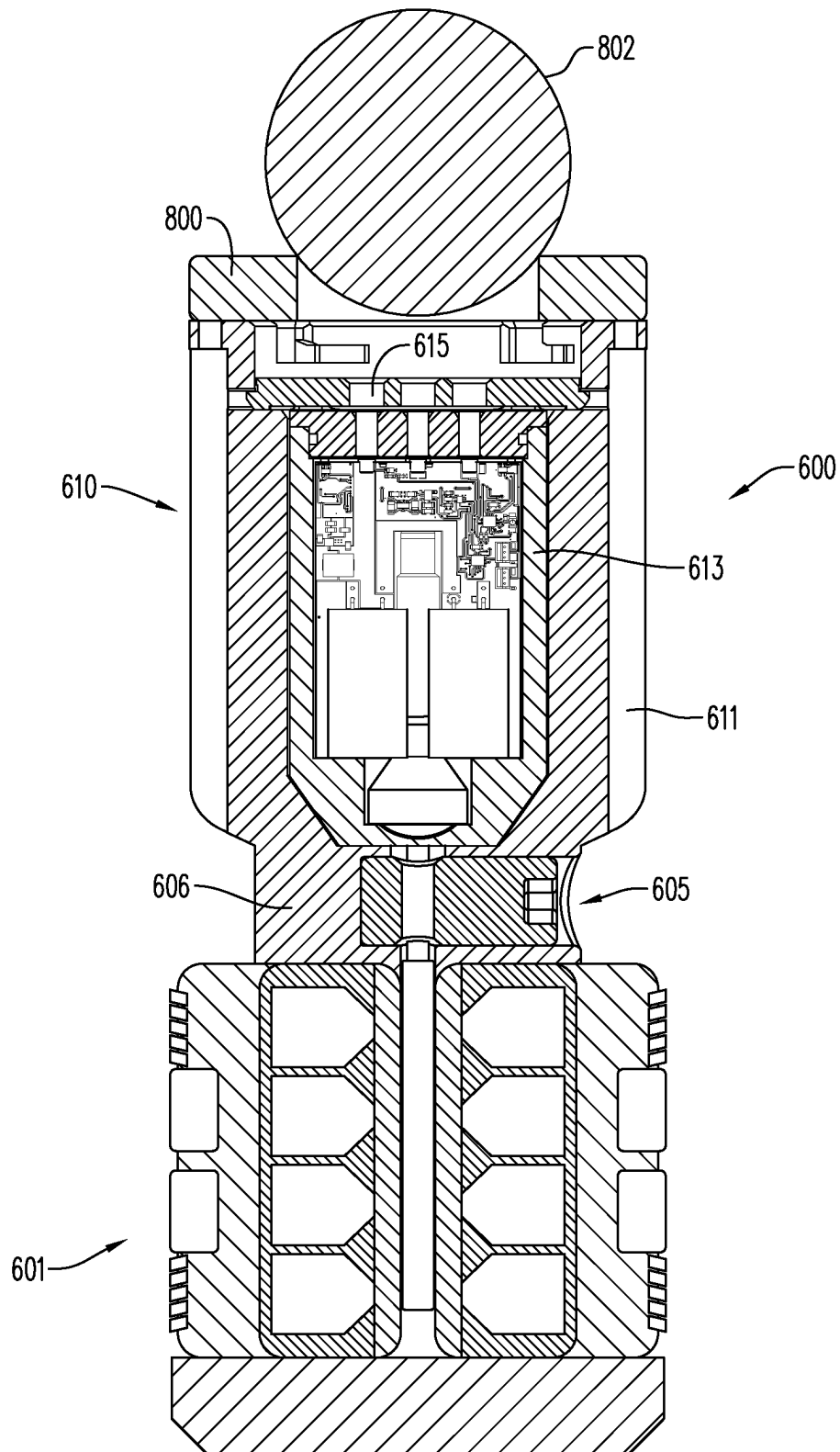


FIG. 8

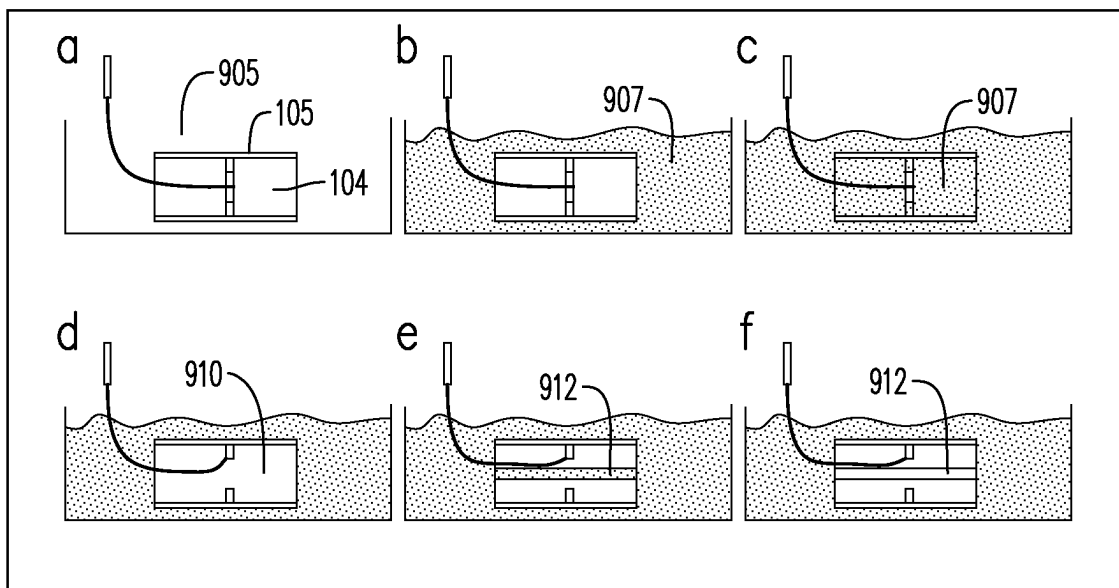


FIG. 9

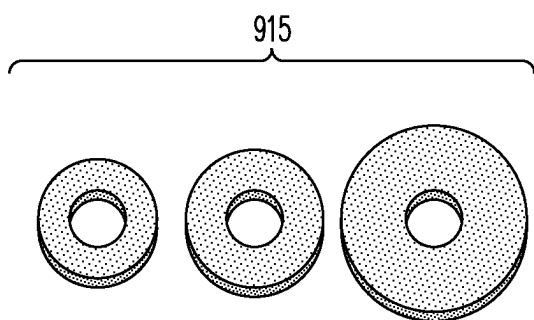


FIG. 10A

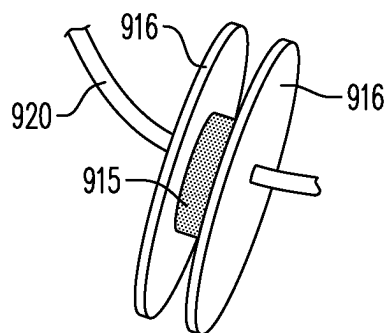


FIG. 10B

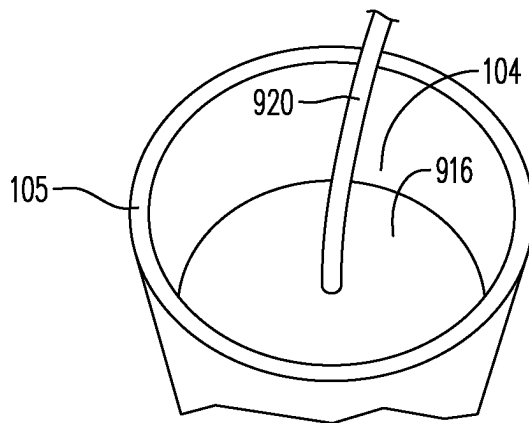


FIG. 11A

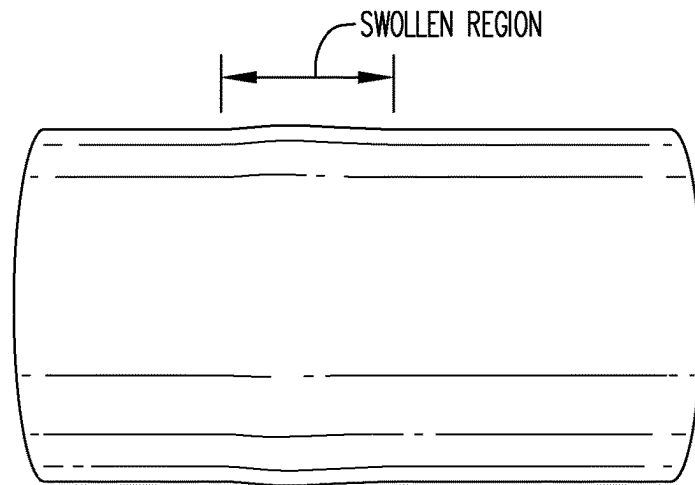


FIG. 11B

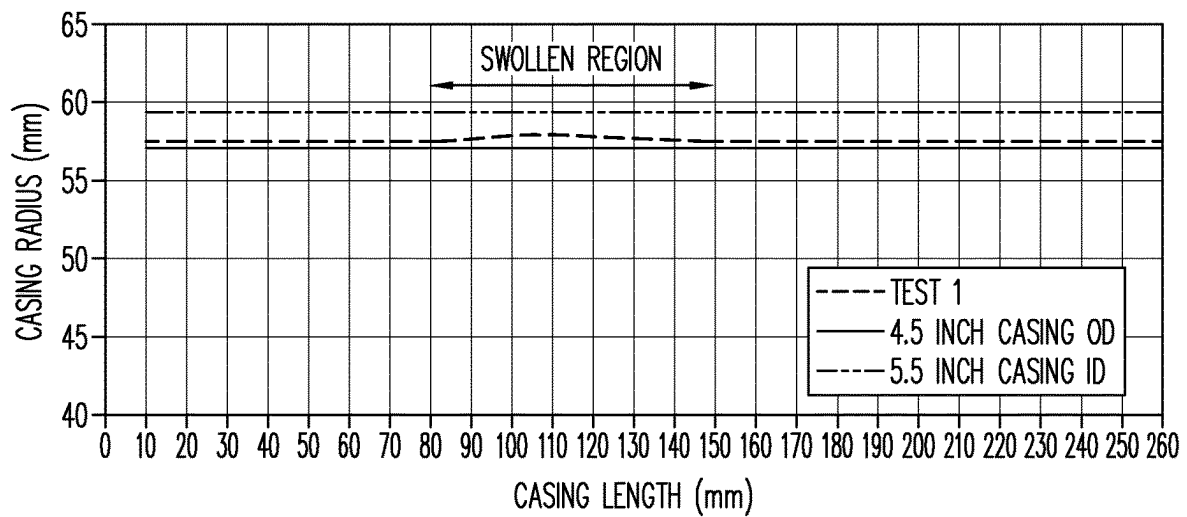


FIG. 11C

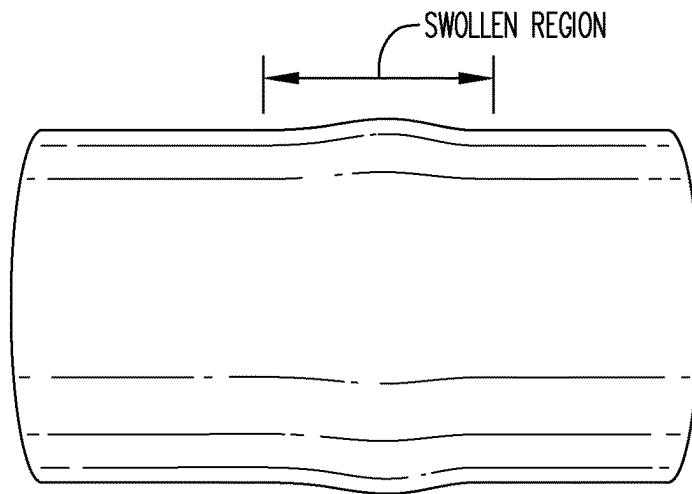


FIG. 11D

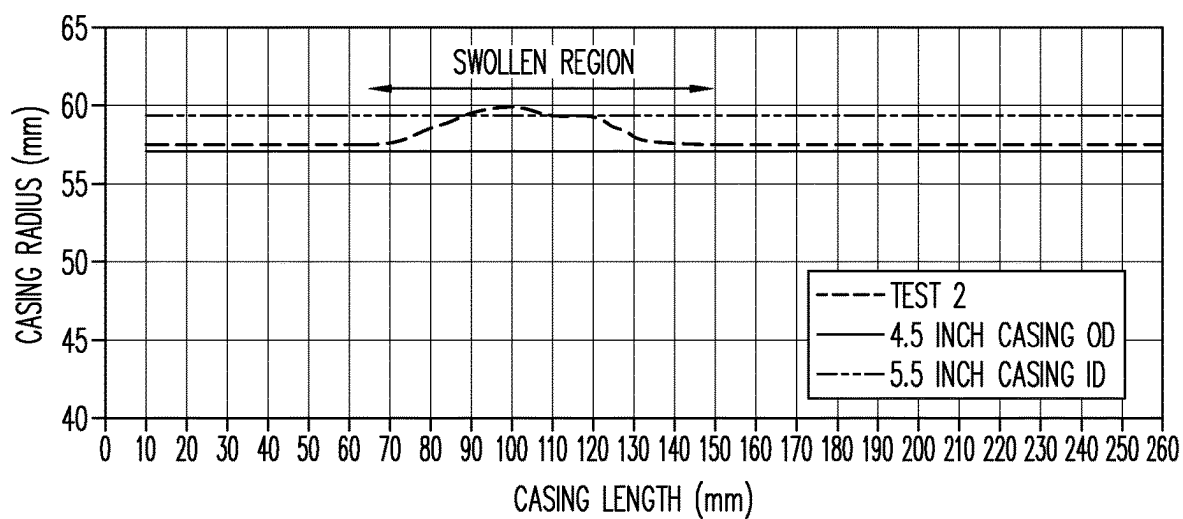


FIG. 11E

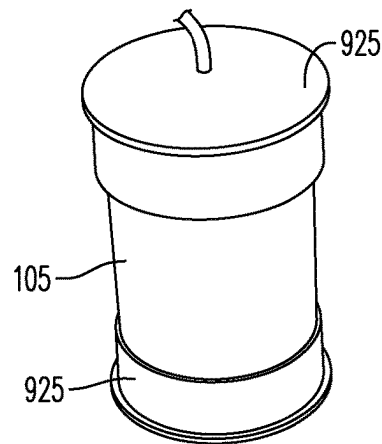


FIG. 12A

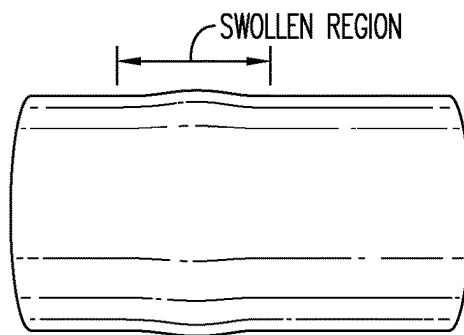


FIG. 12B

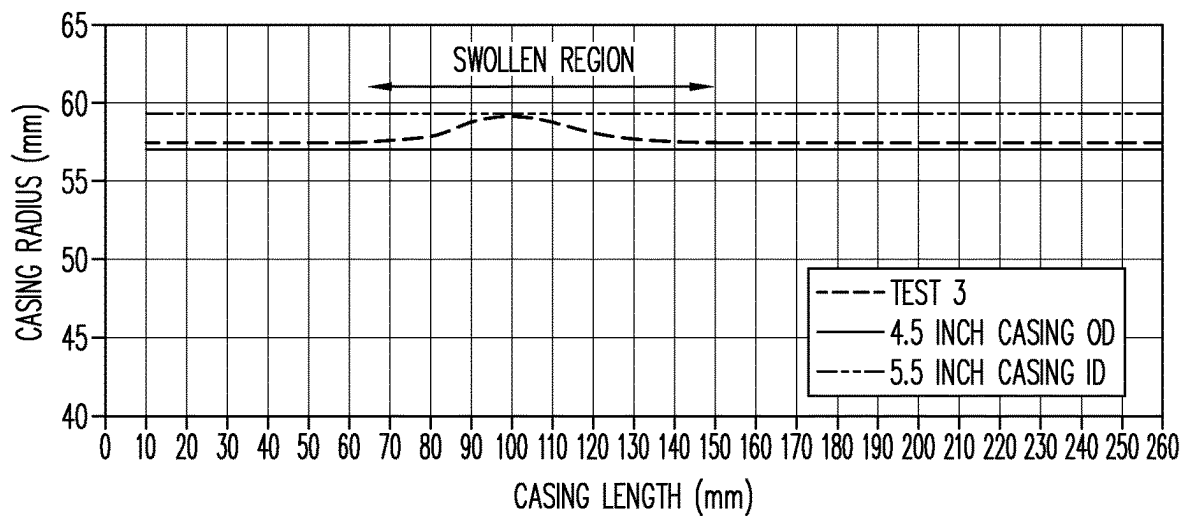


FIG. 12C

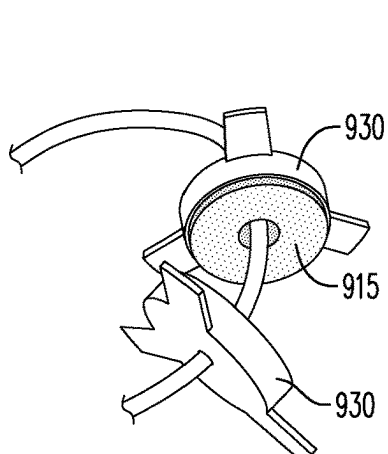


FIG. 13A

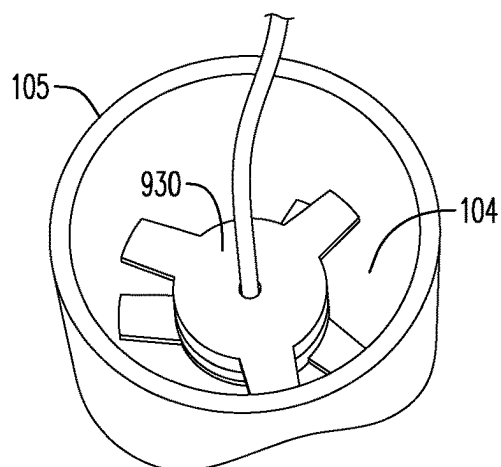


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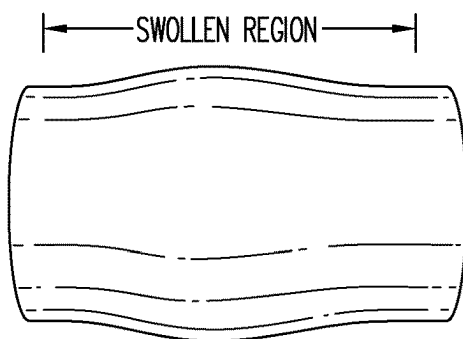


FIG. 13C

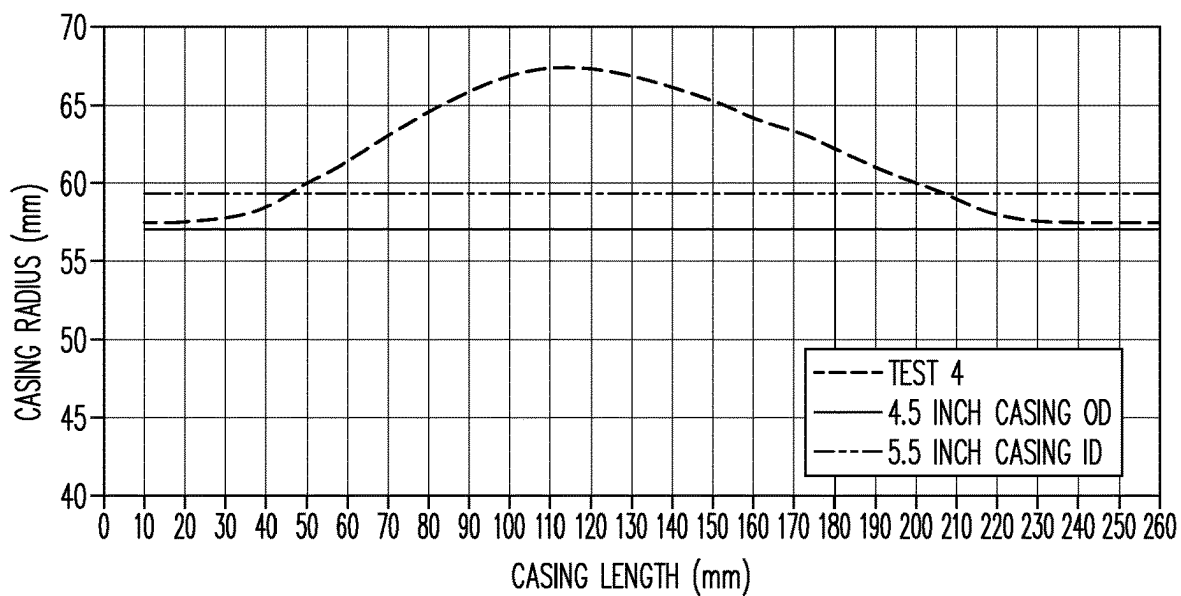
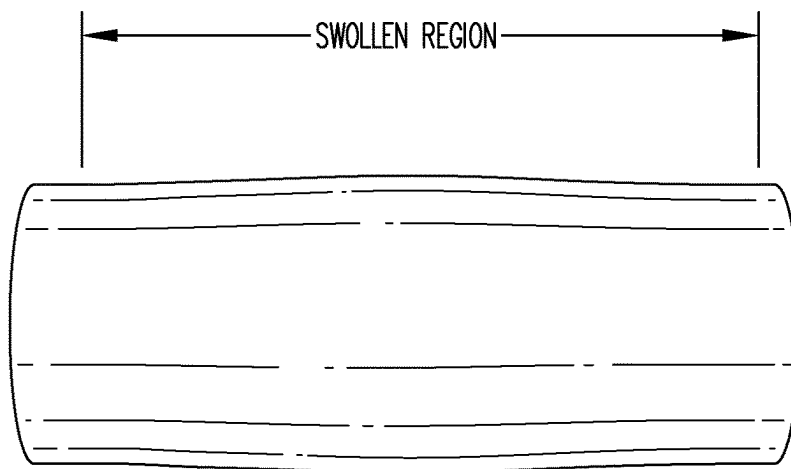
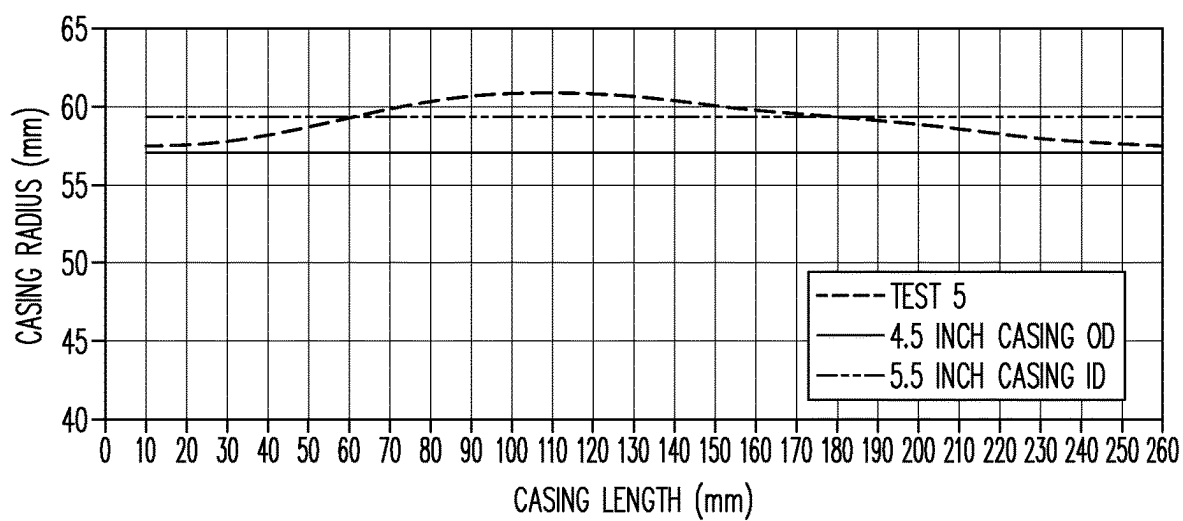


FIG. 13D

**FIG. 13E****FIG. 13F**

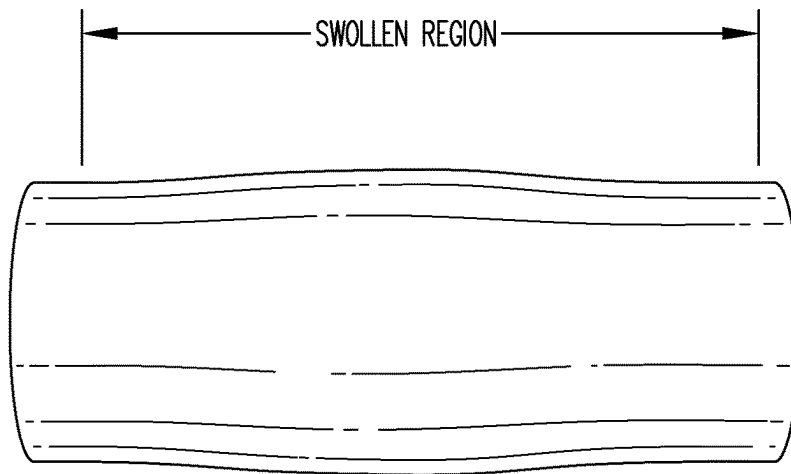


FIG. 13G

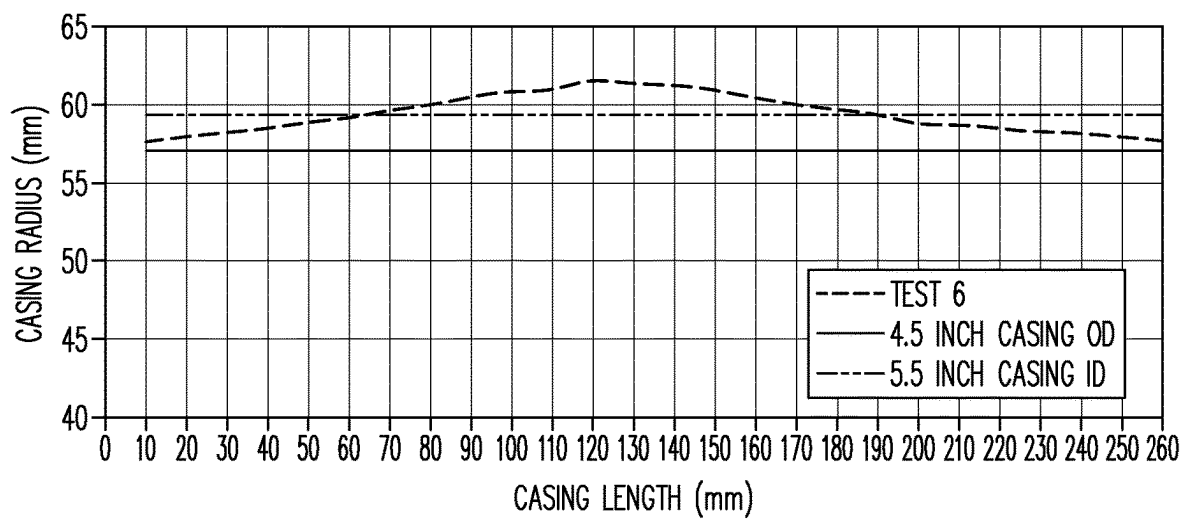


FIG. 13H

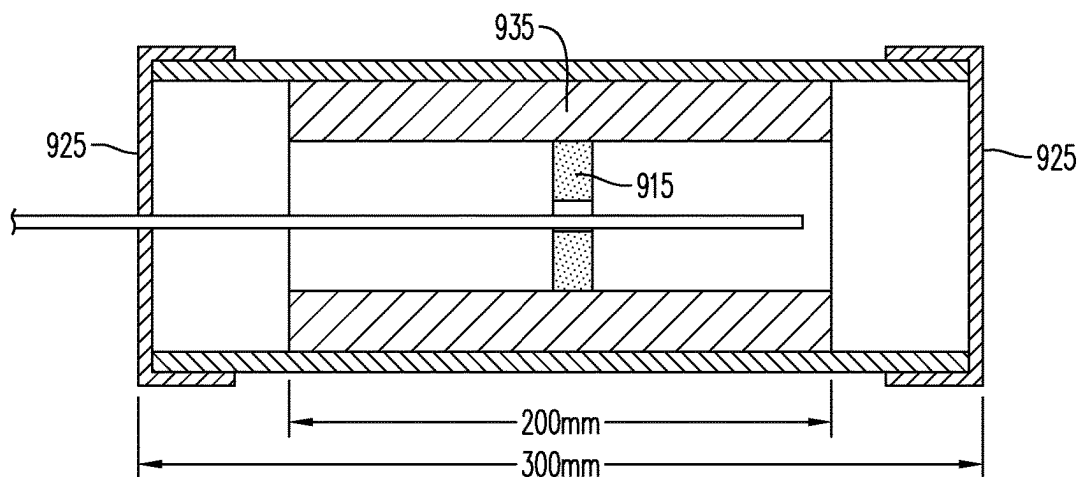


FIG. 14

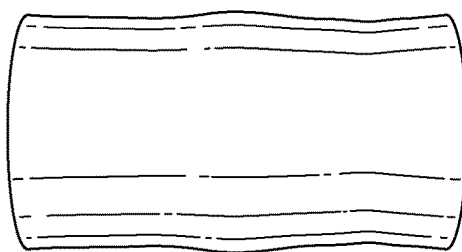


FIG. 15A

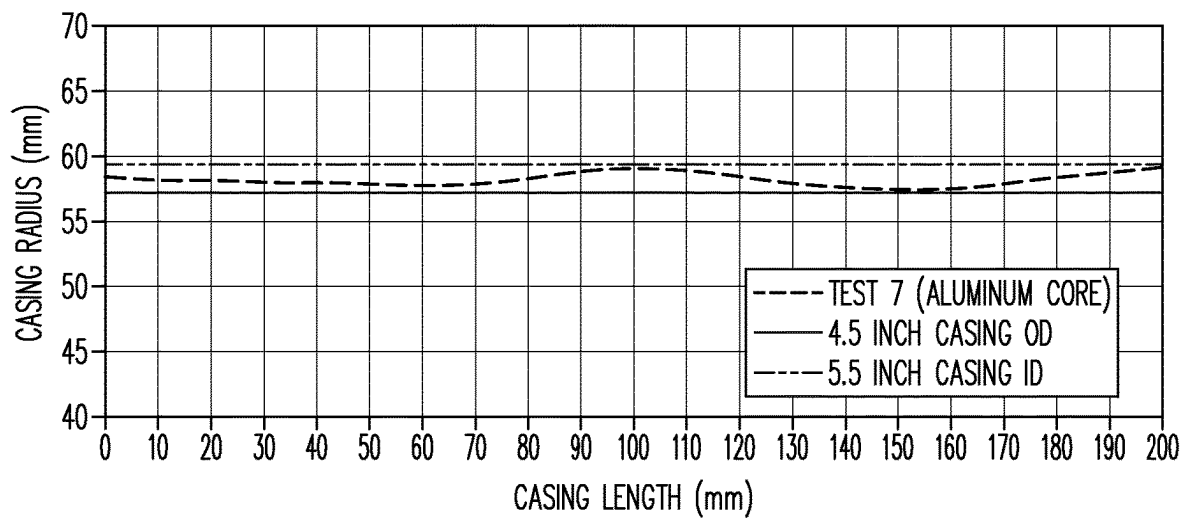
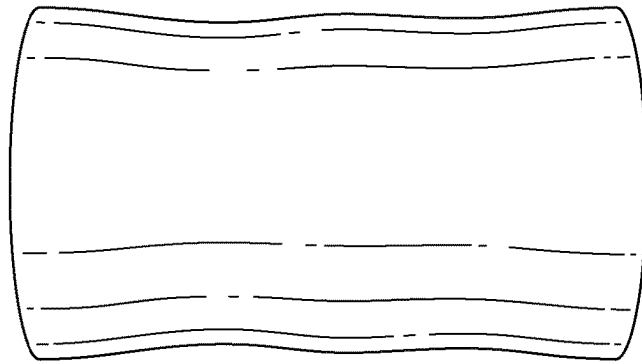
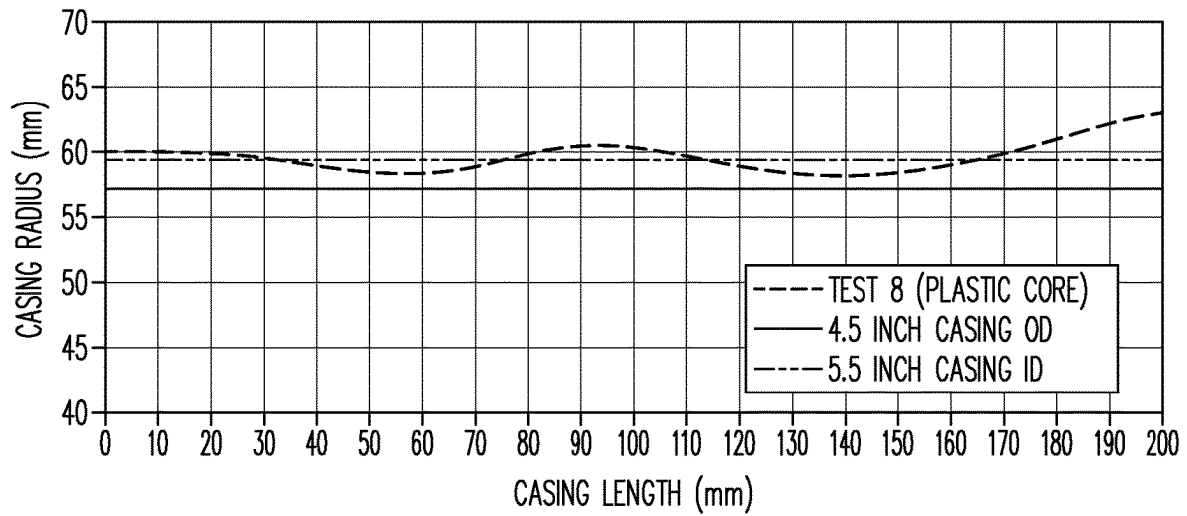
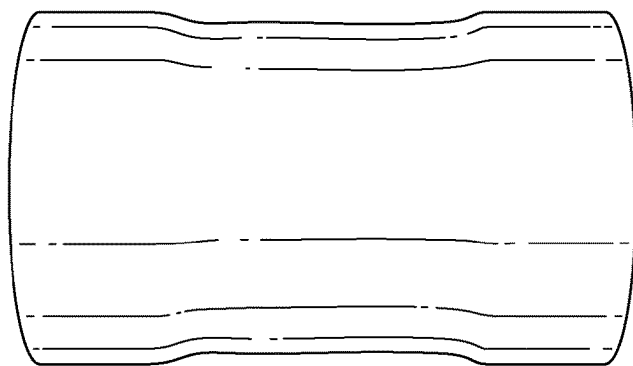
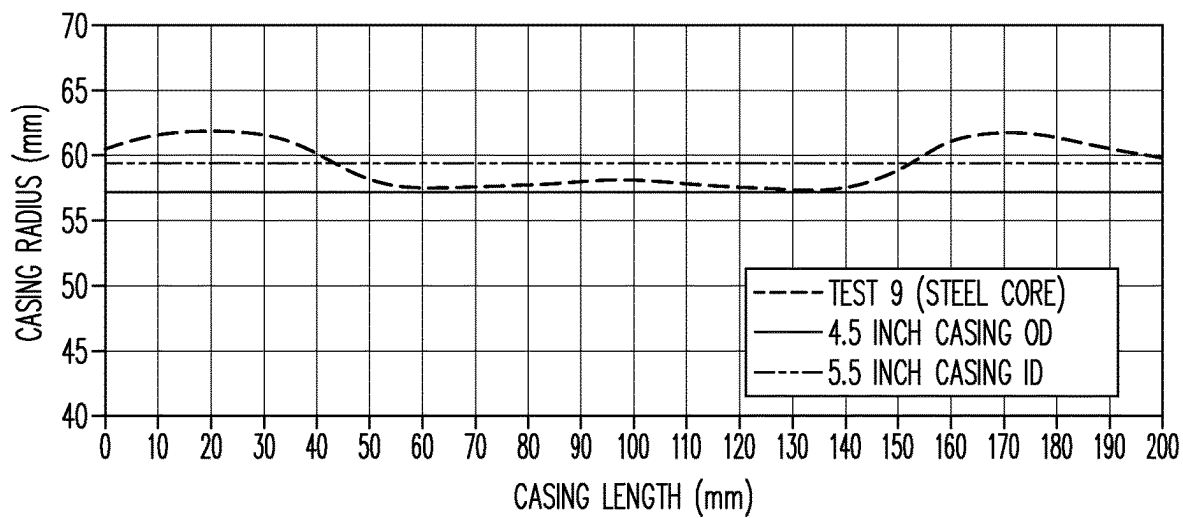
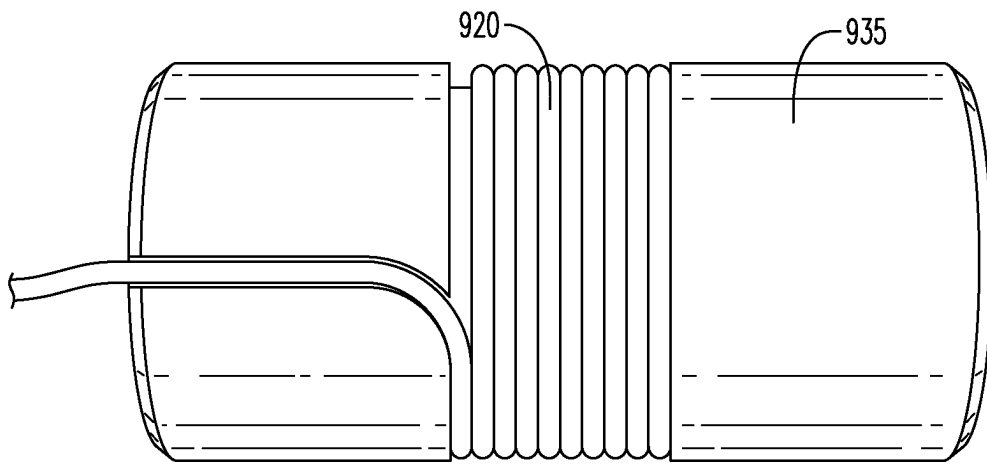
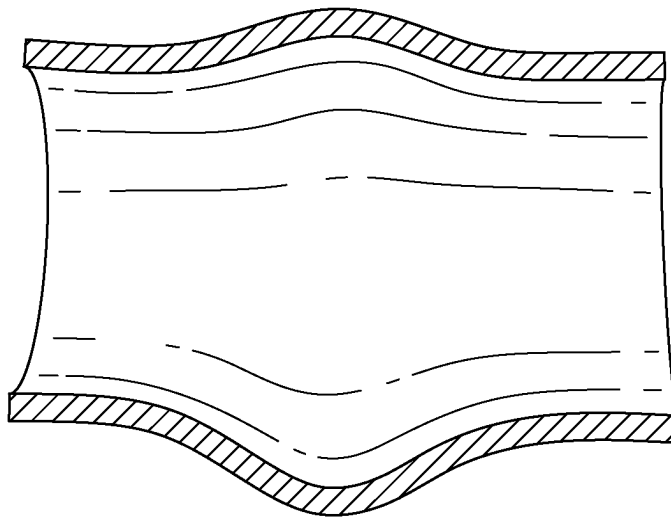


FIG. 15B

**FIG. 15C****FIG. 15D**

**FIG. 15E****FIG. 15F**

**FIG. 16A****FIG. 16B**

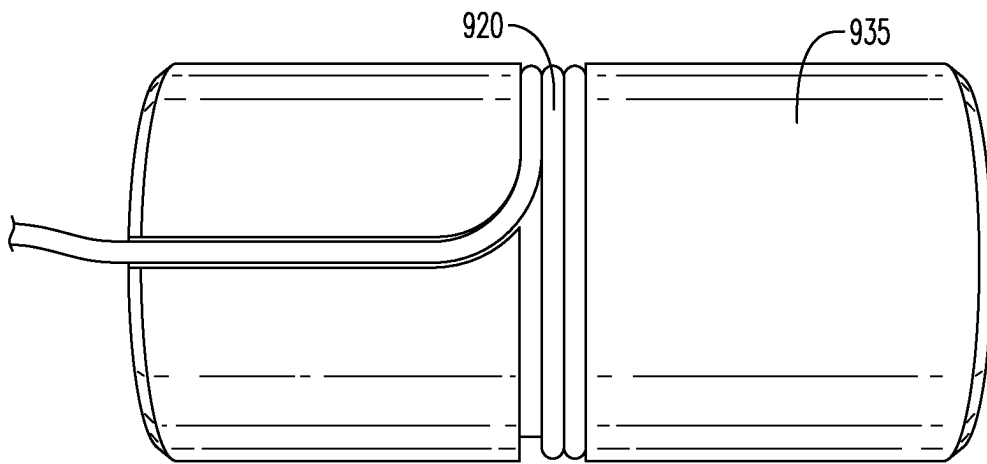


FIG. 16C

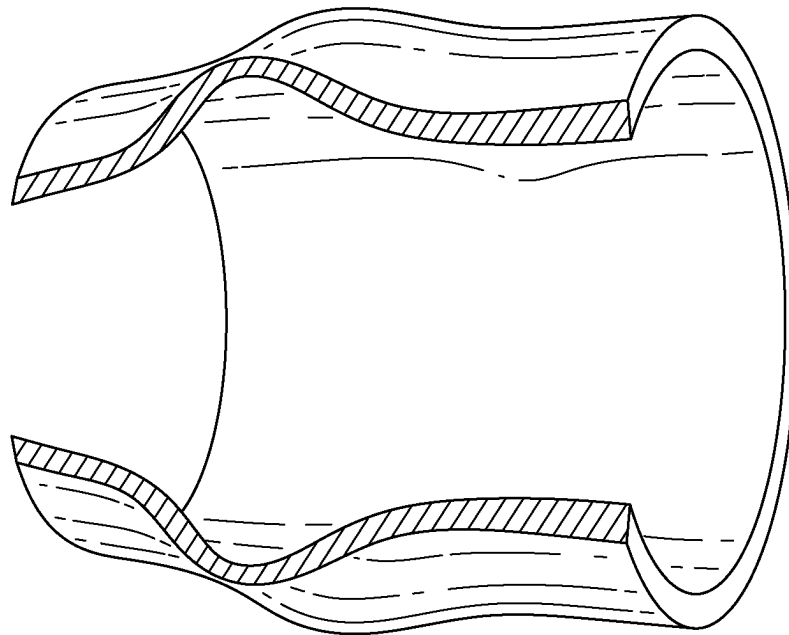


FIG. 16D

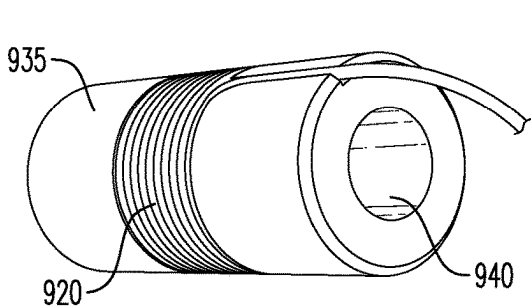


FIG. 17A

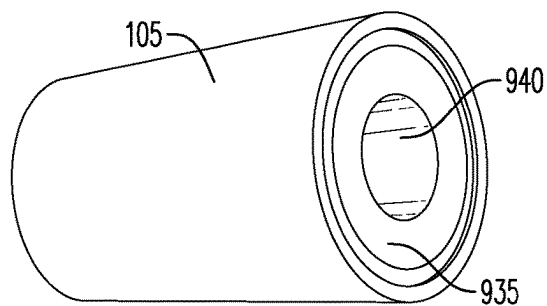


FIG. 17B

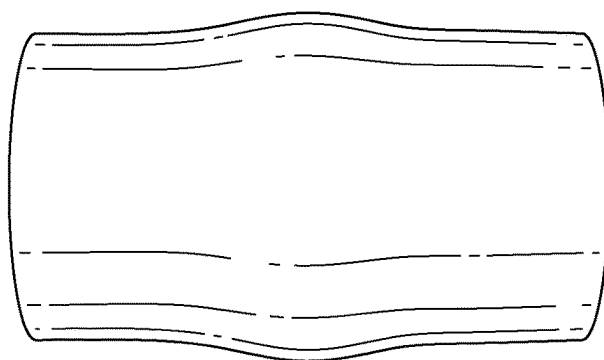


FIG. 17C

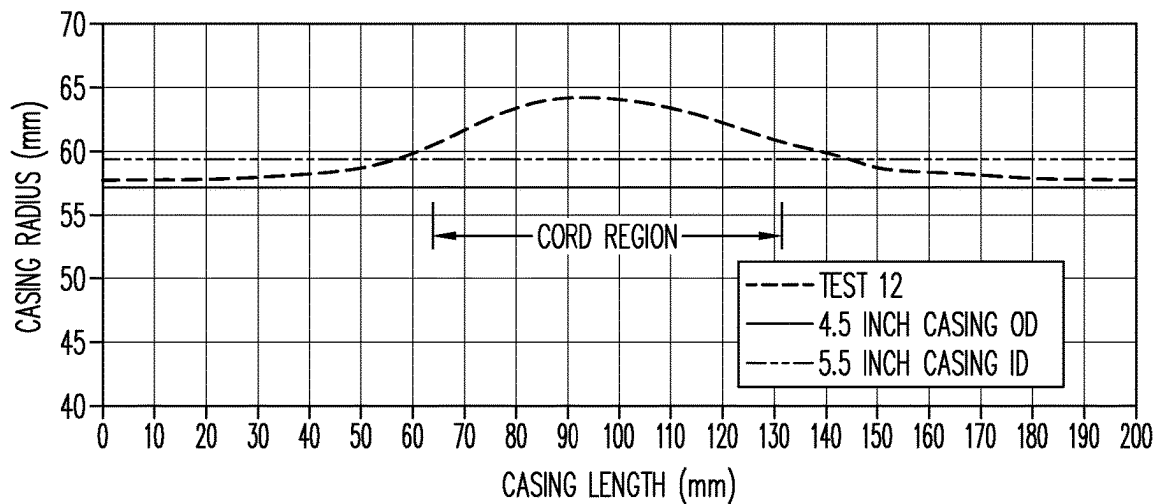
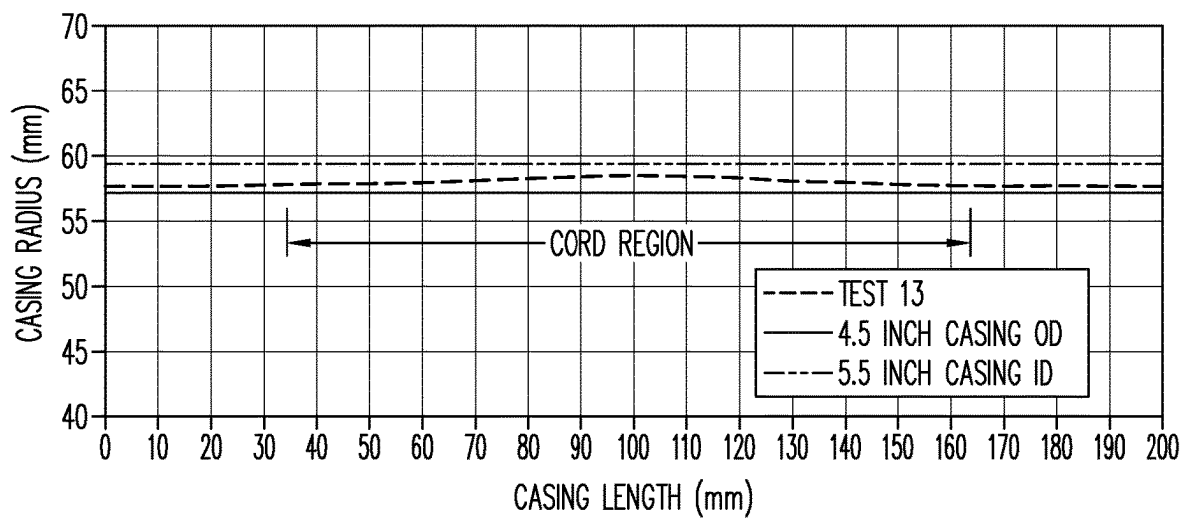
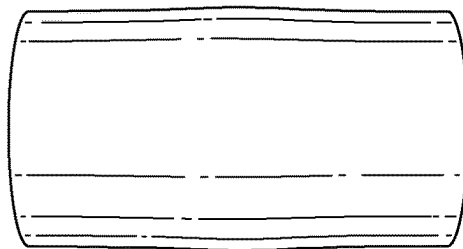
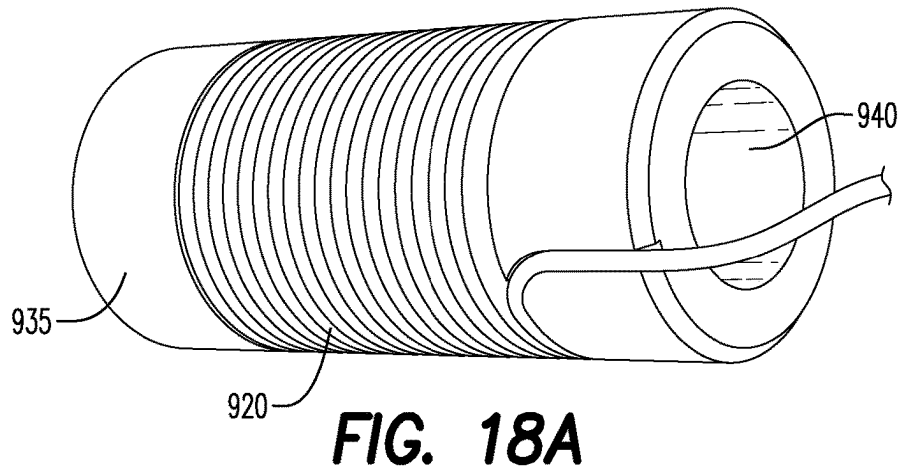


FIG. 17D



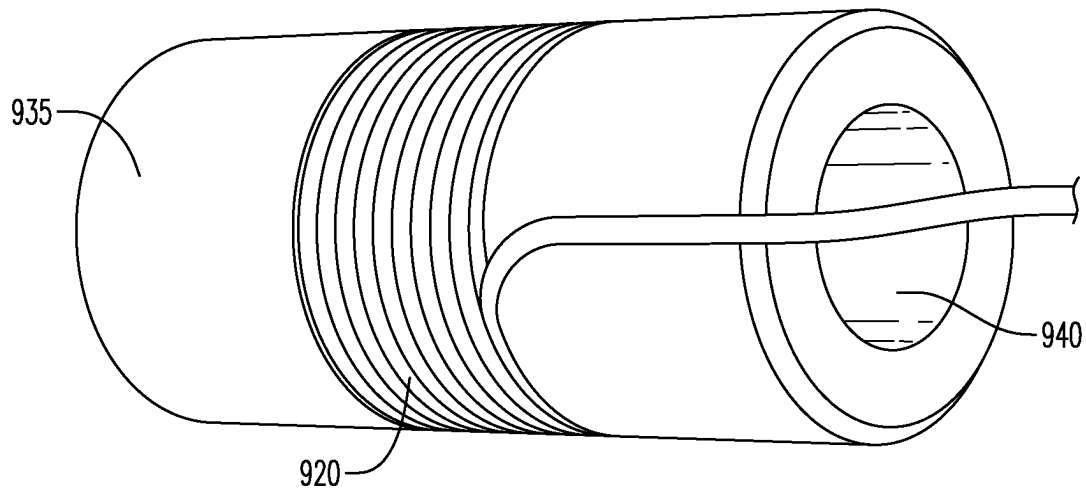


FIG. 19A

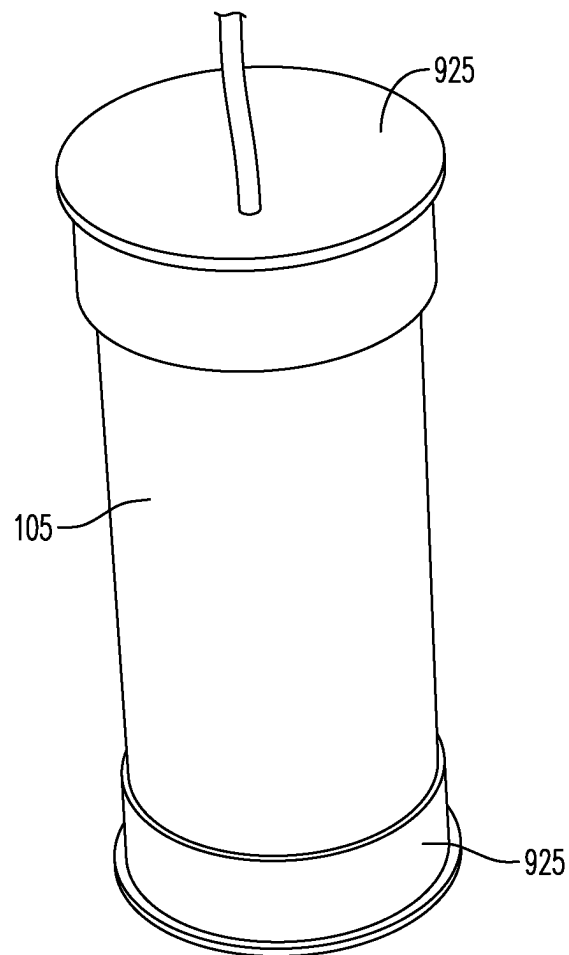
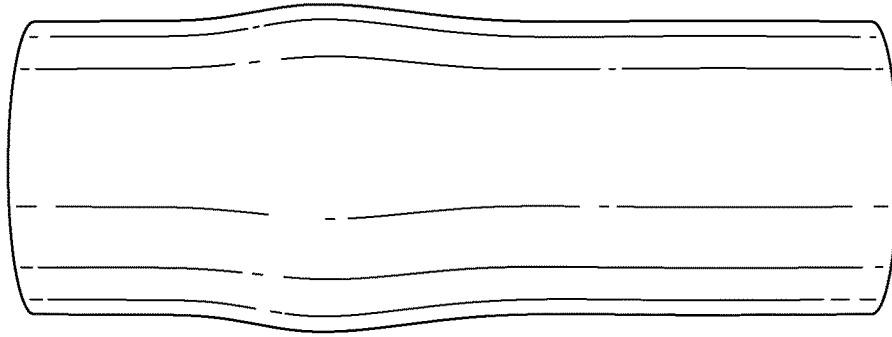
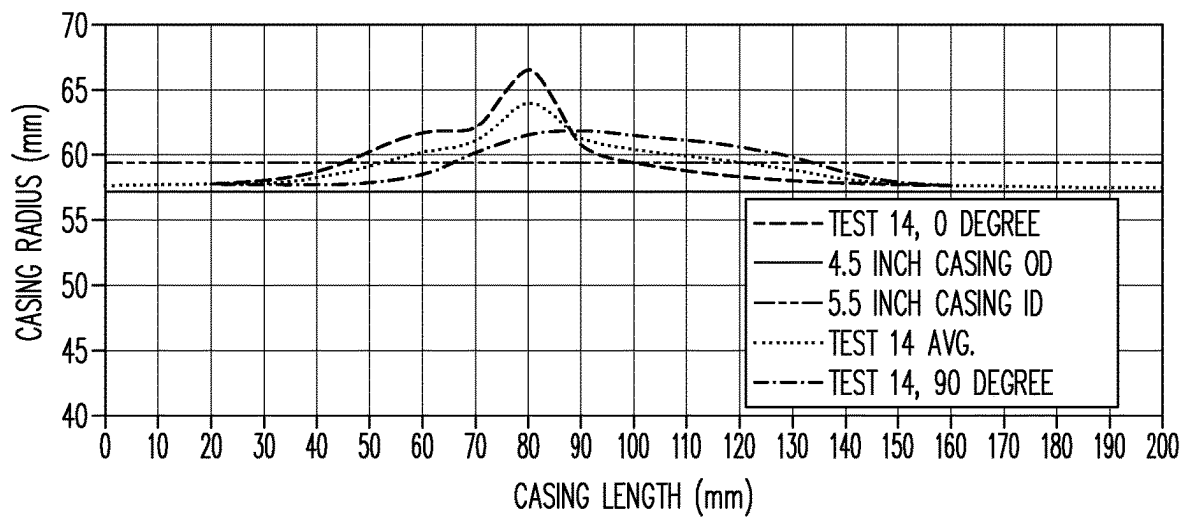


FIG. 19B

**FIG. 19C****FIG. 19D**

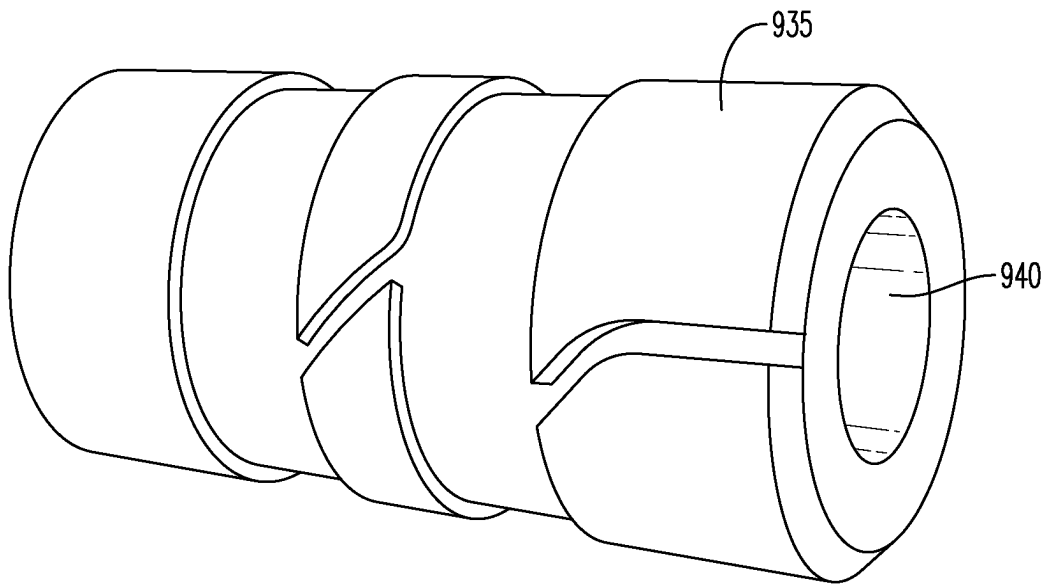


FIG. 20A

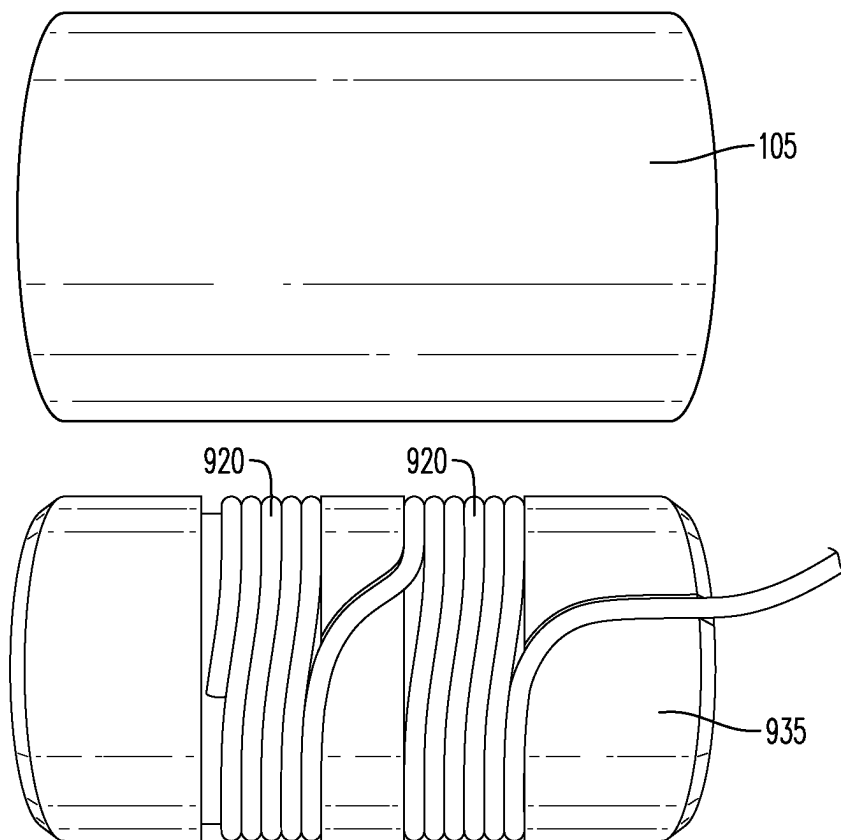
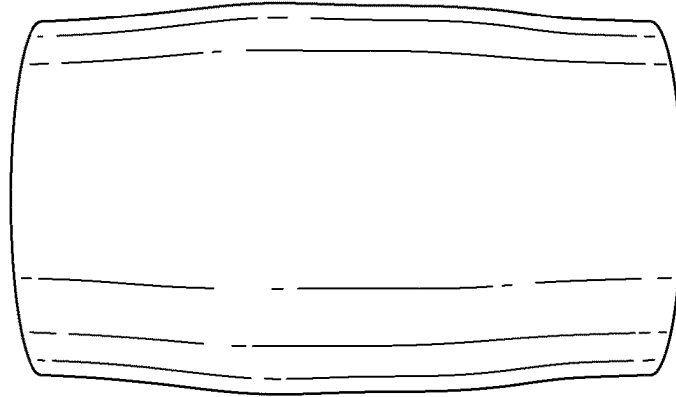
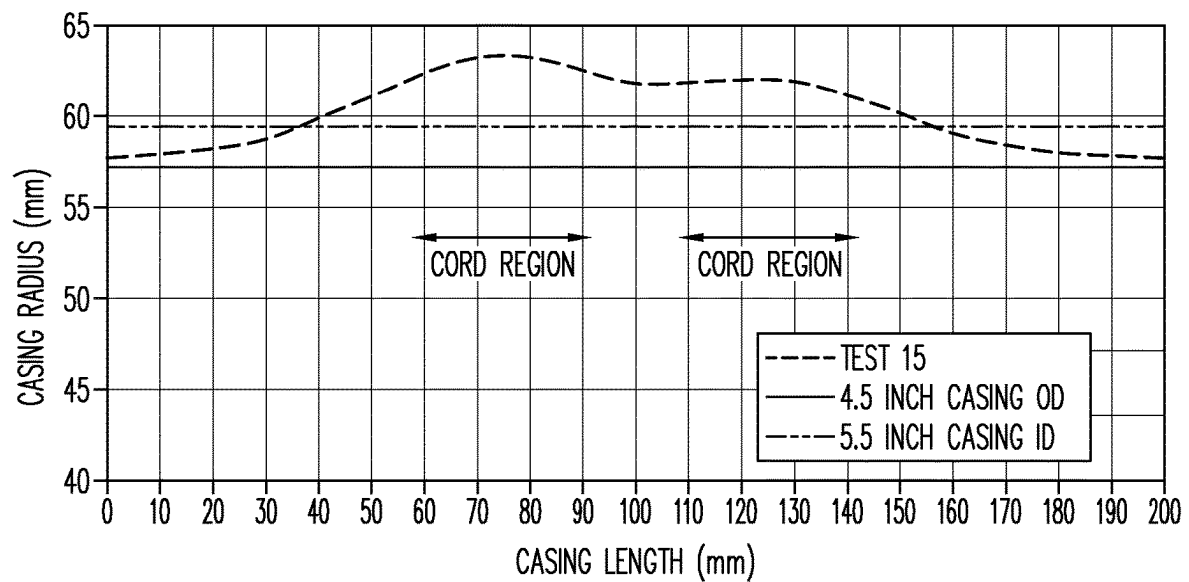


FIG. 20B

**FIG. 20C****FIG. 20D**

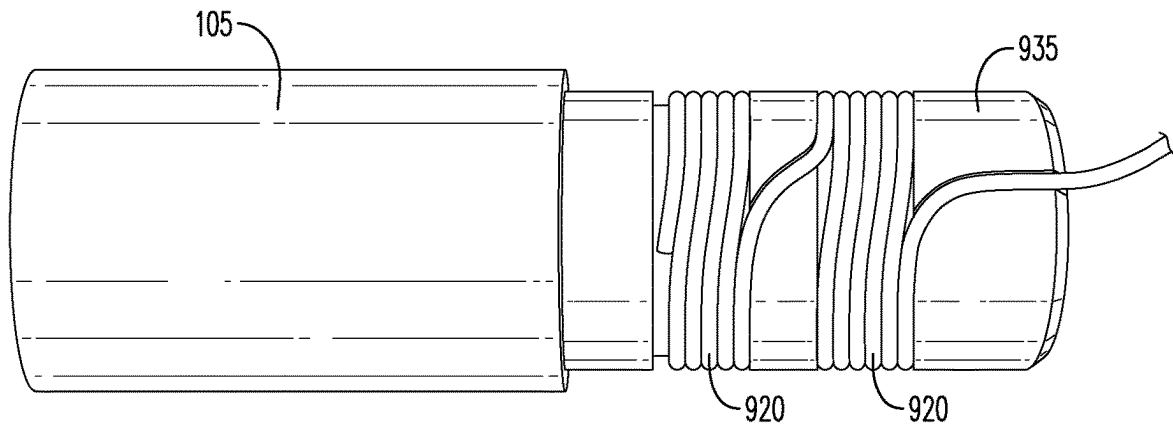


FIG. 20E

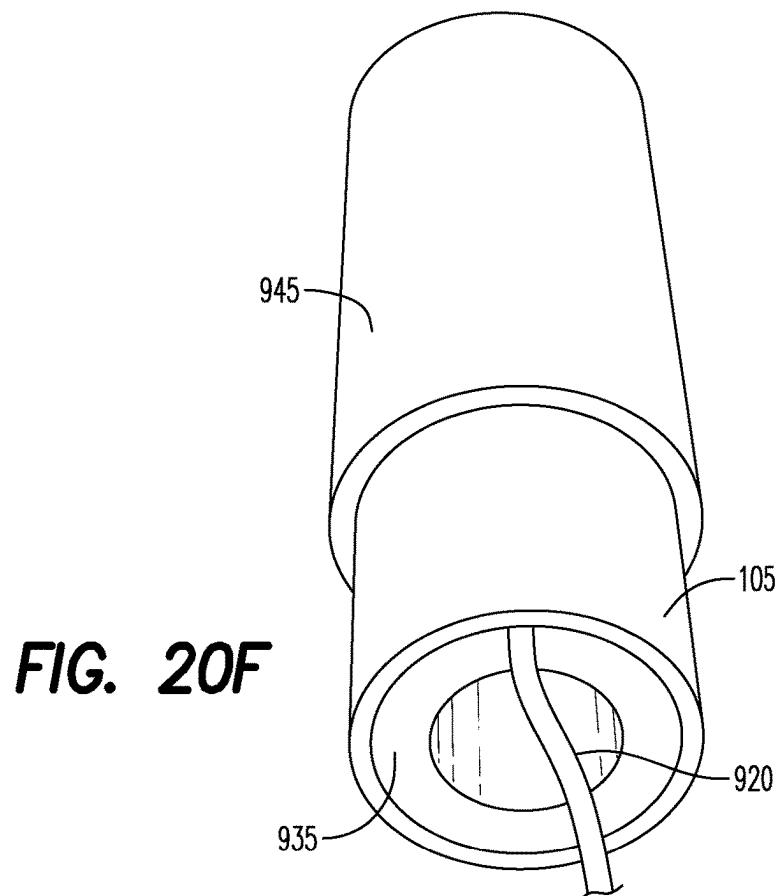


FIG. 20F

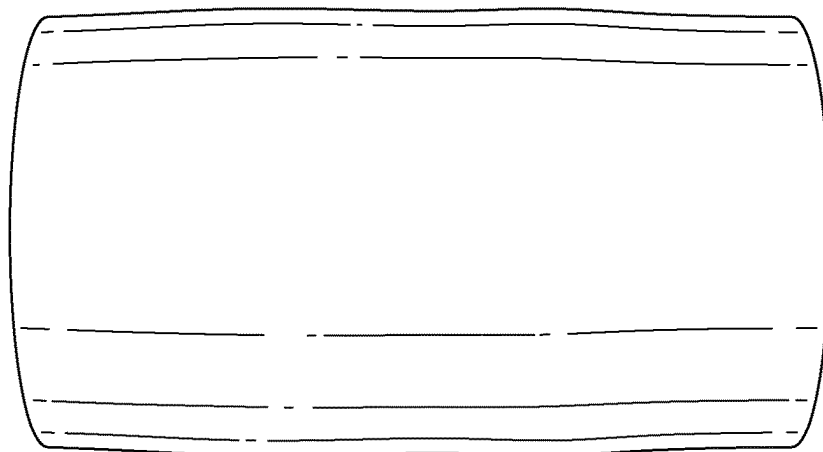


FIG. 20G

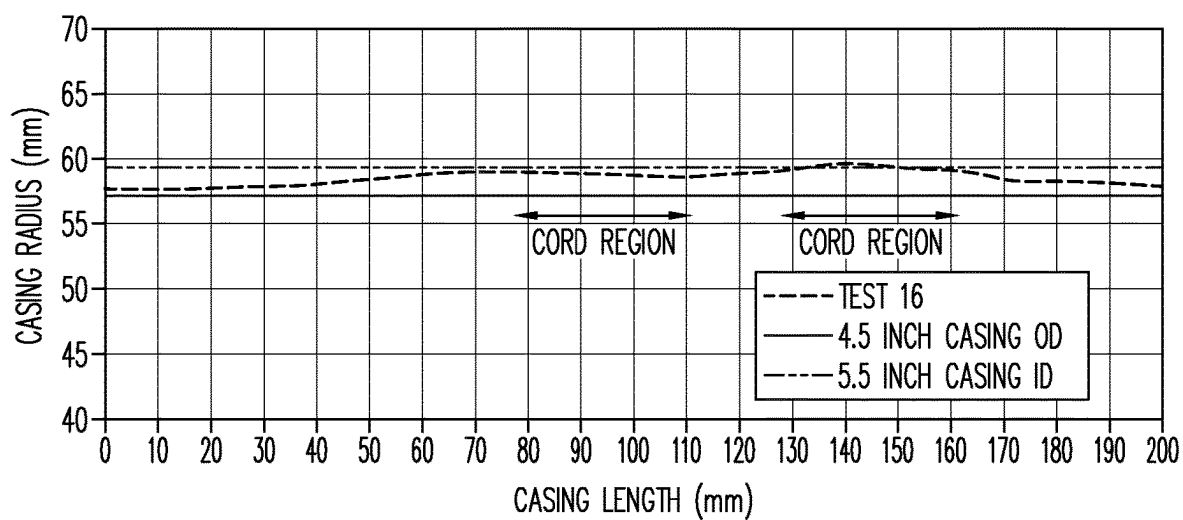


FIG. 20H

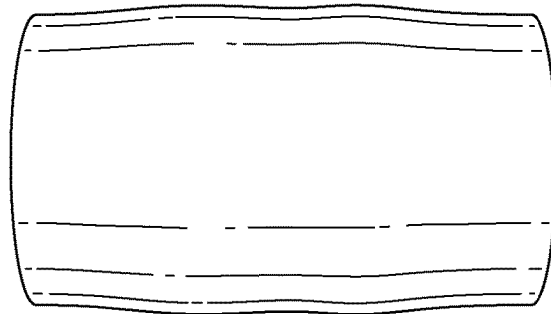


FIG. 20I

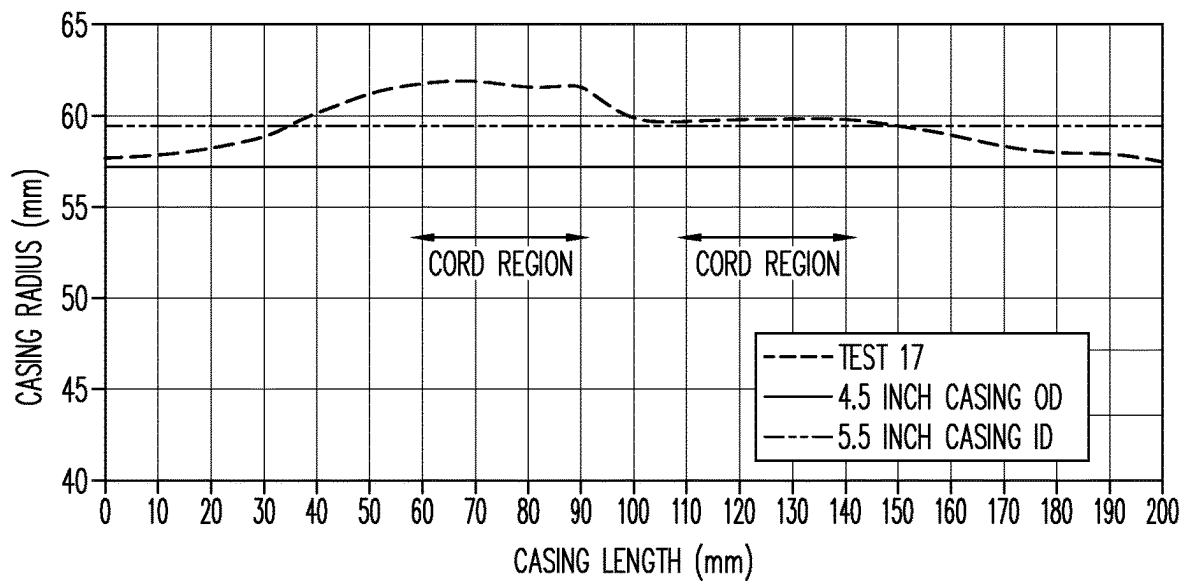


FIG. 20J

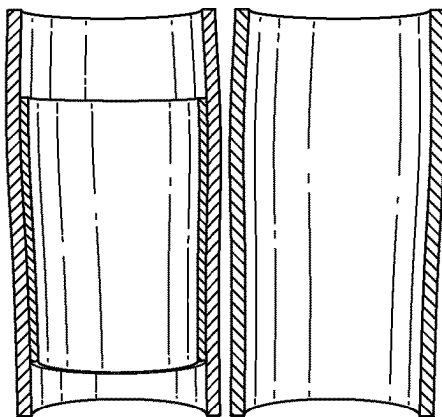


FIG. 20K

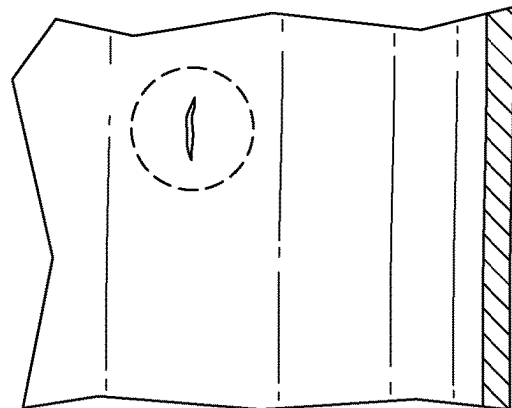


FIG. 20L

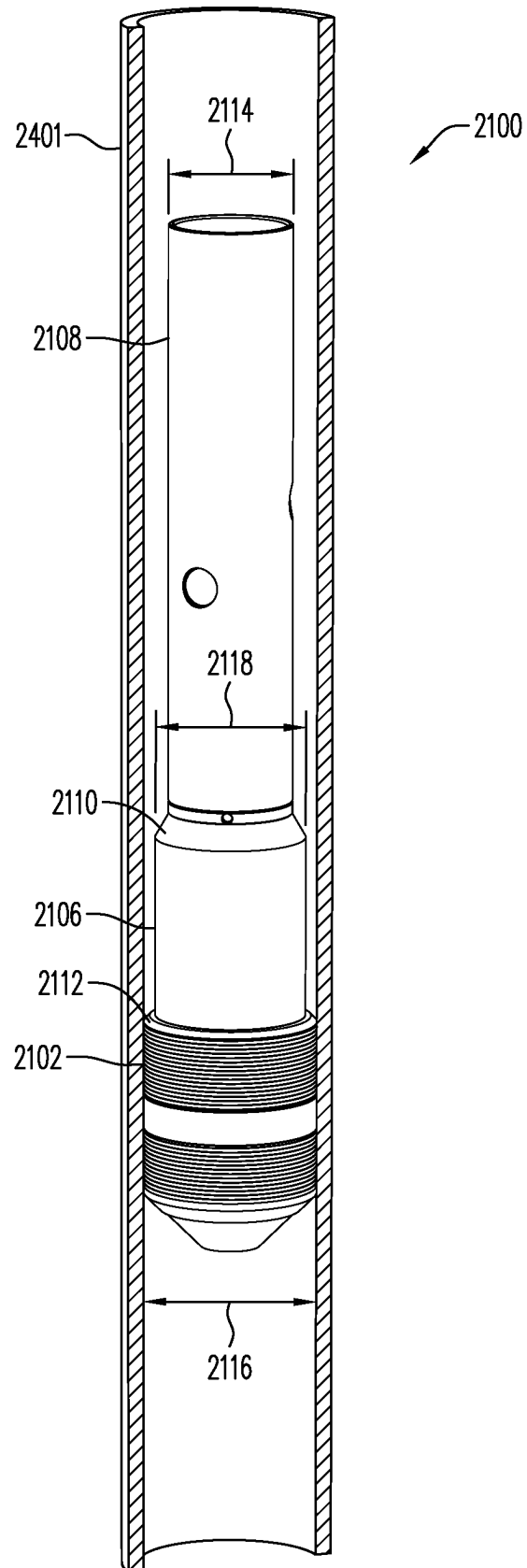
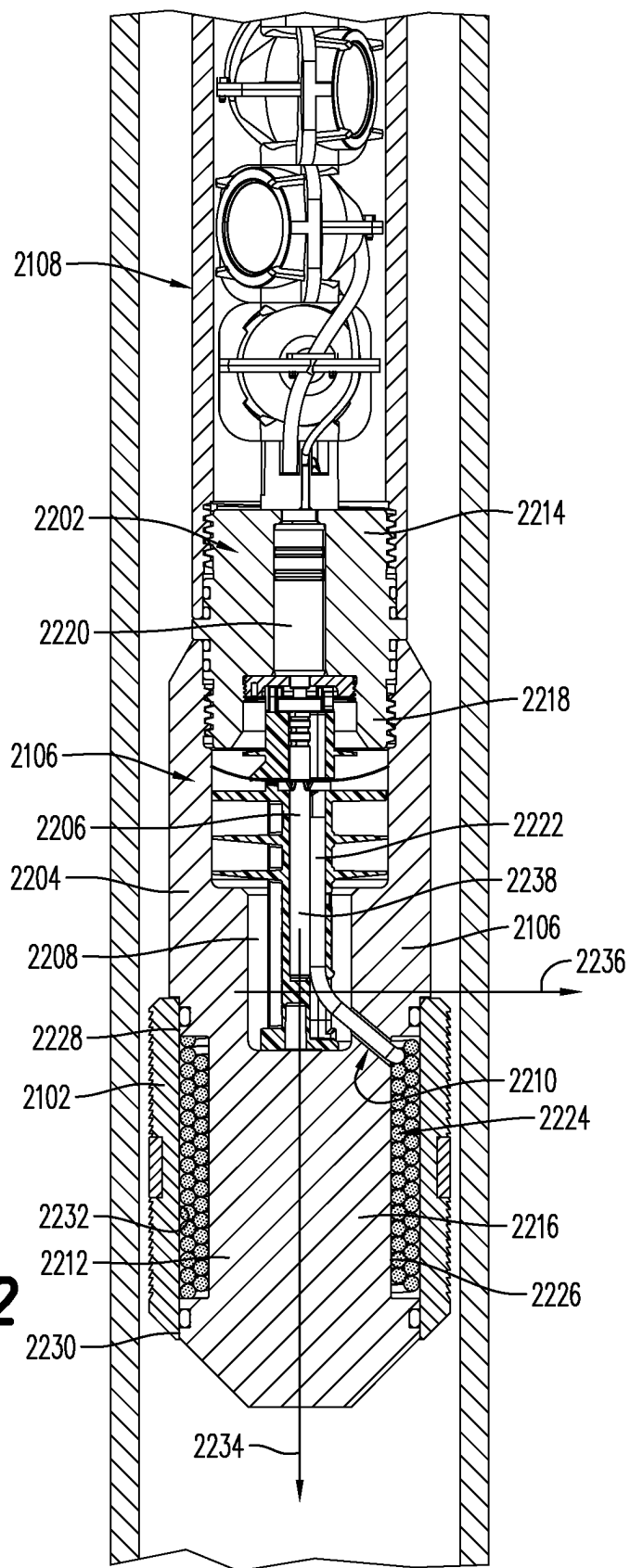


FIG. 21

FIG. 22



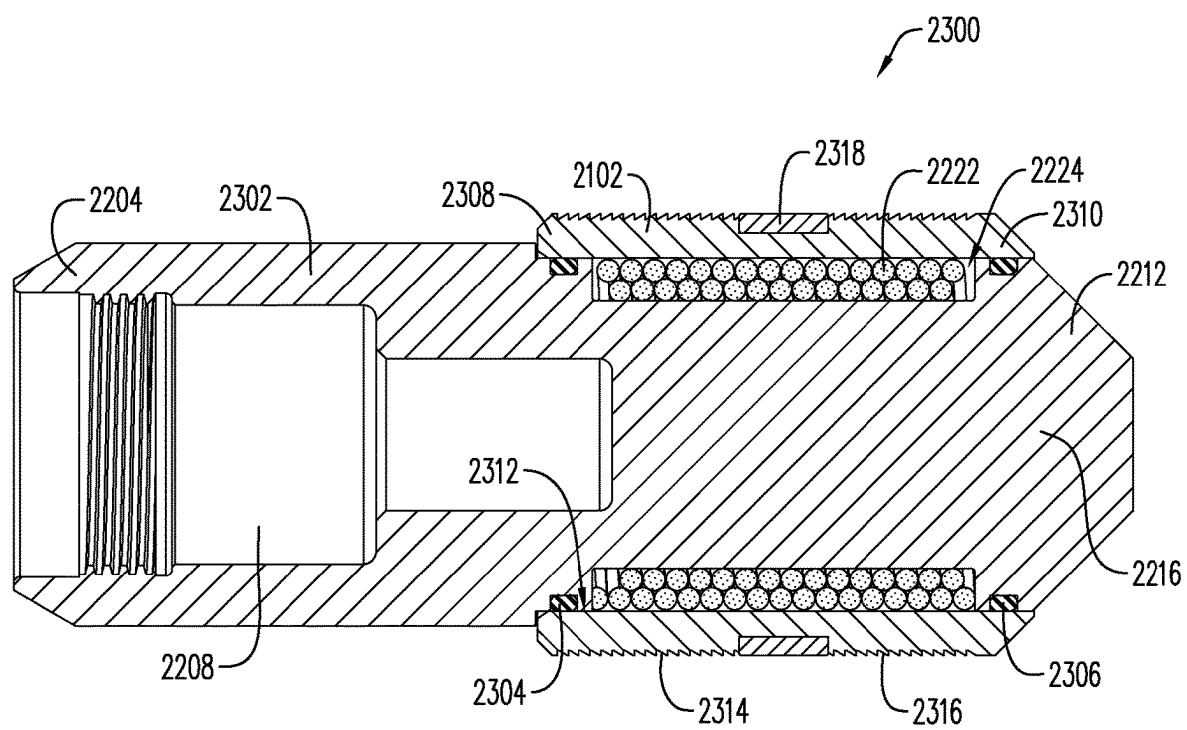


FIG. 23

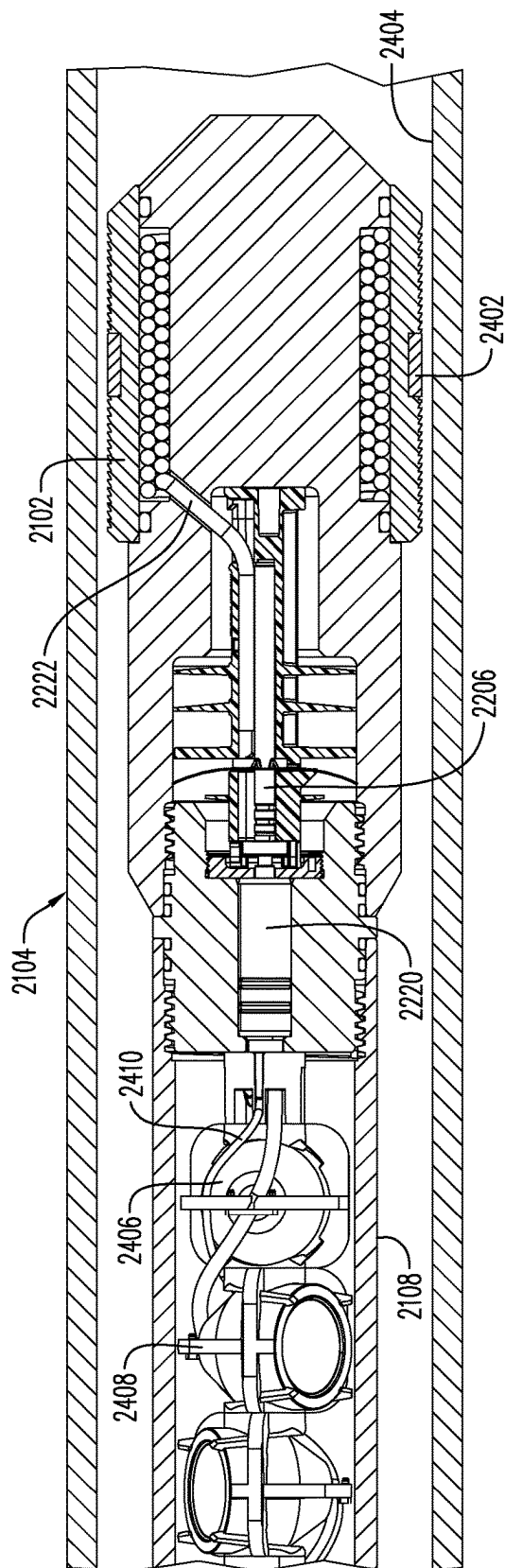


FIG. 24

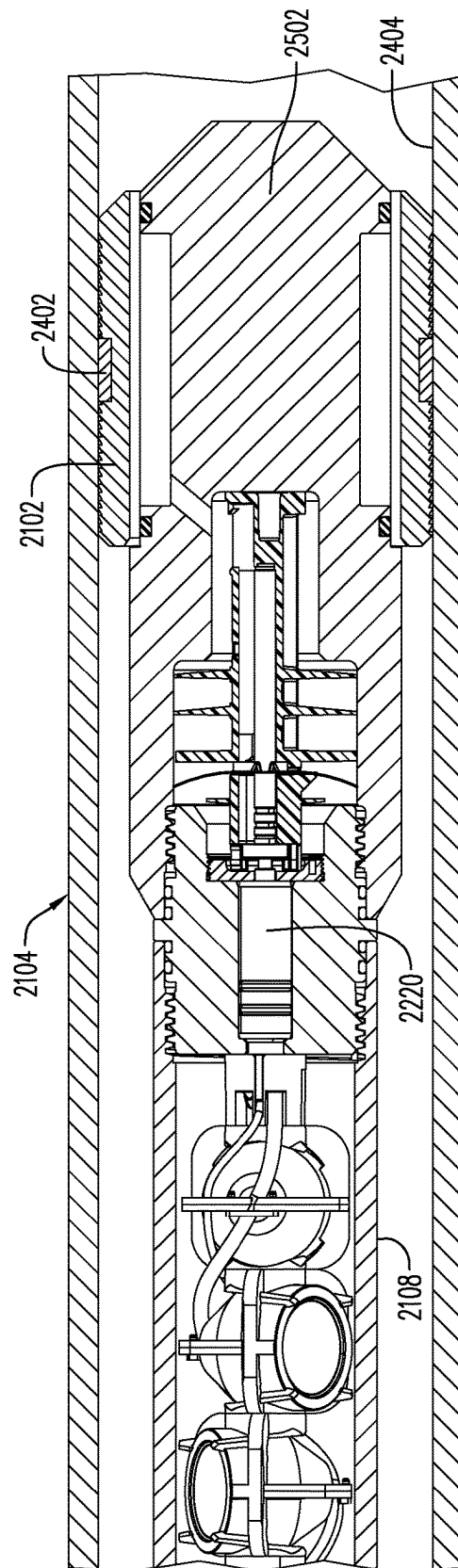


FIG. 25

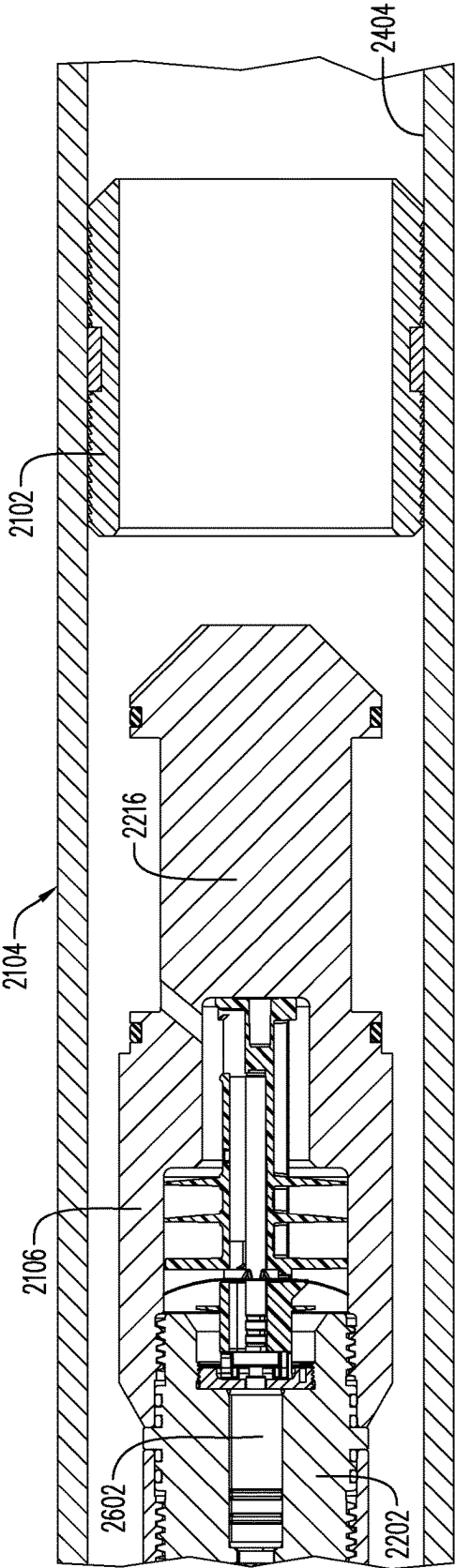


FIG. 26

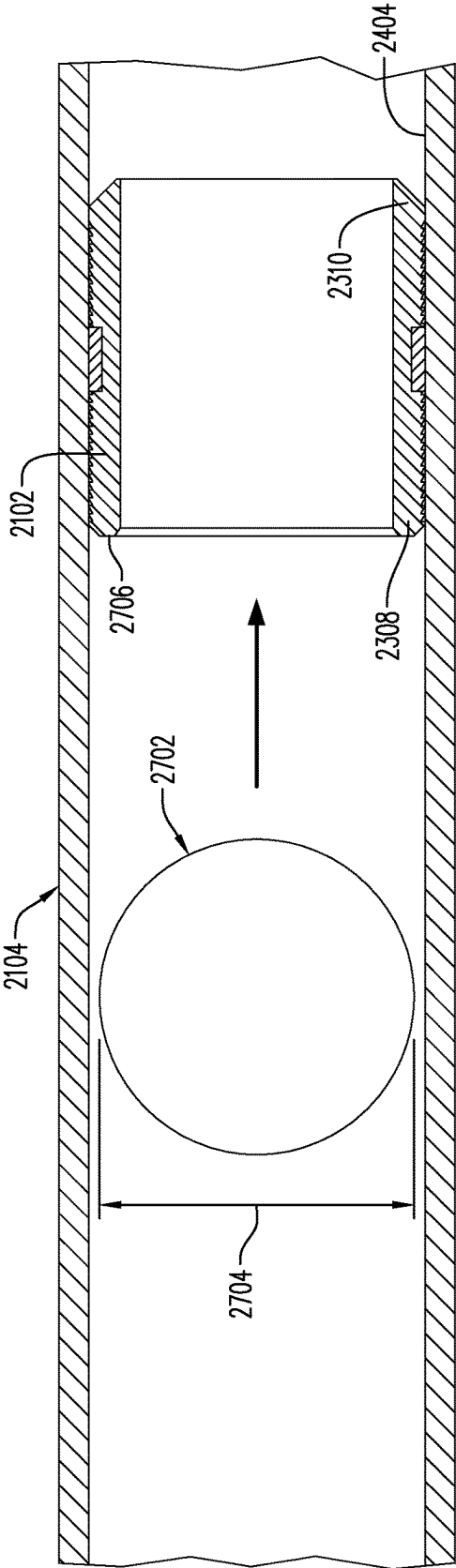


FIG. 27

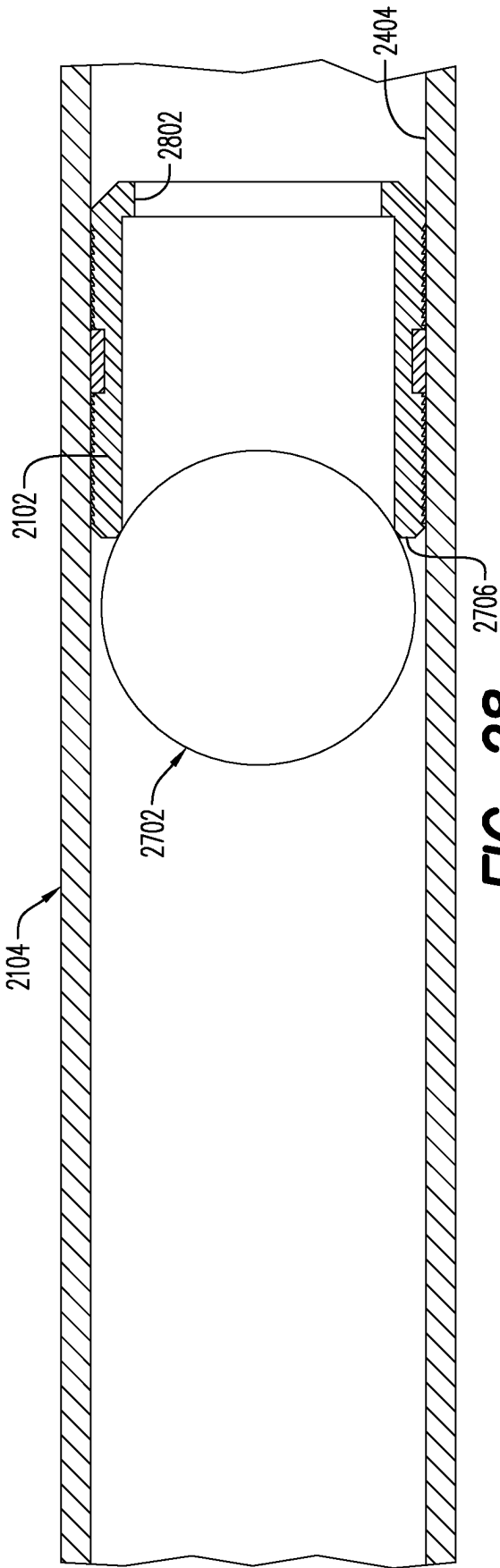


FIG. 28

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**BALLISTICALLY ACTUATED WELLBORE
TOOL****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This application claims the benefit of U.S. Provisional Patent Application No. 63/347,056 filed May 31, 2022 and U.S. Provisional Patent Application No. 63/271,466 filed Oct. 25, 2021, the entire contents of which are incorporated herein by reference in their entireties.

BACKGROUND

Hydraulic Fracturing (or, “fracking”) is a commonly used method for extracting oil and gas from geological formations (i.e., “hydrocarbon bearing formations”) such as shale and tight-rock formations. Fracking typically involves, among other things, drilling a wellbore into a hydrocarbon bearing formation, deploying a perforating gun including shaped explosive charges into the wellbore via a wireline or other methods, positioning the perforating gun within the wellbore at a desired area, perforating the wellbore and the hydrocarbon formation by detonating the shaped charges, and pumping high hydraulic pressure fracking fluid into the wellbore to force open perforations, cracks, and imperfections in the hydrocarbon formation to liberate the hydrocarbons and collect them via a wellbore tubing or casing within the wellbore that collects the hydrocarbons and directs them to the surface.

Various downhole operations may require actuating one or more tools, such as wellbore plugs (bridge plugs, frac plugs, etc.), tubing cutters, packers, and the like as are well known in the art. For example, in an aspect of a fracking operation, a plug-and-perforate (“plug-and-perf”) operation is often used. In a plug-and-perf operation, a tool string including a plug, such as a bridge plug, frac plug, or the like, a setting tool for the plug, and one or more perforating guns are connected together and sent downhole. The plug assembly is located furthest downstream (in a direction further into the wellbore) in the string and is connected to the setting tool which is in turn connected to the bottom (downstream)-most perforating gun. The setting tool is for activating (i.e., expanding) the plug to isolate a portion of the wellbore to be perforated. Isolating these portions, or “zones”, makes more efficient use of the hydraulic pressure of the fracking fluid by limiting the volume that the fracking fluid must fill in the wellbore before it is forced into the perforations.

Using a setting tool for deploying the plug adds length to the tool string as well as potential failure points at the connections to the perforating guns/plug. A typical setting tool may use a pyrotechnic igniter and/or explosive to generate pressure for moving a piston that in turn forces a pressure, which may be a hydraulic pressure, into the plug assembly to expand the plug and shear the plug from the setting tool. Once the plug is expanded it makes contact with an inner surface of the wellbore casing and creates a fluid seal between the plug and the wellbore casing to isolate the zone with respect to the wellbore casing. The setting tool may be retrieved with the spent perforating guns on the tool string, after the perforating operation. Considering that most plugs include a hollow interior for housing components and accepting the pressures that will expand the plug, once the plug is in place a resulting open passage in the plug must be sealed by, e.g., dropping into the wellbore a ball that is sized to set within an opening of the passage of the plug and thereby fully isolate the zone. This process continues for

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each zone of the wellbore. Once the perforating operations are complete and the wellbore is ready for production, the balls and/or plugs remaining in the wellbore must be drilled out to allow hydrocarbons to travel to the surface of the wellbore for collection.

These typical aspects of a plug-and-perf operation create certain undesirable issues for the operation. For example, increased length of the tool string, including the setting tool, affects ease of handling and deployment of the tool string. Components of the plug assembly that remain in the wellbore post-perforation create obstructive debris in the wellbore. And the delay between initiating the setting tool and ultimately expanding the plug by, e.g., at least one mechanical process, may lead to inaccurate positioning of the tool string and perforating guns within the wellbore.

Accordingly, integrated and instantaneously expanding plugs would be beneficial in plug-and-perf operations. Similarly, these principles and certain disadvantages as explained above may be encountered with a variety of wellbore tools that must be actuated within the wellbore, and the benefits associated with, e.g., an instantaneously expanding plug would be similarly applicable and beneficial for any wellbore tool that must be actuated within the wellbore according to particular operations as are known.

BRIEF SUMMARY

Embodiments of the disclosure are associated with a ballistically actuated plug for being deployed in a wellbore casing. The ballistically actuated plug includes a ballistic housing including a first housing portion. The first housing portion includes a chamber, and a second housing portion having a noncompressible core. According to an aspect, an initiator is positioned within the chamber. A detonation extender may be in ballistic communication with the initiator and may be circumferentially disposed around the noncompressible core of the second housing portion. According to an aspect, an outer seal housing is circumferentially disposed around the noncompressible core. The detonation extender is configured, upon detonation, to expand the outer seal housing from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing.

Additional embodiments of the disclosure are associated with a ballistically actuated plug for being deployed in a wellbore casing. The ballistically actuated plug includes a ballistic housing including a first housing portion including a chamber, and a second housing portion extending from the first housing portion. According to an aspect, an initiator is positioned within the chamber, and a detonation extender in side-by-side contact with the initiator extends from the chamber to an outer surface of the second housing portion. An outer seal housing is circumferentially disposed around the second housing portion. According to an aspect, the detonation extender is configured, upon detonation, to expand the outer seal housing from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing.

Further embodiments of the disclosure are associated with a wellbore tool assembly. The wellbore tool assembly includes a perforating gun assembly. A sub-assembly may be secured to the perforating gun assembly. According to an aspect, the sub-assembly includes a first sub portion and a second sub portion. The first sub portion is coupled to the perforating gun assembly. According to an aspect, the wellbore tool assembly further includes a ballistic housing including a first housing portion including a chamber, and a second housing portion having a noncompressible core.

According to an aspect, the second sub portion is coupled to the first housing portion. An initiator is positioned within the chamber, and a detonation extender is in ballistic communication with the initiator and is circumferentially disposed around the noncompressible core. The wellbore tool assembly may further include an outer seal housing circumferentially disposed around the noncompressible core. According to an aspect, the detonation extender is configured, upon detonation, to expand the outer seal housing from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing, and when the outer seal housing is in the expanded form, the ballistic housing is disengaged from the outer seal housing.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

A more particular description will be rendered by reference to exemplary embodiments that are illustrated in the accompanying figures. Understanding that these drawings depict exemplary embodiments and do not limit the scope of this disclosure, the exemplary embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

FIG. 1A is a partial cutaway view of an instantaneously expanding, ballistically actuated plug according to an exemplary embodiment;

FIG. 1B is a partial cutaway view of an instantaneously expanding, ballistically actuated plug according to an exemplary embodiment;

FIG. 2A shows an instantaneously expanding, ballistically actuated plug in an unexpanded form, according to an exemplary embodiment, inside of a wellbore casing;

FIG. 2B shows an instantaneously expanding, ballistically actuated plug in an expanded form, according to an exemplary embodiment, inside of a wellbore casing;

FIG. 2C shows a cross-sectional end view of an exemplary instantaneously expanding, ballistically actuated plug in an expanded form within a wellbore;

FIG. 2D shows a cross-sectional side view of an exemplary instantaneously expanding, ballistically actuated plug in an expanded form and sealed by a frac ball within a wellbore;

FIG. 3 shows a ballistic carrier according to an exemplary embodiment;

FIG. 4 shows a ballistic carrier in a wellbore tool, according to an exemplary embodiment;

FIG. 5A shows an instantaneously expanding, ballistically actuated plug attached to a tool string, according to an exemplary embodiment;

FIG. 5B shows an instantaneously expanding, ballistically actuated plug attached to a tool string, according to an exemplary embodiment;

FIG. 5C shows an exemplary Tandem Seal Adapter (TSA) and bulkhead connection assembly, according to an exemplary embodiment;

FIG. 6 is a cross-sectional side view of an instantaneously expanding, ballistically actuated autonomous plug drone according to an exemplary embodiment;

FIG. 7 is a partial cross-sectional side view of a daisy-chained ballistically actuated autonomous plug drone and wellbore tool assembly, according to an exemplary embodiment;

FIG. 8 is a cross-sectional view of an instantaneously expanding, ballistically actuated autonomous plug drone with frac ball, according to an exemplary embodiment;

FIG. 9 shows various experimental test setups for a ballistically actuated wellbore tool;

FIG. 10A shows explosive pellets for use with a ballistically actuated wellbore tool;

FIG. 10B shows an experimental setup for an explosive pellet as in FIG. 10A;

FIG. 11A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 11B shows a ballistically actuated wellbore tool after an experimental test;

FIG. 11C shows a swell profile for the ballistically actuated wellbore tool of FIG. 11B;

FIG. 11D shows a ballistically actuated wellbore tool after an experimental test;

FIG. 11E shows a swell profile for the ballistically actuated wellbore tool of FIG. 11D;

FIG. 12A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 12B shows a ballistically actuated wellbore tool after an experimental test;

FIG. 12C shows a swell profile for the ballistically actuated wellbore tool of FIG. 12B;

FIG. 13A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 13B shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 13C shows a ballistically actuated wellbore tool after an experimental test;

FIG. 13D shows a swell profile for the ballistically actuated wellbore tool of FIG. 13C;

FIG. 13E shows a ballistically actuated wellbore tool after an experimental test;

FIG. 13F shows a swell profile for the ballistically actuated wellbore tool of FIG. 13E;

FIG. 13G shows a ballistically actuated wellbore tool after an experimental test;

FIG. 13H shows a swell profile for the ballistically actuated wellbore tool of FIG. 13G;

FIG. 14 shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 15A shows a ballistically actuated wellbore tool after an experimental test;

FIG. 15B shows a swell profile for the ballistically actuated wellbore tool of FIG. 15A;

FIG. 15C shows a ballistically actuated wellbore tool after an experimental test;

FIG. 15D shows a swell profile for the ballistically actuated wellbore tool of FIG. 15C;

FIG. 15E shows a ballistically actuated wellbore tool after an experimental test;

FIG. 15F shows a swell profile for the ballistically actuated wellbore tool of FIG. 15E;

FIG. 16A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 16B shows a ballistically actuated wellbore tool after an experimental test;

FIG. 16C shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 16D shows a ballistically actuated wellbore tool after an experimental test;

FIG. 17A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 17B shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 17C shows a ballistically actuated wellbore tool after an experimental test;

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FIG. 17D shows a swell profile for the ballistically actuated wellbore tool of FIG. 17C;

FIG. 18A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 18B shows a ballistically actuated wellbore tool after an experimental test;

FIG. 18C shows a swell profile for the ballistically actuated wellbore tool of FIG. 18B;

FIG. 19A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 19B shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 19C shows a ballistically actuated wellbore tool after an experimental test;

FIG. 19D shows a swell profile for the ballistically actuated wellbore tool of FIG. 19C;

FIG. 20A shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 20B shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 20C shows a ballistically actuated wellbore tool after an experimental test;

FIG. 20D shows a swell profile for the ballistically actuated wellbore tool of FIG. 20C;

FIG. 20E shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 20F shows an experimental setup for a ballistically actuated wellbore tool;

FIG. 20G shows a ballistically actuated wellbore tool after an experimental test;

FIG. 20H shows a swell profile for the ballistically actuated wellbore tool of FIG. 20G;

FIG. 20I shows a ballistically actuated wellbore tool after an experimental test;

FIG. 20J shows a swell profile for the ballistically actuated wellbore tool of FIG. 20I;

FIG. 20K shows the ballistically actuated wellbore tool of FIG. 20I in a casing after the experimental test;

FIG. 20L shows the ballistically actuated wellbore tool of FIG. 20I in a casing after the experimental test;

FIG. 21 is a cross-sectional view of a wellbore casing including a ballistically actuated wellbore tool;

FIG. 22 is a cross-sectional view of a ballistically actuated wellbore tool, according to an aspect;

FIG. 23 is a cross-sectional view of a ballistically actuated wellbore tool including a ballistic housing secured to an outer seal housing, according to an aspect;

FIG. 24 is a cross-sectional view of a ballistically actuated wellbore tool including a detonator and a detonation extender, prior to ballistic activation, according to an aspect;

FIG. 25 is a cross-sectional view of a ballistically actuated wellbore tool including a detonator and a detonation extender, after ballistic activation, according to an aspect;

FIG. 26 is a cross-sectional view of a ballistically actuated wellbore tool, illustrating a ballistic housing being detached from a seal housing after ballistic activation, according to an aspect;

FIG. 27 is a cross-sectional view of the seal housing and a ball traveling towards the seal housing, according to an aspect; and

FIG. 28 is a cross-sectional view of the seal housing of FIG. 27, illustrating the ball seated in the seal housing.

Various features, aspects, and advantages of the exemplary embodiments will become more apparent from the following detailed description, along with the accompanying drawings in which like numerals represent like components throughout the figures and detailed description. The various

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described features are not necessarily drawn to scale in the drawings but are drawn to emphasize specific features relevant to some embodiments.

The headings used herein are for organizational purposes only and are not meant to limit the scope of the disclosure or the claims. To facilitate understanding, reference numerals have been used, where possible, to designate like elements common to the figures.

DETAILED DESCRIPTION

Reference will now be made in detail to various embodiments. Each example is provided by way of explanation and is not meant as a limitation and does not constitute a definition of all possible embodiments.

Embodiments described herein relate generally to devices, systems, and methods for instantaneously setting a plug in a wellbore. For purposes of this disclosure, “instantaneously” means directly resulting from an initiating event, e.g., an explosive event such as detonation of an explosive charge, substantially at the speed of the initiating event. For purposes of this disclosure, the phrases “devices,” “systems,” and “methods” may be used either individually or in any combination referring without limitation to disclosed components, grouping, arrangements, steps, functions, or processes.

For purposes of illustrating features of the embodiments, an exemplary embodiment will now be introduced and referenced throughout the disclosure. This example is illustrative and not limiting and is provided for illustrating the exemplary features of a ballistically actuated plug as described throughout this disclosure. Further, the exemplary embodiment(s) herein are presented representatively and for brevity with respect to a ballistically actuated plug but are not so limited. The exemplary principles and descriptions of a ballistically actuated wellbore tool are applicable not only to, e.g., wellbore plugs, but to any wellbore tool that must be actuated within the wellbore. For example, packers and other known wellbore or annular isolation tools may variously incorporate the disclosed structures, configurations, components, techniques, etc. under similar operating principles.

FIG. 1A and FIG. 1B show exemplary embodiment(s) of a ballistically actuated plug **100** (i.e., instantaneously expanding plug) for being deployed in a wellbore. The exemplary ballistically actuated plug **100** includes, among other things, an outer carrier **105** having a first end **101** and a second end **102** opposite the first end **101** and defining a hollow interior chamber **104** within the outer carrier **105**. In the exemplary embodiments shown in FIG. 1A and FIG. 1B, the hollow interior chamber **104** extends from the first end **101** of the outer carrier **105** to the second end **102** of the outer carrier **105**.

With continuing reference to FIG. 1A and FIG. 1B, and further reference to FIG. 3 and FIG. 4, a ballistic carrier **106** is received and/or positioned within the hollow interior chamber **104** for ballistically actuating a wellbore tool, e.g. the wellbore plug **100**. The ballistic carrier **106** includes a body portion **115** having a first end **107** and a second end **108** opposite the first end **107**. A bore **112** is formed within and defined by the body portion **115** of the ballistic carrier **106** and extends along a length **L** of the ballistic carrier **106**, an initiator **114** is positioned within the bore **112**. In addition, the ballistic carrier **106** includes one or more ballistic components **110** positioned within ballistic slots **109** which are formed in an outer surface **130** of the body portion **115** of the ballistic carrier **106** and extend into the body portion

115 of the ballistic carrier **106**. For purposes of this disclosure, a “ballistic component” is a component that generates one or more of kinetic energy (i.e., propelling physical components), thermal energy, and increased pressures upon initiation such as ignition or detonation of the ballistic component. The ballistic components **110** and the initiator **114** are relatively positioned for allowing the initiator **114** to initiate the ballistic components **110**. While the exemplary embodiments disclosed herein include the ballistic carrier **106** for holding and orienting, e.g., the initiator **114** and the ballistic components **110**, any structure or component consistent with this disclosure may be used for the same purpose. Such components may include, without limitation, a charge tube, strip, or stackable charge carriers. However, a particular orientation of the ballistic components **110** may not be required, in which case any structure or component for relatively positioning the initiator **114** and ballistic components **110** such that the initiator **114** will initiate the ballistic components **110** would be sufficient.

In an aspect of the exemplary embodiment(s), the ballistic carrier **106** may be formed from a substantially fragmentable or disintegrable material such as, without limitation, an injection molded plastic that will substantially fragment and/or disintegrate upon detonation of the ballistic components **110**. The ballistic components **110** in such embodiments should thus have sufficient power for fragmenting and/or disintegrating the ballistic carrier **106**. The ballistic components **110** may include any known explosive or incendiary components, or the like, for use in a wellbore operation. Non-limiting examples include shaped charges, explosive loads, black powder igniters, and the like.

In the exemplary embodiments, the ballistic components **110** may include, without limitation, explosive rings (such as linear shaped charges) in the ballistic slots **109** formed in the ballistic carrier **106**. The ballistic slots **109** may be formed, without limitation, about an entire perimeter or periphery of the ballistic carrier **106** or as pockets therein. The explosive rings may be formed, for example, by pressing explosive powder, and then the explosive rings may be inserted into the ballistic slots **109**. Alternatively, the explosive charges (explosive loads) may be pressed directly into the ballistic slots **109**. In operation, the explosive charge may generate thermal energy and pressure forces for expanding the outer carrier **105** from an unexpanded form **170** to an expanded form **171** (see FIG. 2A and FIG. 2B) upon initiation of the ballistic components **110**. The ballistic components **110** and the outer carrier **105** are together configured for instantaneously expanding the outer carrier **105** from the unexpanded form **170** to the expanded form **171** upon initiation of the one or more ballistic components **110**. For example, expanding the outer carrier **105** occurs upon initiation of the ballistic components **110** and substantially as quickly as the pressure forces generated by initiation of the ballistic components **110** propagate to and act upon the outer carrier **105**. Compare that exemplary operation with conventional plugs that rely on a setting tool and, in-part, on moving mechanical components after initiating, e.g., an explosive charge in the setting tool and before expanding the plug with forces generated by moving the mechanical components.

In an exemplary embodiment, the initiator **114** is a pressure sealed detonating cord. In other embodiments, the initiator **114** may be a detonator such as a wireless detonator as described in U.S. Pat. No. 9,605,937, which is commonly assigned to DynaEnergetics Europe GmbH and incorporated herein by reference in its entirety. In other embodiments, the initiator **114** may be an elongated booster. In other embodiments, the initiator **114** may be one or more detonating

pellets. In other embodiments, the initiator **114** may include two or more of the above components in combination. Where the initiator **114** is a component such as a detonating cord, booster, detonating pellets, or other component that itself requires initiation, such initiation may be provided by, without limitation, a firing head, a detonator, an igniter, or other known devices and/or techniques for initiating a ballistic or incendiary component. Such initiation assembly may be configured or contained in, without limitation, a tandem seal adapter (TSA) (such as described with respect to FIGS. 5A-5C), or other known connectors or assemblies used to house an initiating component and relay an initiation signal or power thereto.

The initiator **114** may be completely or partially contained within the bore **112** of the ballistic carrier **106** according to the exemplary embodiments—at least a portion of the initiator **114** may be positioned within the bore **112** while a portion of the initiator **114** may lie outside of the bore **112** or even the outer carrier **105** according to certain embodiments discussed further below. As mentioned previously, the initiator **114** must at least be capable of initiating, either directly or indirectly (via ballistic components that have been directly initiated), the ballistic components **110** within the hollow interior chamber **104** of the outer carrier **105**.

With continuing reference to FIGS. 1A, 1B, 3, and 4, in the exemplary embodiment(s) the ballistic components **110** are respectively positioned and oriented in the ballistic carrier **106** to fire radially outwardly upon initiation of the ballistic components **110**. For purposes of this disclosure, “radially outwardly” means along a radius from a center point in a direction away from the center point. For example, the ballistic components **110** in the exemplary embodiments will fire in a direction from the bore **112** within the body portion **115** of the ballistic carrier **106** towards the outer carrier **105**. For purposes of this disclosure, a direction in which respective ballistic components **110** “fire” means a direction in which an explosive jet, pressure force, and/or kinetic energy propagate from the respective ballistic component **110** upon initiating the ballistic component **110**. Controlling the direction in which the ballistic components **110** fire may aid in expanding the outer carrier **105** from an unexpanded form **170** to an expanded form **171**, as will be discussed below with respect to FIG. 2A and FIG. 2B. The direction in which the ballistic components **110** fire may be controlled by, e.g., the orientation of the ballistic slots **109**. In the exemplary embodiment(s), the ballistic slots **109** extend radially outwardly in a direction from the bore **112** to the outer carrier **105**—i.e., from a portion of the ballistic slot **109** containing the pressed explosive charge to the opening of the ballistic slot **109** on the outer surface **130** of the body portion **115** of the ballistic carrier **106** from which the explosive jet/energy will be ejected.

In the exemplary embodiments, the ballistic slots **109** may be formed, without limitation, as pockets or depressions extending from the outer surface **130** of the body portion **115** of the ballistic carrier **106** into the body portion **115** of the ballistic carrier **106**, or as channels extending around at least a portion of a circumference of the exemplary cylindrically-shaped ballistic carrier **106**. The exemplary bore **112** may be formed as an axial bore extending along a longitudinal axis **x** through the body portion **115** of the ballistic carrier **106** and adjacent to the ballistic slots **109** at a portion of the ballistic slots **109** containing at least a portion of the pressed explosive charges.

The direction in which the ballistic components **110** fire is not limited by the disclosure—the ballistic components **110** may fire in any direction, uniformly or individually, at

random or according to a particular orientation, provided that the ballistic components **100** are configured with, for example and without limitation, a type and amount of explosive sufficient for generating the energy and forces required for expanding the outer carrier **105**.

In addition, and as will be discussed below, the ballistic components **110** may also be used to fragment and/or disintegrate the ballistic carrier **106** upon setting the ballistically actuated plug **100**. Accordingly, it may be beneficial for at least some of the ballistic components **110** to fire radially inwardly, i.e., in a direction from a point within or at the outer surface **130** of the body portion **115** of the ballistic carrier **106** towards the axis **x**. In an example of such embodiment (not illustrated in the Figures), the ballistic component **110** may be a shaped charge positioned such that an open end (i.e., an end through which the explosive jet is expelled) of the shaped charge is on the outer surface **130** of, or within, the body portion **115** of the ballistic carrier **106**, to direct the explosive jet into the body portion **115** towards the axis **x**. In an aspect of such embodiment, an initiation end (i.e., an end adjacent to an initiator) of the shaped charge may be opposite the open end and adjacent to an initiator outside or on the outer surface **130** of the body portion **115** of the ballistic carrier **106**. In another example of such embodiment (not illustrated in the Figures), a ballistic slot **109** may be formed as a pocket extending from the outer surface **130** of the ballistic carrier **106** into the body portion **115** of the ballistic carrier **106** and past the longitudinal axis **x**, such that a portion of the ballistic slot **109** containing the explosive charge is on a side of the longitudinal axis **x** that is opposite a side into which the ballistic slot **109** extends from the outer surface **130** of the body portion **115** of the ballistic carrier **106**. In an aspect of such embodiment, the bore **112** may be positioned off-center within the body portion **115** of the ballistic carrier **106** and adjacent to the portion of the ballistic slot **109** containing the explosive charge, and the initiator **114** may be positioned within the bore **112**.

In certain embodiments, the ballistic carrier **106** may include a plurality of ballistic components **110** variously configured to fire in different directions from different orientations. In such embodiments, one or more corresponding initiators in, e.g., corresponding bores and/or outside or on the outer surface **130** of the body portion **115** of the ballistic carrier **106** may be respectively positioned for initiating each of the plurality of ballistic components **110**.

In certain embodiments, the ballistic carrier **106** may include a plurality of ballistic components **110** variously configured to fire in different directions. In such embodiments, respective portions of ballistic slots **109** containing the explosive charge may not all be positioned along a single axis or around a single point. In an aspect of such embodiments, the ballistic carrier **106** may include a plurality of initiators respectively positioned within corresponding bores, and the corresponding bores may be respectively positioned adjacent to corresponding respective portions of the ballistic slots **109** containing the explosive charge.

In an aspect, where the ballistic components **110** are explosive charges pressed into the ballistic slots **109** according to the exemplary embodiment(s), the explosive charges may be covered in whole or in part by a liner **131** (FIG. 3). Upon initiation of the explosive charges the liner **131** will collapse and form a jet of material with kinetic energy that may enhance the fragmentation or disintegration of the ballistic carrier **106** according to known principles.

The ballistic components **110** and the outer carrier **105** are together configured for deforming and radially expanding

the outer carrier **105** upon initiation of the ballistic components **110**. For example, the ballistic components **110** may have a certain explosive force and the outer carrier **105** may be formed in a configuration and/or from a material with physical properties sufficient to achieve the desired expansion of the outer carrier **105** upon initiation of the ballistic components **110**. For example, the outer carrier **105** may be formed from a ductile material such as steel having a high yield strength (e.g., >1000 MPa) and impact strength (e.g., Charpy Value >80 J), according to the ASTM-A519 specifications. Other exemplary materials may be aluminum, strong plastics (including injection molded plastics), and the like having the requisite ductility for swelling, resistance to the wellbore environment, and resiliency (i.e., not too brittle) for being drilled out after use.

Accordingly, the exemplary ballistically actuated plug **100** sets by expanding only radially outwardly, without lateral moving parts, into the wellbore casing **300** (FIG. 2B) and does not require a setting tool or moving parts such as pistons with mechanical connections.

As discussed further below, a sufficient degree of “swell”—i.e., the degree to which the size of the outer carrier **105** is expanded upon ballistic actuation—is required for the exemplary instantaneously expanding, ballistically actuated plug **100** to seal within the wellbore in the expanded state **171**. For example, initiation of the ballistic components **110** must cause sufficient controlled plastic deformation of the outer carrier **105** to expand the outer carrier **105** enough for engaging and sealing elements (discussed below) to contact the inner wellbore surface and thereby hold, anchor, and seal the ballistically actuated plug **100** thereto, without causing failure of the ballistically actuated plug **100** by, for example, splitting the outer carrier **105**. Various considerations that may affect swell include the ratio of explosive mass to free volume within the wellbore tool, the material from which the swellable component is formed and properties such as, without limitation, the yield strength of the material, the thickness of the swellable component(s) such as the outer carrier **105**, and the type of ballistic component(s) (e.g., explosive loads, detonating cords, explosive pellets, etc.). Other considerations may be applicable for particular actuatable wellbore tools. In the case of the ballistically actuated plug **100**, for example, the type and position of the ballistic components **110** within the outer carrier **105** may affect the degree of swell at different portions/positions of the outer carrier **105**. These concepts are discussed further below with respect to the test results being provided herein.

With continuing reference to FIG. 1A and FIG. 1B, the exemplary outer carrier **105** includes a plurality of external gripping teeth **124** formed on an outer surface **121** of the outer carrier **105**. The outer carrier **105** is dimensioned such that the gripping teeth **124** will contact an inner surface **301** (FIG. 2B) of a wellbore casing **300** when the outer carrier **105** is in the expanded form. The gripping teeth **124** are shaped to frictionally grip the inner surface **301** of the wellbore casing **300** and thereby position the ballistically actuated plug **100** within the wellbore casing **300** and form a partial or total seal between the gripping teeth **124** and the inner surface **301** of the wellbore casing **300**, when the outer carrier **105** is in the expanded form **171**. By one understood measure in the art, a successful set for a plug in a plug-n-perf operation requires that the plug does not move or exert any significant signs of pressure loss or leakage under 10,000 psi of hydraulic pressure differential.

The exemplary ballistically actuated plug **100** also includes at least one sealing element **122** extending along at

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least a portion of the outer surface 121 of the outer carrier 105. In the exemplary embodiment(s) illustrated in FIG. 1A and FIG. 1B, two sealing elements 122, such as o-rings, extend around a circumference of the outer surface 121 of the outer carrier 105, within a depression 123 formed in the outer surface 121 of the outer carrier 105. Securing the sealing elements 122 within a complimentary receptacle such as depression 123 may help to maintain the position and configuration of the sealing elements 122 as the ballistically actuated plug 100 is pumped down into the wellbore. However, the sealing elements 122 in various embodiments may take any shape or configuration including with respect to fitting the sealing elements 122 on/to the outer carrier 105 or other portions of a ballistically actuated plug consistent with this disclosure.

The sealing elements 122 are formed from a material and in a configuration such that, in operation, the sealing elements 122 will expand along with the outer carrier 105 when the ballistic components 110 are initiated. The outer carrier 105 and the sealing elements 122 are dimensioned such that the sealing elements 122 will contact the inner surface 301 of the wellbore casing 300 and form a seal between the inner surface 301 of the wellbore casing 300 and the sealing elements 122 when the outer carrier 105 is in the expanded form 171.

With further reference to FIG. 1A and FIG. 1B, the exemplary embodiment(s) of the ballistically actuated plug 100 may include a bumper 116 secured to the second end 102 of the outer carrier 105. The ballistically actuated plug 100 is deployed in the wellbore with the second end 102 of the outer carrier 105 and bumper 116 downstream, i.e., further into the wellbore, than the first end 101 of the outer carrier 105. The bumper 116 may provide protection from impacts with the wellbore casing 300 as the ballistically actuated plug 100 is pumped down into the wellbore. The bumper 116 may be made from, without limitation, a plastic or rubber material such that the bumper 116 will absorb impacts on the wellbore casing 300. In an aspect, and with specific reference to FIG. 1B, an exemplary embodiment the bumper 116 may include one or more gills 181 having an inlet 182 in fluid communication with an outlet 183 and a flap 184 covering at least a portion of the outlet 183. As described below, as the ballistically actuated plug 100 is pumped down the wellbore the bumper 116 will be the leading end and wellbore fluid within the wellbore casing 300 will pass through the gills 181, from the inlet 182 to the outlet 183, and the flap 184 will provide additional resistance to the fluid flow as it exits the outlet 183. The flap 184 may be a stationary surface feature that covers a consistent portion of the outlet 183 or it may be, for example and without limitation, a bendable piece of material that is capable of opening and closing to different degrees, based on the velocity of the fluid flow, to dynamically adjust to changing conditions of the wellbore fluid. Generally, the gills 181 may help to stabilize and/or slow the pace of the ballistically actuated plug 100 as it is pumped down the wellbore, thereby decreasing impacts between the ballistically actuated plug 100 against the wellbore casing 300 and providing more control for positioning the ballistically actuated plug 100 at a desired location within the wellbore casing 300. In addition, the gills 181 may decrease fluid consumption for pumping the ballistically actuated plug 100 down into the wellbore, by allowing fluid in front (i.e., downstream) of the ballistically actuated plug 100 to pass through the gills 181 and thereby decreasing the pressure and friction acting against the leading end of the ballistically actuated plug 100 as it is pumped down.

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The bumper 116 may be connected to the second end 102 of the outer carrier 105 using adhesives, tabs, melding, bonding, and the like. In the exemplary embodiment(s) that FIG. 1A and FIG. 1B show, the bumper 116 is annular and a neck portion 160 of the outer carrier 105 extends from the outer carrier 105 and passes through an interior opening 180 of the annular bumper 116. A friction fit between the neck portion 160 and the inner surface (unnumbered) of the bumper 116 bounding the interior opening 180 may further secure the bumper 116 to the outer carrier 105 at the second end 102 of the outer carrier 105.

The neck portion 160 may be integrally (i.e., as a single piece) formed with the outer carrier 105 or bonded or machined on the outer carrier 105, or provided in the disclosed configuration, or other configuration(s) consistent with this disclosure, according to known techniques. For purposes of this disclosure, the “neck portion 160” is so called to aid in the description of the exemplary ballistically actuated plug 100 and without limitation regarding the delineation, position, configuration, or formation of the neck portion 160 with respect to the outer carrier 105 or other components. In the exemplary embodiments, for example, the neck portion 160 is formed integrally with the outer carrier 105, as a portion with a reduced outer diameter as compared to the outer carrier 105. The neck portion 160 includes a first end 161 and a second end 162 opposite the first end 161 and a channel 165 is formed within the neck portion 160 and defined by the neck portion 160. In the exemplary embodiments, the channel 165 extends from a first opening 163 on the first end 161 of the neck portion 160 to a second opening 164 on the second end 162 of the neck portion 160, wherein the channel 165 is adjacent and open to a second end opening 113 of the outer carrier 105, via the first opening 163 of the channel 165. The second end opening 113 of the outer carrier 105 is adjacent and open to the hollow interior chamber 104 of the outer carrier 105, and is effectively a terminus of the hollow interior chamber 104 at the second end 102 of the outer carrier 105.

The second opening 164 of the channel 165 within the neck portion 160 is sealed by a seal disk 118 positioned within the channel 165 and dimensioned to seal the channel 165 by engaging an inner surface (unnumbered) of the neck portion 160 bounding the channel 165. The seal disk 118 may include an additional sealing element, for example, o-ring 120. The ballistic components 110 are configured to dislodge the seal disk 118 from the channel 165 upon initiation of the ballistic components 110. Dislodging the seal disk 118 in combination with fragmenting the ballistic carrier 106 upon initiating the ballistic components 110 provides a flow path for hydrocarbons being recovered through the ballistically actuated plug 100, as explained below with respect to operation of the ballistically actuated plug 100. Accordingly, in the exemplary embodiments the ballistic components 110 are configured for fragmenting or disintegrating the ballistic carrier 106 upon initiation of the ballistic components 110 and the ballistic carrier 106 is formed from a fragmentable or breakable material such as injection molded plastic.

The outer carrier 105 includes a first end opening 103 at the first end 101 of the outer carrier 105 opposite the second end opening 113 at the second end 102 of the outer carrier, and the hollow interior chamber 104 extends from the first end opening 103 to the second end opening 113 and is open to each of the first end opening 103 and the second end opening 113. The first end opening 103 has a rim 103b that defines a passage 103a through the first end opening 103 of the outer carrier 105. In the exemplary embodiment(s), the

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passage 103a has a diameter d_s that is smaller than a diameter d_e (FIG. 4) of the hollow interior chamber 104. Thus, once the ballistic carrier 106 has been fragmented or disintegrated and the seal disk 118 has been dislodged from the channel 165, a flow path exists through the ballistically actuated plug 100 from the second opening 164 of the channel 165 to the first end opening 103 of the outer carrier 105.

In an operation of the exemplary ballistically actuated plug 100, and with reference to FIG. 2A, the ballistically actuated plug 100 in the unexpanded form 170 is pumped downhole via pump-down fluid in the wellbore casing 300 with the second end 102 of the outer carrier 105, including the bumper 116, downstream of the first end 101 of the outer carrier 105, i.e., with the second end 102 of the outer carrier 105 being the leading end in the direction of travel. Upon initiation of the ballistic components 110, the outer carrier 105 expands into its expanded form 171 in which the external teeth 124 and sealing element 122 of the outer carrier 105 engage the inner surface 301 of the wellbore casing 300 in a frictional, sealing engagement.

Similarly to FIG. 2A, in FIG. 2B the ballistically actuated plug 100 in the unexpanded form 170 may be pumped downhole via pump-down fluid in the wellbore casing 300 with the second end 102 of the outer carrier 105, including the bumper 116, downstream of the first end 101 of the outer carrier 105, i.e., with the second end 102 of the outer carrier 105 being the leading end in the direction of travel. Upon initiation of the ballistic components 110, the outer carrier 105 expands into its expanded form 171 in which the external teeth 124 and sealing element 122 of the outer carrier 105 engage the inner surface 301 of the wellbore casing 300 in a frictional, sealing engagement.

With reference to FIG. 2C, a rear cross-sectional view of the ballistically actuated plug 100 in its expanded form 171 is shown from upstream in the wellbore casing 300, towards the first end 101 of the outer carrier 105, and through the outer carrier 105 via the first end opening 103 of the outer carrier 105 and the hollow interior chamber 104 of the outer carrier 105. After the ballistic components 110 have detonated, and the ballistic carrier 106 has been fragmented and the seal disk 118 has been blown out, the hollow interior chamber 104 of the outer carrier 105 is open to a downstream portion of the wellbore casing 300 via the second end opening 113 of the outer carrier 105 and the second end opening 164 of the channel 165 through the neck portion 160. Thus, a flow path through the outer carrier 105 is created for hydrocarbons being recovered to the surface of the wellbore when the well is completed and put into production.

However, before the well is completed and put into production, each zone of the wellbore must be perforated. Typically, each zone of the wellbore is isolated before being perforated, to avoid fluid pressure losses to zones that have already been completed. Accordingly, when a zone upstream of the ballistically actuated plug 100 is to be perforated, a sealing ball, as is known, is dropped down into the wellbore casing 300 to isolate the upstream zone by sealing against an opening of the fluid path that the ballistically actuated plug 100 in the expanded form 171 has created. In the case of the exemplary embodiment shown in FIG. 2C, the ball may have a diameter for seating against the rim 103b of the passage 103a through the first end opening 103, and/or within a portion of the passage 103a of the first end opening 103, or against the second end opening 113 of the outer carrier 105. For example, as shown in FIG. 2D, after the ballistically actuated plug 100 is sealed in its expanded state

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171 against the inner surface 301 of the wellbore casing 300, the flow path through the first end opening 103 and the hollow interior chamber 104 of the outer carrier 105 may be sealed by a frac ball or other sealing component such as the bumper 116 (discussed below) which sets against the rim 103b that circumscribes the opening 103a therethrough, and thereby seals the flow path through the first end opening 103 of the outer carrier 105.

After the well is completed and ready for production, the balls sealing any ballistically actuated plugs 100 (or other plugs) may be drilled out, thus restoring the flow path through the outer carrier 105.

With reference to FIG. 2D, shows a cross-sectional side view of a frac ball secured within the ballistically actuated plug when the plug is in an expanded form. The ball is illustrated as being in a sealing relationship with the opening 103a of the outer carrier 105, and thereby seals the flow path through the first end opening 103 of the outer carrier 105.

FIG. 3 illustrates the ballistic carrier 106 including ballistic slots 109. The ballistic slots 109 are formed in an outer surface 130 of the body portion 115 of the ballistic carrier 106 and extend into the body portion 115 of the ballistic carrier 106. According to an aspect, and as described hereinabove, one or more ballistic components 110 are configured for being positioned within the ballistic slots 109.

As shown in FIG. 4, and with reference back to FIG. 1A and FIG. 1B, the ballistic carrier 106 may be dimensioned for being received within the hollow interior chamber 204 of the actuable wellbore tool 200. For example, an outer diameter d_1 of the ballistic carrier 106 may be sufficient to fit securely and not allow for excessive movement within the hollow interior chamber 204 which may have a diameter d_2 (as previously discussed with respect to FIG. 1A and FIG. 1B).

With reference now to FIG. 4, an alternative exemplary embodiment of the ballistic carrier 106 is shown housed within a hollow interior chamber 204 of a wellbore tool 200 generally. In the exemplary embodiment that FIG. 4 shows, the ballistic carrier 106 is substantially as has been described with respect to FIGS. 1A, 1B, and 3, and common features will not be repeated here. In the exemplary embodiment shown in FIG. 4, each ballistic slot 109 includes an opening 117 extending from the ballistic slot 109 to the axial bore 112 and open to each of the ballistic slot 109 and the axial bore 112. Providing the openings 117 between the respective ballistic slots 109 and the axial bore 112 may improve the reliability of the initiation between the initiator 114 and the ballistic components 110.

With reference now to FIGS. 1A-4, an exemplary method for positioning an instantaneously expanding, ballistically actuated plug within a wellbore includes, without limitation, deploying an instantaneously expanding, ballistically actuated plug 100 according to this disclosure into the wellbore casing 300 to a predetermined or desired position within the wellbore casing 300. Once the ballistically actuated plug 100 is at the predetermined or desired position within the wellbore casing 300, the initiator 114 positioned in the axial bore 112 of the ballistic carrier 106 is initiated. The ballistic component(s) 110 are then initiated by the initiator 114, and the forces generated by the initiation of the ballistic component(s) 110 within the hollow interior chamber 104 of the outer carrier 105 will cause expanding the outer carrier 105 from the unexpanded state 170 to the expanded state 171. Expanding the outer carrier 105 to the expanded state 171 causes the outer carrier 105 to contact the inner surface 301 of the wellbore casing 300 with the gripping teeth 124 on the

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outer surface **121** of the outer carrier **105**, according to the configuration of the outer carrier **105** in the expanded state **171**.

In an aspect of the exemplary method, expanding the outer carrier **105** from the unexpanded state **170** to the expanded state **171** includes expanding the sealing element **122** that extends along the outer surface **121** of the outer carrier **105**, wherein the outer carrier **105** and the sealing element **122** are together dimensioned for contacting and forming a seal between the sealing element **122** and the inner surface **301** of the wellbore casing **300** when the outer carrier **105** is in the expanded state **171**.

In an aspect of the exemplary method, initiating the ballistic component(s) **110** includes firing one or more ballistic component(s) **110** radially outwardly from the axial bore **112**.

In an aspect of the exemplary method, the ballistic carrier **106** is fragmented upon initiating the ballistic component **110**. In a further aspect of the exemplary method, the seal disk **118** is dislodged from the channel **165** within a portion of the outer carrier **105** upon initiating the ballistic component **110**. As a result, an aspect of the exemplary method includes enabling fluid communication through the hollow interior chamber **104** of the outer carrier **105** between a location upstream of the ballistically actuated plug **100** and a location downstream of the ballistically actuated plug **100**.

With reference now to FIGS. 5A-5C, an exemplary configuration and connections of the ballistically actuated plug **100** on a tool string **505** is shown. In the illustrated exemplary embodiment, the ballistically actuated plug **100** is connected to a tandem seal adapter (TSA) **500** as is known. For example and without limitation, the ballistically actuated plug **100** may include a threaded portion (not shown) on an interior surface (i.e., adjacent the passage **103a**) of the rim **103b** of the passage **103a** through the first end opening **103** of the outer carrier **105**. The TSA **500** may include a complimentary threaded portion **515** (FIG. 5C) on a first end **502** of the TSA **500** for connecting to the threaded portion on the rim **103b** of the passage **103a** through the first end opening **103** of the outer carrier **105**, and may also include one or more sealing components, such as o-rings **514** (FIG. 5C), for sealing the interior components of the ballistically actuated plug **100** and TSA **500** from wellbore fluid.

A detonator **501**, for example, a selective switch detonator as previously discussed, may be, as shown in phantom in FIG. 5A, partially held within the TSA **500** and extend into the ballistically actuated plug **100** for initiating the ballistic components **100**. The TSA **500** may be adapted to hold the detonator **501**. Alternatively, the TSA **500** may house a bulkhead **512** (shown in phantom in FIG. 5B), e.g., in an assembly as disclosed in U.S. Pat. No. 9,494,021, commonly assigned to DynaEnergetics Europe GmbH, for transferring a selective detonation signal to the detonator **501** (shown in phantom in FIG. 5B) which may be housed in a detonator holder **511** (shown in phantom in FIG. 5B) within the outer carrier **105** of the ballistically actuated plug **100**.

A cross-sectional view of the exemplary bulkhead **512** configuration in the TSA **500** is shown in FIG. 5C. FIG. 5C shows a cutaway portion of the ballistically actuated plug **100** and perforating gun **510** at the TSA **500** connection. The bulkhead **512** includes a first electrical contact **512a** and a second electrical contact **512b** for relaying an electrical signal or power supply between an upstream source or wellbore tool such as the perforating gun **510** and a downstream wellbore tool such as the ballistically actuated plug **100**. The electrical signal may be, for example, a selective detonation signal. In the exemplary embodiment, the second

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electrical contact **512b** electrically contacts a signal-in connection **513** of the detonator **501** and may relay the electrical signal or power supply therethrough to the detonator **501**. The detonator holder **511** holds the detonator **501** in the ballistically actuated plug **100**, for example in the hollow interior portion **104** of the outer carrier **105**.

The TSA **500** may connect at a second end **503** of the TSA **500** to a wellbore tool **510** such as a perforating gun, which may be connected as part of a tool string **505** to additional wellbore tools further upstream, i.e., in a direction away from the ballistically actuated plug **100**, as is known. In such configuration, the tool string **505** may be run downhole in the wellbore casing **300** such that after the ballistically actuated plug **100** is set within the wellbore casing **300** in its expanded form **171** as described herein, the additional wellbore tool(s) **510** may be initiated for various operations. In an example, and without limitation, the wellbore tool **510** may be a perforating gun that is fired after the ballistically actuated plug **100** is set. In such embodiment, the tool string **505** may be removed (for example, by retracting a wireline (not shown) to which the tool string is attached) after all perforating guns in the tool string **505** have fired, and a ball may then be dropped into the wellbore casing **300** as previously discussed, thereby sealing the flow path through the outer carrier **105** of the ballistically actuated plug **100** in its expanded form **171**. Once the ball has sealed the flow path and isolated the upstream zone, fracking fluid may then be pumped into the wellbore to fracture the hydrocarbon formations via the perforations that the perforating guns created.

In other embodiments, the ballistically actuated plug **100** may be connected to a firing head, as is known, for initiating the ballistically actuated plug **100**. The firing head may initiate, without limitation, a wireless detonator as described in U.S. Pat. No. 9,605,937, discussed above. The firing head may be connected to a wireline serving as a connection to the surface of the wellbore and/or a relay for a power supply or electrical control signals, as is known. In other embodiments, the ballistically actuated plug **100** and detonator **501** or other initiator may be electrically connected to a wireline that connects to, e.g., a top sub or other known connector that electrically connects the wireline to the detonator **501** via, for example, a relay such as the bulkhead **512** discussed with respect to FIG. 5C, or other known techniques. Whether conveyed as a single tool or as part of a tool string, a connector, firing head, etc. connected to the first end **101** of the outer carrier **105** should sufficiently seal the first end opening **103** of the outer carrier **105**, to prevent wellbore fluid and other contaminants from entering the hollow interior chamber **104**.

With reference now to FIG. 6, in an exemplary embodiment the ballistically actuated plug **100** may be a plug drone **600**. For purposes of this disclosure, a "drone" is a self-contained, autonomous or semi-autonomous vehicle for downhole delivery of a wellbore tool. For example, the drone may be sent downhole in the wellbore casing **300** without being attached to a wireline or other physical connection, and/or without requiring communication with the surface of the wellbore to execute a wellbore operation. In the exemplary embodiment FIG. 6 shows, the plug drone **600** includes a ballistically actuated plug section **601** at a first end, a control module section **610** at a second end opposite the first end, and a ballistic interrupt section **605** positioned between and connected to each of the ballistically actuated plug section **601** and the control module section **610**. For purposes of this disclosure, references to a "ballistically actuated plug section," "ballistic interrupt section,"

and “control module section” are to aid in the description of an exemplary plug drone including the relative positioning of various components, without limiting the description to any particular configuration or delineation of an exemplary plug drone or type, configuration, or distribution of components of an exemplary plug drone. The control module section **610**, ballistic interrupt section **605**, and configuration and operation generally of an autonomous wellbore tool including a control module section and ballistic interrupt section may be as described in International Patent Publication No. WO2020/035616 published Feb. 20, 2020, which is commonly owned by DynaEnergetics Europe GmbH and incorporated by reference herein in its entirety.

The ballistically actuated plug section **601** is substantially a ballistically actuated plug **100** as described throughout this disclosure, the description of which will not be repeated here. The ballistically actuated plug section **601** may be connected to the ballistic interrupt section **605** by, without limitation, a threaded engagement (e.g., as discussed with respect to a TSA **500** in FIG. **5**), a friction fit, a weld, a mold, an adhesive, or any other technique consistent with this disclosure. In an aspect, a body **606** of the ballistic interrupt section **605** may be formed from, without limitation, a fragmentable or disintegrable material, such as an injection molded plastic, such that the body **606** of the ballistic interrupt section **605** will substantially disintegrate upon detonation of the ballistic components **110** and/or a donor charge **622** as described below. In an exemplary configuration, the body **606** of the ballistic interrupt section **605** is formed integrally (i.e., as a single piece) with the ballistic carrier **106**, which may also be formed from the disintegrable injection molded plastic as previously discussed.

The ballistic interrupt section **605** includes a ballistic interrupt **640** housed within the body **606** of the ballistic interrupt section **605**. The ballistic interrupt **640** has a through-bore **642** formed therethrough at a position such that the through-bore **642** in the open position, as shown in FIG. **6**, is substantially parallel and coaxial with a ballistic channel **623** that is formed through the body **606** of the ballistic interrupt section **605**, in which the through-bore **642** is positioned. In the open position, the through-bore **642** forms a passage, within the ballistic channel **623**, between the donor charge **622** in the control module section **610** and the initiator **114** in the ballistically actuated plug section **601**. The ballistic channel **623** extends between the control module section **610**, adjacent the donor charge **622**, and the initiator **114** such that, when the ballistic interrupt **640** is in the open position, the ballistic channel **623** and the through-bore **642** together define a path for an explosive jet formed upon detonation of the donor charge **622** to pass through the ballistic channel **623** including the through-bore **642**, and reach the initiator **114** to initiate detonation of the ballistic components **110** in the ballistically actuated plug section **601**. In a closed position (not shown), the ballistic interrupt **640** of the exemplary embodiment is rotated approximately 90 degrees, such that the through-bore **642** is substantially perpendicular to the ballistic channel **623** and closes the ballistic channel **623** to prevent an explosive jet from the donor charge **622** from reaching the initiator **114**. In an aspect, the plug drone **600** is “armed” when the ballistic interrupt **640** is in the open position, and is in a safe, non-armed state when the ballistic interrupt **640** is in the closed position.

The ballistic interrupt **640** may be transported in the closed position and rotated from the closed position to the open position at the wellbore site, to arm the plug drone **600** before deploying the plug drone **600** into the wellbore. The

ballistic interrupt **640** includes a keyway **660** for accepting a tool that may be used to rotate the ballistic interrupt **640** from the closed position to the open position. The ballistic interrupt **640** may be rotated, via the keyway **660**, either manually or automatically in, or with, a device for engaging the keyway **660**. In an exemplary operation, the ballistic interrupt **640** is rotated, and the plug drone **600** is armed, in a launcher (not shown) that arms the plug drone **600** before launching it into the wellbore.

The control module section **610** is generally defined by a control module section body **611** and may be, without limitation, generally circumferentially-shaped and formed about a longitudinal axis **y**. The control module section body **611** may be formed from, without limitation, a fragmentable or disintegrable material, such as an injection molded plastic, such that the control module section body **611** will substantially disintegrate upon detonation of the ballistic components **110** and/or the donor charge **622**. In an aspect, the control module section **610** may be formed integrally (i.e., as a single piece) with the ballistic interrupt section **605**.

The control module section **610** includes a Control Interface Unit (CIU) **613** that may be, for example, a programmable onboard computer as described below or in International Patent Publication No. WO2020/035616 published Feb. 20, 2020, which is commonly owned by DynaEnergetics Europe GmbH and incorporated by reference herein in its entirety. The CIU **613** is housed within a control module housing **614** positioned within a hollow interior portion **612** of the control module section **610** and defined by the control module section body **611**. Charging and programming contacts **615** include pin contact leads **616** electrically connected to the CIU **613**, for example, to a programmable electronic circuit which may be contained on a Printed Circuit Board (PCB) **617**. The pin contact leads **616** may be exposed through, and sealed within, apertures **618** through a sealing access plate **619** that closes the hollow interior portion **612** of the control module section **610**. The charging and programming contacts **615** may be used for charging a power source of the CIU **613** and/or programming onboard circuitry by, for example and without limitation, connecting the charging and programming contacts **615** to a power supply and/or control computer at the surface of the wellbore, before deploying the plug drone **600** into the wellbore.

The CIU **613** may contain such electronic systems such as power supplies, programmable circuits, sensors, processors, and the like for detecting a position, orientation, or location of the plug drone **600** and/or the condition of the wellbore around the plug drone **600**, for powering the onboard computer systems and/or trigger/arming components, and for triggering initiation of the plug drone **600** as described below. In an aspect, the CIU **613** may include capacitor and/or battery power sources **620**, a detonator **621**, and a donor charge **622**. The detonator **621** is positioned for initiating the donor charge **622** upon receiving a signal (e.g., from the programmable electronic circuit) to detonate the plug drone **600**. The detonator **621** may include a Non-Mass Explosive (NME) body and the donor charge **622** may, in an aspect, be integrated with the explosive load of the detonator **621**. In an aspect of integrating the donor charge **622** with the explosive load of the detonator **621**, the amount of explosive may be adjusted to accommodate the donor charge **622** and the size and spacing of components such as a ballistic channel **623** along which a jet from the donor charge **622** propagates upon detonation of the donor charge **622**.

In an aspect, the CIU **613** may include the PCB **617** and a fuse for initiating the detonator **621** may be attached directly to the PCB **617**. In an aspect of those embodiments, the detonator **621** may be connected to a non-charged firing panel—for example, a selective detonator may be attached to the PCB **617** such that upon receiving a selective detonation signal the firing sequence, controls, and power may be supplied by components of the PCB **617** or CIU **613** via the PCB **617**. This can enhance safety and potentially allow shipping the fully assembled plug drone **600** in compliance with transportation regulations if, as discussed above, the ballistic interrupt **640** is in the closed position. Connections for the detonator **621** (and associated components) on the PCB **617** may be, without limitation, sealed contact pins or concentric rings with o-ring/groove seals to prevent the introduction of moisture, debris, and other undesirable materials.

In alternative embodiments, the CIU **613** may be configured without a control module housing **614**. For example, the CIU **613** may be contained within the hollow interior portion **612** of the control module section **610** and sealed from external conditions by the control module section body **611** itself. Alternatively, the CIU **613** may be housed within an injection molded case and sealed within the control module section body **611**. The injection molded case may be potted on the inside to add additional stability. In addition, or alternatively, the control module housing **614** or other volume in which the CIU **613** is positioned may be filled with a fluid to serve as a buffer. An exemplary fluid is a non-conductive oil, such as mineral insulating oil, that will not compromise the CIU components including, e.g., the detonator **621**. The control module housing **614** may also be a plastic carrier or housing to reduce weight versus a metal casing. In any configuration including a control module housing **614** the CIU components may be potted in place within the control module housing **614**, or alternatively potted in place within whatever space the CIU **613** occupies.

The detonator **621** and the donor charge **622** are contained within the control module housing **614** and the donor charge **622** is substantially adjacent to and aligned with the ballistic channel **623** along the axis *y* which is further aligned with the initiator **114**. Upon detonation of the detonator **621**, the donor charge **622** is initiated and the explosive jet from the donor charge **622** will pierce a portion **624** of the control module housing **614** that is positioned between the donor charge **622** and the ballistic channel **623** and propagate into the ballistic channel **623**. When the ballistic interrupt **640** is in the open position, the explosive jet will reach the initiator **114** which will in turn initiate the ballistic components **110** to expand the outer carrier **105** of the ballistically actuated plug section **601** in the same manner as described throughout this disclosure for a ballistically actuated plug **100**.

In an aspect of the exemplary plug drone(s) described above, the bumper **116** on the ballistically actuated plug section **601** may act as, or be replaced by, a frac ball for sealing a plug as previously discussed. For example, the frac ball, which may be the bumper **116**, may be attached to the ballistically actuated plug section **601** of a second plug drone **600** that is deployed into the wellbore after a first plug drone has previously been set in the wellbore casing **300** with the outer carrier **105** in the expanded form **171**. When the second plug drone **600** is actuated, the frac ball—made from a resilient material—is detached from the second plug drone **600** and propelled downstream towards the expanded plug. The frac ball is dimensionally configured to seal the expanded plug as previously discussed. Accordingly, one plug may be sealed as another is set upstream in the next

zone to be perforated. However, the frac ball may also be attached to any wellbore tool, or may itself be the wellbore tool, for autonomous deployment on a ballistically actuated drone. In embodiments where the bumper **116** serves as a frac ball, e.g., to seal a plug that has been set downstream, the bumper **116** may not be annularly shaped but have, for example, a solid front portion such that the interior opening **180** of the bumper **116** is closed at one end to prevent the flow of fluid therethrough.

With reference now to FIG. 7, an alternative exemplary configuration of a drone according to the disclosure includes a daisy-chained, ballistically actuated, autonomous wellbore tool assembly **700** including a single CIU **613** connected to and controlling each of a first wellbore tool **601** and a second wellbore tool **510**. In the exemplary embodiment shown in FIG. 7, the first wellbore tool may be a ballistically actuated plug **601** according to the exemplary embodiments described herein. The CIU **613** may be positioned within a control module section **610** connected to or integral with a ballistic interrupt section **605** that includes a ballistic interrupt **640** as previously shown in and described with respect to FIG. 6. In the exemplary embodiment, the second wellbore tool **510** may be a perforating gun assembly (or, perforating assembly section of the wellbore tool assembly) such as described in International Patent Publication No. WO2020/035616 published Feb. 20, 2020, which is commonly owned by DynaEnergetics Europe GmbH and incorporated by reference herein in its entirety. The perforating gun assembly **510** may include one or more shaped charges **701**. In the exemplary embodiment shown in FIG. 7, the CIU **613** and the ballistic interrupt **640** control operation of each wellbore tool in the daisy-chained string. The different tools or sections of the assembly may be, without limitation, integrally formed as a single piece of a common material or separate components that are joined by known techniques such as molding, threaded connectors, welding, positive locking engagements, friction fits, and the like.

In an exemplary operation of a plug drone **600** as described with respect to FIG. 6, the plug drone **600** may be transported to a wellbore site with the ballistic interrupt **640** in the closed position. The plug drone **600** may then be connected, via the charging and programming contacts **615**, to a power supply and/or computer interface at the wellbore site, to charge the power source **620** of the plug drone **600** and provide deployment and detonation instructions to onboard electronic circuitry. The ballistic interrupt **640** may be rotated from the closed position to the open position when the plug drone **600** is ready for deployment.

Once deployed in the wellbore, the plug drone **600** may use onboard sensors to determine a speed, orientation, position, and the like of the plug drone **600** within the wellbore. The plug drone **600** may transmit to a surface controller information determined by the sensors, for generating a wellbore topography profile. The plug drone **600** may also use, for example and without limitation, temperature and pressure sensors to determine a temperature and pressure of the wellbore around the plug drone **600** and may transmit to the surface controller a profile of such wellbore conditions.

Upon reaching a predetermined location within the wellbore as determined by, without limitation, an elapsed time from deployment, a distance traveled, a location as determined from, e.g., casing collar locators (CCLs) or other known position-sensing devices, an orientation of the plug drone **600**, and the like, the CIU **613** may trigger the detonator **621** to detonate and thereby initiate the donor charge **622**, which will detonate and form an explosive jet

that will propagate through the ballistic channel **623** and initiate the initiator **114**. The initiator **114** will in turn initiate the ballistic components **110** and cause the ballistically actuated plug section **601** to expand and engage the inner surface **301** of the wellbore casing **300** at a desired location, at which the plug will be set. Instructions regarding, e.g., the predetermined location and/or conditions at which the plug drone **600** should detonate may be programmed into the CIU **613**, via the charging and programming contacts **615**, by a computer interface at the surface of wellbore, before the plug drone **600** is deployed in the wellbore. While the above sensor-based type initiation is particularly useful in the exemplary plug drone **600** in which no physical connection with the surface is maintained after the plug drone **600** is deployed into the wellbore, such techniques are not limited to use with an autonomous tool and may also contribute to automating deployment and actuation of non-autonomous wellbore tools such as those attached to wirelines or tool strings.

In the exemplary embodiments, the ballistic carrier **106** in the ballistically actuated plug section **601**, the body **606** of the ballistic interrupt section **605**, and the control module section body **611** are each made from a frangible or disintegrable material that will substantially fragment or disintegrate upon detonation of the detonator **621**, donor charge **622**, and/or ballistic components **110**. The CIU **613** and other internal components of the plug drone **600** may be similarly fragmented into debris that will be carried away from the plug drone **600** upon expansion. Accordingly, the plug drone **600** post expansion will substantially resemble the configuration of the ballistically actuated plug **100** in the expanded form **171**, as shown and described with respect to FIG. 2C. Isolation of an upstream wellbore zone and completion of the zone may then proceed as previously discussed.

A method of transporting and arming the exemplary plug drone **600** for use at the wellbore site may include transporting the plug drone **600** in a safe state to the wellbore site and arming the ballistically actuated plug drone **600** at the wellbore site. The safe state of the plug drone **600** is when the ballistic interrupt **640** is in the closed position and arming the plug drone **600** includes moving the ballistic interrupt **640** from the closed position to the open position. The method may also include programming the CIU **613** of the plug drone **600** and/or charging a power source **620** of the plug drone **600**, at the wellbore site.

With reference back to FIG. 7, an exemplary method for performing a plug-n-perf operation using the exemplary ballistically actuated, autonomous wellbore tool assembly **700** may be according to similar principles as for use of the plug drone **600** and incorporating, e.g., the perforating step. For example, the method may include deploying the ballistically actuated, autonomous wellbore tool assembly **700** into the wellbore and, first, initiating detonation of one or more shaped charges in the perforating gun assembly **510** by, for example, providing an explosive jet from the donor charge **622** to initiate a booster and/or detonating cord (or other initiator) in the perforating gun assembly **510** for initiating the shaped charge(s) **701**. The ballistically actuated plug **601** may be initiated prior to initiating the perforating gun assembly, without limitation, one or a combination of a separate initiation signal that the CIU **613** may send through a relay through the perforating gun assembly **510** to a separate initiator in the ballistically actuated plug **601**, a ballistic energy transfer, such as, e.g., a booster, donor charge, or combination of the two and/or other initiating components, from the initiator in the perforating gun assem-

bly **510** to an initiator of the ballistically actuated plug **601**, and a portion of the same initiator in the perforating gun assembly **510**, such as a detonating cord, that extends into the ballistically actuated plug **601**. Accordingly, an explosive component of the ballistically actuated plug **601** will be initiated and thereby expand the ballistically actuated plug **601** to an expanded state **171** before or after the perforating has been performed further upstream. The body portions **606**, **611** of the various sections of the ballistically actuated, autonomous wellbore tool assembly **700** may be formed from a fragmentable or disintegrable material such that during the actuation processes those body portions **606**, **611** and other components are fragmented or destroyed and the debris is allowed to pass downstream through the flow path formed by the ballistically actuated plug **601** in the expanded state **171**. A frac ball or other sealing element may then be provided to seat against and seal the flow passage through the expanded plug, as previously discussed, and isolate the perforated zone.

With reference now to FIG. 8, an exemplary embodiment of a plug drone **600** such as shown in and discussed above with respect to FIG. 6 may include a frac ball **802** (or similar component) connected to the control module section **610** by a connector **800** that may be any structure consistent with this disclosure. For example, the connector **800** may be, without limitation, an integrally formed extension of the control module section body **611** or may be connected to the control module section body **611** by any known technique such as threading, adhesives, positive locking engagements, resilient retaining structures, and the like. The connector **800** may retain the frac ball **802** by any known technique such as magnetically, frictionally, by resilient retainers, and the like. Other connectors generally of any configuration, operating principle, or otherwise may be used consistent with this disclosure. The plug drone **600** in the exemplary embodiment of FIG. 8 is deployed and actuated within the wellbore as previously described with respect to, e.g., FIG. 6. The control module section body **611** and ballistic interrupt section body **606** may be formed from frangible or disintegrable materials, as discussed above. Upon actuating the tool, i.e., initiating the detonator **621**, the donor charge **622**, and the initiator **114** and expanding the ballistically actuated plug **601** to the expanded state **171**, the control module section body **611** and ballistic interrupt section body **606** may be fragmented/disintegrated by the ballistic, thermal, and/or kinetic energies, and the CIU **613** and remaining components may also be destroyed/fragmented, and the debris washed downstream through the open hollow interior chamber **104**. The frac ball **802** may then advance into and seat against the first end opening **103** of the outer carrier **105**, to seal the expanded plug and isolate a perforating zone as previously discussed.

In an aspect, one or more of the frac ball **802** and various components of the plug drone **600** (or actuatable wellbore tool, generally) may be formed from known degradable materials that will dissolve in the wellbore fluid and therefore not require drilling out.

In an aspect, the exemplary plug drone **600** including the frac ball **802** carried thereon may be part of a daisy-chained assembly **700** including a perforating gun **510** as shown in and described with respect to FIG. 7. The frac ball **802** may be, without limitation, positioned and carried between the perforating gun **510** and the ballistically actuated plug section **601**.

Tests

With reference now to FIGS. 9-20L, a test setup, components, and results for evaluating the effect of certain

variables in a ballistically actuated plug design on the swell induced in the outer carrier are shown. The tests included, among other things, various setups, explosive weights for ballistic components, kinds of explosive products for the ballistic components, and materials for the outer carrier. For example, as shown in FIG. 9, two different fluids, air **905** and water **907**, were used as the medium both within (**104**) and outside of the outer casing **105**. The test setups illustrated in FIG. 9, and explained in greater detail below, are: a) air filled plug in air; b) air filled plug in water; c) water filled plug in water; d) cord on a noncompressible core **910** in water; e) cord on a hollow core **912**, filled with water, in water; and f) cord on a hollow core **912**, filled with air, in water.

With reference to FIGS. **10A-11A**, explosive pellets **915** such as the pressed rings discussed with respect to the ballistic carrier **106** are shown as used in tests a)-c). The explosive pellets **915** included different outside diameters (OD) and explosive loads as indicated in the test results below. All of the pellets were formed from octahydro-1,3,5,7-tetranitro-1,3,5,7-tetrazocine (High Melting Explosive (HMX)). The pellets **915** were positioned approximately in the middle of the hollow interior **104** of the outer carrier **105** and held in place between pellet holder plates **916**. A detonating cord **920** was passed through the center of the plates **916** and pellet **915** to initiate the pellet **915**. This test setup was used in tests 1 and 2. The test conditions, including the casing (outer carrier **105**) size, outer and inner media, explosive mass of the pellet **915**, diameter of the pellet **915**, and max swell observed in each of tests 1 and 2 are shown in Table 1 below. Except where otherwise noted, the tests were performed with a 4.5" casing that was a steel pipe with min. tensile strength=95,000 psi, min. yield strength=550 MPa, and max. hardness=240 HBW.

TABLE 1

Test Nr	Casing	Outer Medium	Inner Medium	Explosive mass	Pellet Diameter	Max Swell
Test 1	4.5" L80	air	air	22.7 g	39 mm	1.4 mm
Test 2	4.5" L80	air	air	50 g	55 mm	5.4 mm

FIG. **11B** shows the casing and swell profile observed after test 1.

FIG. **11C** shows the casing and swell profile observed after test 1.

FIG. **11D** shows the casing and swell profile for test 2.

FIG. **11E** shows the casing and swell profile for test 2.

With reference now to FIG. **12A**, test 3 included the same setup for the explosive pellet **915** as in tests 1 and 2 except that the outer carrier **105** was closed completely with two caps **925** and the whole system was submerged in water to evaluate the influence of a surrounding medium. The properties and max swell in test 3 are shown in Table 2 below. FIGS. **12B** and **12C** show the casing and swell profile after test 3.

TABLE 2

Test Nr	Casing	Outer Medium	Inner Medium	Explosive mas	Pellet Diameter	Max Swell
Test 3	4.5" L80	water	air	50 g	55 mm	44 mm

With reference now to FIGS. **13A** and **13B**, the influence on swell of an inner medium was evaluated in tests 4-6, otherwise using the same test setup as in tests 1-3. As air is very compressible, one theory was that changing the inner medium to water would significantly influence the swell. The pellet **915** was sealed with a silicone and centered inside the outer carrier **105** using a plastic fixture **930**. Similar to test 3, the ends of the outer carrier were capped (not shown) after the hollow interior **104** was filled with water, and the system was submerged in water. The properties and max swell in tests 4-6 are shown in Table 3 below.

TABLE 3

Test Nr	Casing	Outer Medium	Inner Medium	Explosive mass	Pellet Diameter	Max Swell
Test 4	4.5" L80	water	water	50 g	55 mm	20 mm
Test 5	4.5" L80	water	water	22.7 g	38 mm	7.4 mm
Test 6	4.5" L80	water	water	22.7 g	55 mm	8.6 mm

According to the results of tests 1-6, it is believed that each of changing the inner medium from air to water and especially providing water within the outer carrier such that water is between the explosive and the outer carrier, increasing the explosive mass, and increasing the pellet diameter have a significant impact for increasing the amount of swell. Changing the outer medium from air to water slightly decreased the swell.

FIG. **13C** shows the casing and swell profile after test 4.

FIG. **13D** shows the casing and swell profile after test 4.

FIG. **13E** shows the casing and swell profile after test 5.

FIG. **13F** shows the casing and swell profile after test 5.

FIG. **13G** showS the casing and swell profile after test 6.

FIG. **13H** showS the casing and swell profile after test 6.

With reference now to FIGS. **14-15E**, tests 7-9 were performed to evaluate the impact of decreasing the free inner volume of the outer carrier **105** with an inner core **935** of varying material. For each test, a 50 g pellet **915** (53 mm OD) was positioned in the middle of the inner core **935** within the outer carrier **105**.

In test 7, the inner core **935** was an aluminum pipe. FIG.

15A shows the carrier and swell profile after test 7.

FIG. **15B** also shows the carrier and swell profile after test 7.

In test 8, the inner core **935** was a plastic tube. FIG. **15C** shows the carrier and the swell profile after test 8.

FIG. **15D** also shows the carrier and the swell profile after test 8.

In test 9, the inner core **935** was a steel tube. FIGS. **15E** and **15F** show the carrier and the swell profile after test 9.

As shown in FIGS. **15B**, **15D**, and **15F**, the swell induced by each of tests 7-9 is not uniform, and the maximum swell achieved in the middle of the casing was by the plastic tube.

With reference now to FIG. **16A**, test 10 replaced the explosive pellet with about 9 rows of detonating cord **920** wrapped around an inner core **935** of polyvinyl chloride (PVC) that was inserted into the carrier. The detonating cord in these and other tests include HMX explosive material. The resulting explosive weight was about 48.06 g.

As shown in FIG. **16B**, this arrangement cut the carrier in half such that a swell measurement was not possible.

With reference now to FIG. **16C**, for test 11 a similar setup as in test 10 was used but the length of detonating cord **920** (number of rows) was decreased and the thickness of the

cord was increased. The resulting explosive weight was about 51.66 g. As shown in FIG. 16D, this arrangement cut the carrier in half such that a swell measurement was not possible.

Based on the results from tests 10 and 11, it is believed that the free space in the carrier may play an important role in swelling the carrier such that decreasing the free space in the carrier could have a severe impact on the carrier.

With reference now to FIG. 17A and FIG. 17B, to avoid rupturing the carrier as in tests 10 and 11, test 12 was designed with a PVC having an inner diameter (ID) of 50 mm and an inner free space 940. The total explosive weight from the detonating cord 920 was approximately 48 g and the inner free space 940 had a diameter of 50 mm.

The tests noted above re FIG. 17A and FIG. 17B were performed with air as the inner and outer media.

FIG. 17C shows the carrier and the swell profile after test 12, and a substantially uniform swell in the carrier.

FIG. 17D also shows the carrier and the swell profile after test 12, and a substantially uniform swell in the carrier.

With reference now to FIG. 18A, test 13 included a test setup similar to test 12 but with an increased length of detonating cord 920 including dummy cord to space out the explosive detonating cord 920. The explosive weight was approximately 48 g.

FIGS. 18B and FIG. 18C each show the carrier and swell profile after test 13.

As shown in FIG. 17D and FIG. 18C, the PVC core with free space filled with air seems to induce a more uniform swell and prevents the rupturing observed in tests 11 and 12 with a solid PVC core. In addition, increasing the width of the cord axially along the inner core apparently significantly decreases the maximum swell.

With reference now to FIG. 19A, test 14 used approximately 48.06 g explosive weight of detonating cord 920 and a PVC core with an ID of 62 mm, and therefore increased free space 940 compared to tests 12 and 13. The PVC core was filled with water. The outer carrier 105 was sealed with caps 925.

FIG. 19B similarly shows the set up for test 14.

FIGS. 19C and 19D show the carrier and swell profile after test 14. After test 14, the swell was not completely round and somewhat inconsistent. The swell had certain areas with an oval profile.

Accordingly, as shown in FIG. 19D, the circumference of the carrier after test 14 was measured on two different axes: 0 degrees and 90 degrees. The average circumference value (charted in FIG. 19D) is the average of the 0-degree and 90-degree measurements.

Filling the casing with water (test 14) instead of air (tests 12 and 13) seems to have increased the maximum swell, likely due to the water as an inner medium. Test 13 showed the least amount of swell of tests 12-14, likely due to the explosive sections of the detonating cord being spaced further apart.

With reference now to FIG. 20A, tests 15-17 investigated the possibility of increasing the swell length (i.e., axially along the carrier) in a 4.5" carrier 105.

FIG. 20B illustrates that the setup included wrapping the detonating cord 920 in two different rows around the PVC inner core 935 with an inner free area 940. In test 15, approximately 58.5 g of explosive weight was used between the two rows of detonating cord 920.

FIG. 20C shows the carrier and swell profile after test 15, and the increased axial region that experienced swell versus previous tests.

FIG. 20D also shows the carrier and swell profile after test 15, and the increased axial region that experienced swell versus previous tests.

With reference now to FIG. 20E, test 16 used a similar setup with respect to the inner core 935 as in test 15, but in test 16 the total explosive weight was increased to 61.2 g and the 4.5" outer carrier 105 was inserted into and shot within a 5.5" casing 945 representing a wellbore casing within which the carrier/wellbore tool would be actuated.

FIG. 20F also shows that test 16 used a similar setup with respect to the inner core 935 as in test 15, with a similar adjustment in total explosive weight as in FIG. 20E.

FIG. 20G shows the carrier and swell profile after test 16, after which the carrier was capable of removal from the casing 945.

FIG. 20H also shows the carrier and swell profile after test 16, after which the carrier was capable of removal from the casing 945.

With reference now to FIGS. 20I-20L, test 17 used a similar setup to test 16 but the explosive weight from the detonating cord was approximately 115 g. As shown in FIG. 20L, test 17 also caused an open crack on the outer surface of the carrier.

FIG. 20I shows the carrier and swell profile after test 17, in which the carrier got stuck in the casing as shown in FIG. 20K. The swell was measured after cutting the casing open and removing the carrier from within.

FIG. 20J also shows the carrier and swell profile after test 17, in which the carrier got stuck in the casing as shown in FIG. 20K. As in FIG. 20I, the swell was measured after cutting the casing open and removing the carrier from within.

According to tests 15-17, two rows of detonating cord on the inner core apparently induce a wider (i.e., along a greater axial length of the carrier) swell compared to one row of cord. Increasing the explosive weight apparently increases the maximum swell and the fixation of the carrier in the wellbore casing.

Test 18 evaluated a different 4.5" carrier grade and used a similar setup with detonating cord 920 wrapped around an inner core 935 as in tests 15-17, and the inner core 935 was placed in a carrier 105 made from D10053 ST 37 steel and shot in a 5.5" casing. The total explosive weight from the detonating cord was approximately 54 g. The carrier became completely trapped in the casing and swell was not measured.

Overall, according to the test results, using the detonating cord as the explosive material instead of the explosive pellet results in an increase in the swollen region. Other suggestions from the testing include: 1) the inner and outer medium fluid directly affect the amount of swell and the shape of the swell; 2) increasing explosive weight (while keeping other conditions constant) increases the amount of swell; 3) the amount of free volume in the carrier affects the swell; 4) using water instead of air between the explosive and the carrier, within the carrier, increases the swell; 5) the material of the inner core (e.g., to reduce free volume in the carrier) affects the swell; 6) the grade of steel from which the carrier is formed affects the amount of swell; and 7) where two rows of detonating cord are used on a PVC inner core, the row at which initiation starts induces a greater swell than the other row.

In other testing done with a setup including a PVC inner core with inner free volume such as in test 12, except with water as an inner medium and an outer medium, results showed or suggested, among other things, that doubling the thickness of the outer carrier wall from 7 mm to 14 mm

decreased swell by approximate 58% but prevented the outer carrier wall from cracking and substituting steel for the PVC as the inner core material increased the swell by approximate 131%.

FIG. 21 is a cross-sectional view of a wellbore casing 2104 including a ballistically actuated wellbore tool positioned in the wellbore casing 2104. According to an aspect, the ballistically actuated wellbore tool includes a ballistically actuated plug 2100. According to an aspect, the ballistically actuated plug 2100 includes a ballistic housing 2106. The ballistic housing 2106 may be connected to a perforating gun housing 2108 on a first end 2110 of the ballistic housing 2106 and an outer seal housing 2102 on a second end 2112 of the ballistic housing 2106. The ballistically actuated plug 2100 includes the outer seal housing 2102 connected to the ballistic housing 2106. The ballistic housing 2106 may be attached to a perforating gun housing 2108 or another wellbore tool.

As seen in FIG. 21, for example, an outer diameter (a third outer diameter) 2118 of the ballistic housing 2106 may be less than an outer diameter (second outer diameter) 2116 of the outer seal housing 2102, and an outer diameter (a first outer diameter) 2114 of the perforating gun housing 2108 may be less than the second outer diameter 2116 of the ballistic housing 2106. The second outer diameter 2116 of the outer seal housing 2102 may be the largest of the outer diameters because the ballistically actuated plug 2100 is contemplated to expand in an outward configuration in order to form a sealing engagement with the wellbore casing 2104.

FIG. 22 depicts a cross-sectional view of the wellbore casing 2104, with the ballistically actuated plug 2100 being disposed therein. While FIG. 22 illustrates the ballistically actuated plug 2100 being in a relatively centralized position in the wellbore casing 2104, it is contemplated that the ballistically actuated plug 2100 may be decentralized, that is, the ballistically actuated plug 2100 may be positioned in the wellbore casing 2104 such that the clearance between the ballistically actuated plug 2100 and the wellbore casing 2104 varies along the length of the ballistically actuated wellbore tool.

According to an aspect, the ballistic housing 2106 includes a first housing portion 2204 including a chamber 2208. The chamber 2208 may be configured as an open space within which an item or other physical structure may be positioned. The chamber 2208 may be a drilled out portion of the first housing portion 2204 of the ballistic housing 2106. According to an aspect, the chamber 2208 is a multi-bore space (i.e., consisting of bores having multiple different inner diameters) for receiving one or more ballistic components. For example, the chamber 2208 includes an initiator 2206. The initiator 2206 may be a detonator such as a wireless detonator as described in U.S. Pat. No. 9,605,937, which is commonly assigned to DynaEnergetics Europe GmbH and incorporated herein by reference in its entirety. According to an aspect, the initiator 2206 includes a wireless detonator that is capable of being positioned or placed into a perforating gun assembly with minimal effort, by means of placement/positioning within a detonator positioning assembly, for wireless connection to adjacent electrical components in the gun assembly. In an embodiment, the detonator positioning assembly includes the detonator positioned within the detonator positioning assembly, which is then positioned within the chamber 2208 of the ballistic housing 2106. The detonator may be configured to electrically contactably form an electrical connection without the need of manually and physically connecting, cutting or crimping

wires as required in a wired electrical connection. In other words, the detonator may be a wirelessly-connectable selective detonator.

As illustrated in at least FIG. 22, and with reference to FIGS. 24-26, embodiments of the disclosure are associated with a wellbore tool assembly including a perforating gun bulkhead assembly 2220, and a sub-assembly 2202 including a first sub portion 2214 and a second sub portion 2218. The first sub portion 2214 may be coupled to the perforating gun housing 2108. According to an aspect, the second sub portion 2218 is coupled to the first housing portion 2204. As would be understood by one of ordinary skill in the art, an electrical feed-through assembly may be positioned in the sub (i.e., in a bore formed through the sub). The electrical feed through assembly is illustrated as being a bulkhead assembly 2220, however, other electrical feed through mechanisms are contemplated. The bulkhead assembly 2220 is illustrated as being connected to the initiator 2206, which is positioned in the ballistic housing 2106 of the ballistically actuated wellbore tool, and to a detonating cord, electrical feedthrough, and shaped charge positioning device provided in the adjacent perforating gun housing 2108.

According to an aspect, the ballistic housing 2106 further includes a second housing portion 2212. The second housing portion 2212 extends from the first housing portion 2204 and is configured to extend through an opening 2312 (or a central channel) of the outer seal housing 2102 and has a noncompressible core 2216. The noncompressible core 2216 may include a core of solid material as opposed to a hollow core (or a portion of a structure that includes an empty chamber or an empty space). In at least one other configuration, the noncompressible core 2216 may include a space or chamber that is filled with a liquid. According to an aspect, the noncompressible core 2216 has an outer diameter that varies based on the amount of detonating cord or explosive layers that is utilized. According to an aspect, the noncompressible core 2216 may include protrusions that act as spacers. The detonating cord or explosive layers may be disposed in between these protrusions or may be disposed circumferentially around the protrusions. The noncompressible core 2216 helps to ensure that the ballistic housing 2106 remains intact after detonation of an initiator 2206 and the detonation extender 2222 expands and deforms the outer seal housing 2102.

In an aspect, the ballistic housing is formed from a single monolithic piece of metal. The ballistic housing includes a recess 2224 formed in an outer surface 2226 of the second housing portion 2212. The recess 2224 may be a reduced diameter portion of the second housing portion 2212 and in an aspect, may extend circumferentially around at least the noncompressible core 2216 of the second housing portion 2212. The recess 2224 extends between a first retention portion 2228 (first retention shoulder) and a second retention portion 2230 (second retention shoulder) of the second housing portion 2212. Each of the first retention portion 2228 and the second retention portion 2230 may form a surface on which the outer seal housing 2102 can be retained. The recess 2224 may be configured as a gap that extends between the first retention portion 2228 and the second retention portion 2230. According to an aspect and as illustrated in FIG. 22, FIG. 23, and FIG. 24, the first retention portion 2228 is in engagement with an inner surface 2232 of the outer seal housing 2102 at the first end 2308 of the outer seal housing 2102, and the second retention portion 2230 is in engagement with the inner surface of the outer seal housing 2102 at the second end 2310 of the outer seal housing 2102.

According to an aspect, a channel **2210** extends from the chamber **2208** of the first housing portion **2204** to the recess **2224**. The channel **2210** may extend in a direction that is away from the chamber **2208** and towards the noncompressible core **2216**. According to an aspect, the channel extends in a direction between a longitudinal axis **2234** and an axial direction **2236** (radial axis) of the first housing portion **2204**. According to an aspect, the channel **2210** may extend at an angle of less than 90-degrees away from the noncompressible core **2216** and may be open to an external area defined by the recess **2224** outside of the second housing portion **2212**.

According to an aspect, a detonation extender **2222** is in ballistic communication with the initiator **2206**. The detonation extender **2222** may be in side-by-side contact with the initiator **2206** in such a configuration that the side-by-side contact ensures that the ballistic component or explosive material of the initiator **2206** can initiate the detonation extender **2222**. The detonation extender **2222** may further extend into the channel **2210** and at least around a portion of the noncompressible core **2216**. As illustrated in FIG. **22** and FIGS. **23-24**, for example, the detonation extender **2222** extends into the recess **2224** formed at the second housing portion **2212**. According to an aspect, the detonation extender **2222** is a detonating cord or a plurality of compressed explosive pellets. The detonation extender **2222** may be circumferentially disposed around the noncompressible core **2216**. According to an aspect, the detonation extender **2222** is positioned in the recess **2224**. When the detonation extender **2222** includes a detonating cord, the detonating cord is wrapped around the noncompressible core **2216** to form at least two layers of detonating cord. The two layers of detonating cord may be wrapped around the noncompressible core **2216** in a staggered configuration.

As described hereinabove, the detonation extender **2222** is positioned so that it is in relatively close contact with or physically touching a shell **2238** of the initiator **2206**. The detonation extender **2222** may include any sort of explosive mechanism (compressed explosive powder, a gel explosive, a detonating cord, or the like). While the detonation extender **2222** is illustrated as being in a side-by-side contact with the detonator, it is contemplated by the detonation extender **2222** may be in an end-to-end arrangement. To be sure, the initiator **2206** and detonation extender **2222** may be arranged in any configuration that allows the detonation extender **2222** to be initiated by the initiator **2206**.

At least a portion of the detonation extender **2222** may be positioned in the ballistic channel **2210** that extends from the chamber **2208** of the ballistic housing **2106** to an exterior surface or portion of the ballistic housing **2106**. The detonation extender **2222** may also be disposed between a gap formed between the ballistic housing **2106** and the outer seal housing **2102**. The gap is formed by the ballistic housing **2106** being positioned within a space extending from a first end of the outer seal housing **2102** to a second end of the outer seal housing **2102**. According to an aspect, the inner diameter of the outer seal housing **2102** may be larger than at least a portion of the outer diameter of the ballistic housing **2106**. At least another portion of the second housing portion **2212**, positioned in the space of the outer seal housing **2102** may have an outer diameter that (with or without o-rings) facilitates frictional engagement with the ballistic housing **2106** and the outer seal housing **2102**.

According to an aspect, the outer seal housing **2102** is circumferentially disposed around the noncompressible core **2216**. The detonation extender **2222** may be configured, upon detonation, to expand the outer seal housing **2102** from

an unexpanded form to an expanded form to secure the outer seal housing **2102** within the wellbore casing.

FIG. **23** illustrates the outer seal housing **2102** being circumferentially disposed around the second housing portion **2212**. The outer seal housing **2102** includes a first end **2308**, and a second end **2310** opposite the first end **2308**. According to an aspect, an opening **2312** extends between the first end **2308** and the second end **2310** of the outer seal housing **2102**. The recess **2224** formed in the outer surface **2226** of the second housing portion **2212** is positioned in the opening **2312**. In other words, the recess **2224** is circumferentially covered by the outer seal housing **2102**.

The outer seal housing **2102** includes an external surface **2314** having an outwardly facing surface feature **2316**. The outwardly facing surface feature **2316** may be configured as a plurality of external gripping teeth formed on the external surface **2314**. The outer seal housing **2102** may be dimensioned such that the external gripping teeth will contact the wellbore casing's **2104** inner surface **2404** when the outer seal housing **2102** is in its outwardly expanded form. According to an aspect, the gripping teeth may be shaped to frictionally grip the inner surface **2404** of the wellbore casing **2104** and thereby position the ballistically actuated plug **2100** within the wellbore casing **2104** and form a partial or total seal between the gripping teeth and the inner surface of the wellbore casing **2104** when the outer seal housing **2102** is in the expanded form. According to an aspect, the external surface **2314** includes a plurality of the outwardly facing surface features **2316** that are each configured to engage an inner surface **2404** of the wellbore casing **2104**.

According to an aspect, the outer seal housing **2102** may be a generally tubular structure that is made of a metal. According to an aspect, the outer seal housing **2102** is made from steel. The outer seal housing **2102** may be made from copper, steel or any other metal capable of being deformed or transition from an unexpanded to an expanded configuration.

One or more sealing mechanisms may be provided on the ballistic housing **2106**. For example, as illustrated in FIG. **23**, a first sealing mechanism **2304** is positioned around the outer surface **2226** of the second housing portion **2212** and a second sealing mechanism **2306** is positioned around the outer surface **2226** of the second housing portion **2212**. The first and second sealing mechanisms are spaced apart from each other, with the recess **2224** extending between them. As further illustrated, the first sealing mechanism **2304** and the second sealing mechanism **2306** are both spaced apart from the recess **2224**. According to an aspect, at least one of the first sealing mechanism **2304** and the second sealing mechanism **2306** includes an o-ring.

A seal receptacle **2318** may extend around the external surface **2314** of the outer seal housing. According to an aspect, the seal receptacle **2318** is configured to receive a sealing band **2402** (FIG. **24**). When the outer seal housing **2102** is in its expanded configuration, the sealing band **2402** engages the inner surface **2404** of the wellbore casing **2104**. According to an aspect, the sealing band **2402** may be compressed between the outer seal housing **2102** and the inner surface **2404** of the wellbore casing **2104** when the outer seal housing **2102** is in its expanded configuration.

As shown in FIG. **24**, FIG. **25**, FIG. **26**, and FIG. **27**, embodiments may include a method of sealing a wellbore casing **2104**. FIG. **24** is a cross-sectional view of a ballistically actuated wellbore tool including an initiator **2206** and a detonation extender **2222**, prior to ballistic activation. As illustrated, the initiator **2206** and the detonation extender **2222** are present, and the outer seal housing **2102** is in an

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unexpanded configuration and engaged with the ballistic housing **2106**. According to an aspect, and as illustrated in FIG. **25**, when the outer seal housing **2102** is in the expanded form, the ballistic housing **2106** is disengaged from the outer seal housing **2102**.

FIG. **25** is a cross-sectional view of the ballistically actuated wellbore tool, after ballistic activation of the initiator **2206** and the detonation extender **2222**, according to an aspect. The outer seal housing **2102** is illustrated in an expanded configuration. In the expanded configuration, the plurality of outwardly facing surface features **2316** and the sealing band **2402** of the outer seal housing **2102** is in contact with the inner surface **2404** of the wellbore casing **2104**. While the plurality of outwardly facing surface features **2316** are depicted as teeth or a jagged surface, it is contemplated that the textured outer surface may include anything that roughens or enhances frictional engagement between the outer surface of the outer seal housing **2102** and the inner surface of the wellbore casing **2104**. A sealing element or the sealing band **2402**, if provided around the body of the outer seal housing **2102**, is also compressed between the body of the outer seal housing **2102** and the inner surface **2404** of the wellbore casing **2104**.

FIG. **26** is a cross-sectional view of the ballistically actuated wellbore tool, illustrating the ballistic housing **2106** being detached from the outer seal housing **2102** after ballistic activation. In the detached configuration, the ballistic housing **2106** and any other perforating tools attached thereto is pulled in an upward direction towards a surface of the wellbore. The perforation tools may then be activated at the upward position in the wellbore, and may subsequently be withdrawn from the wellbore using a wireline or any other retrieval mechanism.

FIG. **27** is a cross-sectional view of the outer seal housing **2102** and a ball **2702** traveling towards the outer seal housing **2102**. The ball **2702** may have an outer diameter **2704** that enables it to be seated or disposed against a rim **2706** of the outer seal housing **2102**. According to an aspect, the ball **2702** may be dropped into the wellbore casing **2104** after ballistic activation and detachment of the ballistic housing **2106** from the outer seal housing **2102**, thereby sealing a flow path through the outer seal housing **2102** in its expanded form. Once the ball **2702** has sealed the flow path and isolated the upstream zone, fracking fluid may then be pumped into the wellbore casing **2104** to fracture the hydrocarbon formations via the perforations that the perforating gun(s) created.

According to an aspect, the outer seal housing **2102** may be dimensioned to receive frac balls of different sizes or different maximum outer diameters.

Example Specifications for Outer Seal Housing					
Casing OD (inches)	Casing ID	Casing Wall Thickness	Running OD Range (before activation)	Open Flow ID Range (after Activation)	% of open flow as compared to unrestricted casing
3.5	2.992	0.254	2.7 to 2.8	2.0 to 2.5	45% to 70%
4.5	4.090	0.205	3.5 to 3.9	2.7 to 3.5	44% to 73%
	4.052	0.224			44% to 75%
	4.000	0.250			46% to 77%
	3.920	0.290	3.3 to 3.7	2.8 to 3.6	44% to 84%
	3.826	0.337			46% to 89%
5.0	4.408	0.296	3.7 to 4.0	2.9 to 3.7	40% to 70%
	4.276	0.362			43% to 75%
	4.126	0.437			46% to 80%

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-continued

Example Specifications for Outer Seal Housing					
Casing OD (inches)	Casing ID	Casing Wall Thickness	Running OD Range (before activation)	Open Flow ID Range (after Activation)	% of open flow as compared to unrestricted casing
5.5	4.950	0.275	4.4 to 4.7	3.7 to 4.5	56% to 83%
	4.892	0.304			57% to 85%
	4.778	0.361	4.0 to 4.3	3.5 to 4.2	54% to 77%
	4.670	0.415			56% to 81%
	4.548	0.476			59% to 85%

FIG. **28** is a cross-sectional view of the seal housing of FIG. **27**, illustrating the ball **2702** seated in the opening **2312** at the first end **2308** of the outer seal housing **2102**, at the rim **2706**. While the ball **2702** is shown as seated at the first end **2308** of the outer seal housing **2102**, it is contemplated that the ball may be seated within any area along the length of the opening **2312**. For example, a ball **2702** having a smaller outer diameter than the inner diameter of the outer seal housing **2102** at the first end **2308** may be seated in a second seat **2802** of the outer seal housing **2102** that has a lesser inner diameter than the inner diameter at the first end **2308**. In other words, the outer seal housing **2102** may have a stepped inner surface that is able to provide a seat for receiving the outer seal housing **2102**. It is further contemplated that the outer seal housing **2102** may have a general conical configuration/inner surface profile along the opening **2312**, with the larger inner diameter being at the first end **2308** and the smaller inner diameter being at the second end **2310** of the outer seal housing **2102**.

This disclosure, in various embodiments, configurations and aspects, includes components, methods, processes, systems, and/or apparatuses as depicted and described herein, including various embodiments, sub-combinations, and subsets thereof. This disclosure contemplates, in various embodiments, configurations and aspects, the actual or optional use or inclusion of, e.g., components or processes as may be well-known or understood in the art and consistent with this disclosure though not depicted and/or described herein.

The phrases “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together.

Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” or “approximately” is not to be limited to the precise value specified. Such approximating language may refer to the specific value and/or may include a range of values that may have the same impact or effect as understood by persons of ordinary skill in the art field. For example, approximating language may include a range of $\pm 10\%$, $\pm 5\%$, or $\pm 3\%$. The term “substantially” as used herein is used in the common way understood by persons of skill in the art field with regard to patents, and may in some instances function as approximating language. In some instances, the approxi-

imating language may correspond to the precision of an instrument for measuring the value.

In this specification and the claims that follow, reference will be made to a number of terms that have the following meanings. The terms “a” (or “an”) and “the” refer to one or more of that entity, thereby including plural referents unless the context clearly dictates otherwise. As such, the terms “a” (or “an”), “one or more” and “at least one” can be used interchangeably herein. Furthermore, references to “one embodiment”, “some embodiments”, “an embodiment” and the like are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term such as “about” is not to be limited to the precise value specified. In some instances, the approximating language may correspond to the precision of an instrument for measuring the value. Terms such as “first,” “second,” “upper,” “lower” etc. are used to identify one element from another, and unless otherwise specified are not meant to refer to a particular order or number of elements.

As used herein, the terms “may” and “may be” indicate a possibility of an occurrence within a set of circumstances; a possession of a specified property, characteristic or function; and/or qualify another verb by expressing one or more of an ability, capability, or possibility associated with the qualified verb. Accordingly, usage of “may” and “may be” indicates that a modified term is apparently appropriate, capable, or suitable for an indicated capacity, function, or usage, while taking into account that in some circumstances the modified term may sometimes not be appropriate, capable, or suitable. For example, in some circumstances an event or capacity can be expected, while in other circumstances the event or capacity cannot occur—this distinction is captured by the terms “may” and “may be.”

As used in the claims, the word “comprises” and its grammatical variants logically also subtend and include phrases of varying and differing extent such as for example, but not limited thereto, “consisting essentially of” and “consisting of.” Where necessary, ranges have been supplied, and those ranges are inclusive of all sub-ranges therebetween. It is to be expected that the appended claims should cover variations in the ranges except where this disclosure makes clear the use of a particular range in certain embodiments.

The terms “determine”, “calculate” and “compute,” and variations thereof, as used herein, are used interchangeably and include any type of methodology, process, mathematical operation or technique.

This disclosure is presented for purposes of illustration and description. This disclosure is not limited to the form or forms disclosed herein. In the Detailed Description of this disclosure, for example, various features of some exemplary embodiments are grouped together to representatively describe those and other contemplated embodiments, configurations, and aspects, to the extent that including in this disclosure a description of every potential embodiment, variant, and combination of features is not feasible. Thus, the features of the disclosed embodiments, configurations, and aspects may be combined in alternate embodiments, configurations, and aspects not expressly discussed above. For example, the features recited in the following claims lie in less than all features of a single disclosed embodiment, configuration, or aspect. Thus, the following claims are

hereby incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment of this disclosure.

Advances in science and technology may provide variations that are not necessarily express in the terminology of this disclosure although the claims would not necessarily exclude these variations.

What is claimed is:

1. A ballistically actuated plug for being deployed in a wellbore casing, comprising:

- a ballistic housing comprising a first housing portion including a chamber, and a second housing portion having a noncompressible core;
 - an initiator positioned within the chamber;
 - a detonation extender in ballistic communication with the initiator and circumferentially disposed around the noncompressible core;
 - an outer seal housing circumferentially disposed around the noncompressible core, and
 - a recess formed in an outer surface of the second housing portion;
 - a first sealing mechanism positioned around the outer surface of the second housing portion; and
 - a second sealing mechanism positioned around the outer surface of the second housing portion;
- wherein the detonation extender is configured, upon detonation, to expand the outer seal housing from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing;
- the recess is configured to receive the detonation extender;
- the recess extends between the first sealing mechanism and the second sealing mechanism; and
- the first sealing mechanism and the second sealing mechanism are both spaced apart from the recess.

2. The ballistically actuated plug of claim 1, wherein a channel extends from the chamber of the first housing portion to the recess, such that at least a portion of the detonation extender is positioned in the channel.

3. The ballistically actuated plug of claim 2, wherein the detonation extender is at least one of detonating cord and a plurality of compressed explosive pellets.

4. The ballistically actuated plug of claim 2, wherein the detonation extender is a detonating cord wrapped around the noncompressible core to form at least two layers of detonating cord.

5. The ballistically actuated plug of claim 4, wherein the two layers of detonating cord are wrapped around the noncompressible core in a staggered configuration.

6. The ballistically actuated plug of claim 1, wherein the ballistic housing is formed from a single monolithic piece of metal.

7. The ballistically actuated plug of claim 1, wherein the outer seal housing further comprises:

- a first end;
 - a second end opposite the first end; and
 - an opening extending between the first end and the second end,
- wherein the recess formed in the outer surface of the second housing portion is positioned in the opening.

8. The ballistically actuated plug of claim 1, wherein the outer seal housing has an external surface, and the external surface comprises a plurality of outwardly facing surface features configured to engage an inner surface of a wellbore casing.

9. A ballistically actuated plug for being deployed in a wellbore casing, comprising:

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a ballistic housing comprising a first housing portion including a chamber, and a second housing portion extending from the first housing portion;
 an initiator positioned within the chamber;
 a detonation extender in side-by-side contact with the initiator and extending from the chamber to an outer surface of the second housing portion;
 an outer seal housing circumferentially disposed around the second housing portion;
 a recess formed in the outer surface of the second housing portion;
 a first sealing mechanism positioned around the outer surface of the second housing portion; and
 a second sealing mechanism positioned around the outer surface of the second housing portion,
 wherein the detonation extender is configured, upon detonation, to expand the outer seal housing from an unexpanded form to an expanded form to secure the outer seal housing within the wellbore casing;
 the recess is configured to receive the detonation extender;
 the recess extends between the first sealing mechanism and the second sealing mechanism, and
 the first sealing mechanism and the second sealing mechanism are both spaced apart from the recess.

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10. The ballistically actuated plug of claim 9, wherein the second housing portion comprises a noncompressible core.

11. The ballistically actuated plug of claim 9, wherein the detonation extender extends from the chamber and is circumferentially disposed around the second housing portion.

12. The ballistically actuated plug of claim 11, wherein a channel extends from the chamber of the first housing portion to the recess, such that at least a portion of the detonation extender is positioned in the channel.

13. The ballistically actuated plug of claim 12, wherein the detonation extender is a detonating cord wrapped around the second housing portion to form at least two layers of detonating cord, and wherein the two layers of detonating cord are wrapped around the second housing portion in a staggered configuration.

14. The ballistically actuated plug of claim 9, wherein the outer seal housing has an external surface, and the external surface comprises:

a plurality of outwardly facing surface features; and
 a sealing band extending around the external surface of the outer seal housing,
 wherein the plurality of outwardly facing surface features and the sealing band are configured to engage an inner surface of a wellbore casing.

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